BAYMAX Experiment Notebook

This folder explains the experiments, rather than merely listing result filenames. Each numbered note covers one training experiment, inference experiment or clearly identified measurement/audit investigation. Multiple evaluations of the same checkpoint can live in its note; a 30-case diagnostic does not become a separate trained model.

Start with a question or follow the stages below. Numbers within folders are reading order, not invented run IDs or a guarantee of exact historical chronology. Every evaluation table links to its original JSON and records its saved date.

Latest completed runs: unique 2k training pool at two epochs and the same 2k pool at four epochs. The four-epoch run records 73/150 successes, macro-F1 0.681 and zero invalid outputs. The owner closed the project on 2026-10-17. These are single-seed development results with remaining clarification errors, not a production claim. The grounding revision did not improve the selected unweighted compact/name-first model; it is retained as the corrected 1k parent for the expansion, not promoted as the reference. The historical summary below describes earlier experiments.

Where We Stand

The selected development reference is cleaned anchored SFT at two epochs. It is the reference for balanced routing, not the winner of every metric.

Candidate	Task successes / 150	Routing macro-F1	Invalid outputs / 150	Premature ACT on boundary cases
Anchored two-epoch SFT	44	0.514	2	31
DPO from that reference	53	0.376	5	56
2:1 SFT	29	0.477	8	22
DPO from 2:1 SFT	31	0.453	7	34
DPO with completed ACT rejections	46	0.482	7	45
Rank-16 two-epoch SFT	46	0.450	7	42
Lower-learning-rate two-epoch SFT	45	0.436	15	37
QKVO two-epoch SFT	52	0.437	14	40

Those are measured development results, not evidence of a robust overall win. First DPO has the highest task completion but substantially worse abstention. Completing its missing rejections recovers some balance, yet remains below original SFT macro-F1. Rank 16 also worsened routing and validity despite two additional task successes. Lower learning rate did not improve balance either. QKVO completes more tasks but has worse

routing and output validity: retain it as a task-completion candidate, not the balanced reference. A's original rank-8 recipe remains the reference. No model switch or new training method was implemented by this documentation task.

The full recipes, matrices, sources and caveats are in the individual notes. For the most recent sequence, read the two-epoch reference, four-epoch trial, matched contrasts, 2:1 trial, and the three DPO notes in that order.

How an Experiment Flows

1. Author or import training examples under the general policy and tool contracts.
2. Protect source train/validation/test membership and contrast/content groups.
3. Mix and sample a deterministic raw training cap.
4. Render the chat template, tokenize, mask non-assistant labels and truncate.
5. Inspect the usable artifact that actually reaches the trainer.
6. Train a new versioned adapter.
7. Run the chosen adapter on a separate evaluation suite through fake adapters.
8. Read task outcome, routing, output validity and wording metrics separately.

Pipeline fixes at steps 2-5 or 7-8 can change the validity of the evidence without creating a better trained model. That distinction explains several apparent jumps in the history.

Vocabulary

Term	Meaning in these notes
ACT	Select supported productive tools when the request is safe and complete.
CLARIFY	Supported safe action needs missing/ambiguous information or required confirmation.
REFUSE	Action/capability is unsafe, unavailable, unsupported or out of scope.
Meta-tool	request_clarification(question) or refuse_request(reason); not an external mutation.
xLAM	External Salesforce function-calling dataset hosted on Hugging Face.
BAYMAX-control	Authored training data; "control" is the dataset name, not necessarily a baseline model.
Mixed training	xLAM plus authored BAYMAX examples.
SFT	Learn supplied correct assistant answers using target-token cross-entropy.
LoRA	Train small adapter weights rather than updating every base-model weight.

Term	Meaning in these notes
DPO	Learn to prefer a chosen answer over a rejected answer for the same prompt.
Anchored sampling	Keep reference examples/positions and replace only explicitly eligible rows.
Matched contrast	Closely related requests/states/tools with a genuine decision-changing difference.
Raw cap	Number selected before tokenization and assistant-label filtering.
Usable rows	Examples retaining trainable assistant labels after preprocessing.
Epoch	One pass over the usable training artifact; not one optimizer step.
Exact duplicate	Identical example content; not a test of semantic or template similarity.
Reference freeze	Record hashes to detect drift; it does not lock or copy files.

Metrics Are Different Questions

- **Task success:** did all scenario checks pass, including arguments and message criteria?
- **Routing macro-F1:** average ACT, CLARIFY and REFUSE F1 equally. F1 combines precision (correctness of predictions of that class) and recall (coverage of expected cases of that class). Invalid/unknown outputs miss their expected class.
- **Decision accuracy:** correctness among recorded known decisions. Read it with coverage; unknown outputs are not included in its conditional denominator.
- **Clarification/refusal accuracy:** required question/reason keyword matches. A correct label can still have an incomplete or misleading explanation.
- **Invalid-output rate:** strict parser rejections, separate from valid but inappropriate decisions or tool-execution failures.
- **Hallucination rate:** currently unavailable tool names, not every invented person, date or success claim. Invalid parsing can leave no accepted calls to count.
- **Latency/cost:** local reported API cost zero is not zero electricity, hardware or training cost. Replayed results inherit original inference timing.

Tables use ratios from zero to one, rounded to three decimals, unless they show explicit counts. `n/a` means unavailable or inapplicable, not zero.

Which Evaluation Suite?

Suite	Size	Purpose
v1	50	Original benchmark, used by early trials.
Combined v1 + v2	150	Established development comparison. Current decision support is ACT 81, CLARIFY 40, REFUSE 29.

Suite	Size	Purpose
Template shift	30	Separate wording/structure diagnostic: ten cases per decision.
Refusal subset	7	Small single-class diagnostic, not balanced overall evaluation.
Smoke subsets	1-4	Sanity checks; not estimates of full benchmark quality.

Historical policy-era fixtures/scoring differ from current measurements. Do not combine the template-shift and established-suite rates into one score. Their repeatedly inspected status means neither is an untouched final benchmark.

Early Training

The first external-only, synthetic-only and mixed trials. Historical provenance is incomplete.

- External Tool Data Only: First Local Fine-Tune
- BAYMAX Control Data Only
- Control Data With Contrast Pairs
- Control Data With Targeted Refusals
- Original External and BAYMAX Training Mix
- Expanded Control Generator: Early Mixed Trial
- Weighted Control Examples: Early Oversampling Trial
- Clarification and Refusal as Explicit Meta-Tools

Format, Prompt and Sampling

Input alignment, weighting, unique-first caps and inference-only prompt changes.

- Aligning Training and Evaluation Formats
- Extra Clarification and Refusal Repeats
- Legacy Boundary-Weighting Trial
- Fixed Unique Boundary Pool at 1,000 Rows
- Increasing the Cap to 1,500 Without Repeating Boundaries
- First Policy Prompt: Evaluation Without Retraining
- Expanded Policy Prompt: Evaluation Without Retraining
- Returning to the Recorded Training Prompt

Data Quality and Class Mix

Changes in class quantities, partition integrity, policy, credentials and target compatibility.

- One Hundred ACT, CLARIFY and REFUSE Examples

- Early Balanced BAYMAX Data at Two Epochs
- Schema-Diverse Control Examples
- Increasing Productive BAYMAX Examples to Two Hundred
- Productive Midpoint: One Hundred Fifty BAYMAX ACT Examples
- Content-Group Splits With Seventy Percent ACT
- Corrected Policy, Credential Filtering and Representative Validation
- Corrected Policy With Two Hundred Productive Examples
- Target-Contract Cleaning With Resampled External Rows
- Target-Contract Cleaning With Anchored External Rows

Training Duration and Matched Contrasts

Fresh one/two/four-epoch comparisons, matched groups and the 2:1 SFT ratio.

- Anchored Data at Two Epochs: Selected Reference
- Anchored Data at Four Epochs
- Matched Contrasts at Fixed Class Counts
- Two-to-One Productive-to-Boundary SFT Ratio

SFT Hyperparameters

Controlled adapter/training changes from the selected SFT reference. B, C and E are completed. Increasing rank, lowering the learning rate and adding attention projections did not beat A's balanced routing. E improved task completion, with output-contract failures documented separately.

- Experiment B: Rank 16 at the Fixed Two-Epoch SFT Recipe
- Experiment C: Lower Learning Rate at Two Epochs
- Experiment E: LoRA on All Four Attention Projections

Output-Contract Learning

Separate structural expression from policy judgment. Preparation is implemented; the next run changes only assistant call-key order using E's unchanged recipe.

- Name-First Tool Calls: Serialization-Only SFT on E
- Held-Out SFT Clarification: Ten-Example Sanity Check
- SFT Versus Benchmark: Prompt Audit and JSON-Presentation Check
- Compact JSON: Manual Review of Ten Clarification Questions
- Held-Out SFT: Question Quality Versus Supplied Targets
- Capped Training Data: Ten Clarification Targets
- Generator 15: Urgency Without an Invented Deadline
- Capped Training Questions: Second Ten-Target Batch

- Training Labels: Counts and Decoded-Target Check
- In-Sample Clarification: Frozen Training-Input Probe
- Existing Checkpoint: Training Loss by Decision Class
- Tool Names Versus Clarification and Refusal Text
- First Tool-Name Loss Weight: Three Instead of One
- First Tool-Name Weighting: Paired Lost and Gained Successes
- First Tool-Name Loss Weight: Two Instead of Three
- Held-Out SFT: Plan Completeness and Explanation Grounding
- Training Targets: Missing Fields and State Grounding
- Controlled Training-Data Revision: Policy and Grounding Variation

Preference Training

The original DPO stage, alternative SFT initialization, and completed ACT rejections.

- DPO From the Selected Two-Epoch SFT Reference
- DPO From the Two-to-One SFT Model
- Completing Missing Productive Rejections for Boundary Decisions

Measurement and Data Audits

Baseline interpretation and engineering corrections. These are not new model-training wins.

- Untrained Qwen: What the Baseline Actually Measures
- Effective Token Distribution and Training-Cap Audit
- Strict Parser Replay: Correcting Measurement, Not Training
- Training Data Contract and Policy Review
- Decision Metrics Separate From Explanation Wording
- Clarification Output Contract: Target Audit and Failure Diagnostics

Rules for Reading Comparisons

1. Compare the same scenario suite, prompt, generation settings, parser and scorer.
2. Distinguish a new trained checkpoint from an inference-only prompt test or strict replay of saved generations.
3. Check actual usable rows and tokens. A 1,000-row raw cap can deliver 974 rows.
4. A 200/100/100 BAYMAX quota plus 600 xLAM ACT examples is 800/100/100 overall, not a balanced three-class mixture. Multi-tool sequences still count as ACT.
5. Matching broad quantities does not guarantee matching domain/subtype coverage.
6. A fresh four-epoch run is not resuming a saved two-epoch adapter. A changed cosine horizon also changes per-step learning rates.

7. A 705-pair and an 800-pair DPO run at one epoch have different training exposure.
8. Single-seed changes are observations, not statistical proof or established causes.
9. Do not turn an inspected evaluation failure into a fixture-specific training target.
10. Do not assign today's config defaults to an old checkpoint with missing provenance.

Reproducibility and Sources

The external source revision used by the pinned pipeline is `26d14ebfe18b1f7b524bd39b404b50af5dc97866`. The later matched training prompt is identified by `a31446de6c8b67f1fc8ba8fc09ed8e2d7425ba832334dc8fd6fe2d55ef978468`.

Configs can declare context 2,048/batch two/accumulation eight while owner low-VRAM commands actually use 1,024/one/16. Later notes state actual reported overrides. Earlier notes explicitly retain missing information instead of filling it by guess.

Original evidence remains in the result folder. That older catalog predates some recent runs; use these notes for their explanations. FILE_RENAMES.json resolves historical names without changing result bytes. Config links identify recipes; generated model/data links are local-only and may not exist on another checkout.

Methodology remains defined by BENCHMARKING.md and SCORING.md. Detailed investigations remain in TRAINING_DATA_AUDIT.md and DPO_EXPERIMENT.md. Some older status paragraphs describe preparations that later completed; measured result artifacts take priority.

This notebook covers all 65 saved Phase 2 evaluation JSON files present on 2026-10-10, plus early external-only local results and supporting audits. No raw result, training example, evaluation target or model weight was changed. No long training/evaluation/audit command was run. These notes are not staged, committed or pushed; the owner's restriction on uploading Phase 2 documents remains.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

External Tool Data Only: First Local Fine-Tune

Experiment index

Question and Context

Could ordinary function-calling SFT make local Qwen better at BAYMAX tasks before adding domain-specific clarification and refusal training?

This is the xLAM-only starting point. Its files live under Phase 1 because that is where the first local comparison was stored; it belongs in the training history nonetheless.

Setup

- Base model: Qwen/Qwen2.5-1.5B-Instruct.
- External source: Salesforce/xLAM-function-calling-60k, downloaded through Hugging Face.
- Owner's medium-run logs describe a 1,000-row training cap, 200 validation rows, one epoch, context 1,024, batch one and gradient accumulation 16.
- LoRA requested: rank 8, alpha 16, q_proj/v_proj, dropout 0.05, no bias.
- The original medium run reported NaN validation loss. A later fixed run was evaluated; the saved result names do not establish which precise changes explain the fix.
- Bare Qwen is the comparison, not the scripted baseline or hosted GPT model.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Bare Qwen: full suite	50	11/50 (0.220)	0.540	0.869	0.000	0.071
xLAM-only LoRA: full suite	50	12/50 (0.240)	0.480	0.854	0.028	0.000
xLAM-only LoRA: legacy full suite	50	12/50 (0.240)	0.480	0.854	0.028	0.000
Bare Qwen: one-case check	1	0/1 (0.000)	1.000	0.500	n/a	n/a
Bare Qwen: four-case smoke	4	0/4 (0.000)	0.250	0.875	0.000	n/a

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
xLAM-only LoRA: one-case check	1	0/1 (0.000)	1.000	0.750	n/a	n/a
xLAM-only LoRA: three-case smoke	3	0/3 (0.000)	0.000	1.000	0.000	n/a

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Result Provenance

- Bare Qwen: full suite: 2026-08-11 UTC, 50 cases.
- xLAM-only LoRA: full suite: 2026-08-11 UTC, 50 cases.
- xLAM-only LoRA: legacy full suite: 2026-08-11 UTC, 50 cases.
- Bare Qwen: one-case check: 2026-08-11 UTC, 1 cases.
- Bare Qwen: four-case smoke: 2026-08-11 UTC, 4 cases.
- xLAM-only LoRA: one-case check: 2026-08-11 UTC, 1 cases.
- xLAM-only LoRA: three-case smoke: 2026-08-11 UTC, 3 cases.

What We Learned

The full-suite gain was one additional task: 12/50 versus 11/50 for bare Qwen. Tool-call accuracy fell, and clarification/refusal remained very weak. This was evidence that training and local inference could run, not a convincing policy-learning result.

The one-, three- and four-case files are smoke diagnostics. They should not be compared to the full-suite rates as if they were additional full experiments.

Limits

These are historical parser/prompt measurements. Exact base revision and complete runtime provenance are not recoverable from these result files alone. The legacy and later full artifacts have the same aggregate success but different latency; that does not prove identical generations.

The NaN loss was a training diagnostic, not evidence that all later checkpoints were invalid. Do not infer a causal fix from filenames alone.

Supporting Files

- qwen2_5_1_5b_lora_v1.json

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

BAYMAX Control Data Only

Experiment index

Question and Context

Would training directly on BAYMAX's tools and boundary decisions help more than generic external tool data?

"Control" here means the authored BAYMAX-control dataset. It does not mean an untrained experimental control.

Setup

Synthetic training requests covered productive actions, clarification and refusal, using BAYMAX interfaces rather than evaluation fixtures. This checkpoint was evaluated without mixing xLAM into its training source.

The current control config is a navigation aid, not proof of the exact settings used by this historical checkpoint. The saved evaluation files do not record a complete frozen training recipe.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Control-only	50	2/50 (0.040)	0.500	0.578	0.000	0.071
Control-only / early parser fix	50	10/50 (0.200)	0.520	0.806	0.083	0.071

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Result Provenance

- Control-only: 2026-08-22 UTC, 50 cases.
- Control-only / early parser fix: 2026-08-22 UTC, 50 cases.

Interpretation

The original evaluation recorded 2/50 successes. The early-parser-fix artifact recorded 10/50. That large difference makes measurement behavior part of the explanation: it cannot be credited wholly to learning.

Even under the later measurement, tool selection and boundary behavior were weak. Purely authored data did not establish a robust replacement for external function-calling data.

Limits

These two artifacts belong to one checkpoint investigation, not proof of two different training recipes. The precise parser-fix execution provenance is incomplete; unlike later strict rescoring, the filename alone does not establish that inference was not rerun.

Older synthetic partitioning and LoRA-initialization issues limit retrospective claims of clean controlled comparisons.

Supporting Files

- qwen2_5_1_5b_lora_baymax_control_v1.json

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

Control Data With Contrast Pairs

Experiment index

Question and Context

Could related ACT and non-ACT examples teach the model the difference between a complete request and one that should not be acted on?

This was an early contrast-pair trial, before the later anchored matched-group experiment.

Setup

The training source remained BAYMAX synthetic control data. Contrast pairs were added to make behavioral alternatives more explicit.

The historical result does not pin the selected pairs, effective token distribution, checkpoint hash or exact prompt. Do not substitute today's generator behavior for the missing historical recipe.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Control contrast pairs	50	12/50 (0.240)	0.530	0.767	0.111	0.071

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Result Provenance

- Control contrast pairs: 2026-08-22 UTC, 50 cases.

Interpretation

The run completed 12/50 tasks. Clarification and refusal keyword scores were still low. This supported further investigation of boundary training, but did not demonstrate that contrast pairs generalize reliably.

Limits

Comparisons with the previous control evaluation are historical, not a verified single-variable ablation. The later matched experiment changes group selection and coverage much more explicitly; it is not a repeat of this trial.

Supporting Files

The result links above are the surviving primary evidence. A separate immutable training record was not established for this historical experiment.

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

Control Data With Targeted Refusals

Experiment index

Question and Context

Could explicit refusal examples reduce unsupported actions when early control training still rarely refused?

Setup

Refusal-focused synthetic examples were added to the control-data path. A full 50-case evaluation and later seven-case prompt diagnostics survive.

The seven-case outputs are refusal-only diagnostics, not 50-case benchmarks. Some of those tests changed the inference prompt or parser; their numbers must not be attributed solely to training.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Targeted refusals	50	12/50 (0.240)	0.500	0.759	0.111	0.143
Targeted refusals / prompt-only	7	0/7 (0.000)	0.429	1.000	n/a	0.000
Control / policy prompt	7	0/7 (0.000)	0.429	1.000	n/a	0.071
Control / policy prompt / early parser fix	7	1/7 (0.143)	0.429	1.000	n/a	0.143

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Result Provenance

- Targeted refusals: 2026-08-22 UTC, 50 cases.
- Targeted refusals / prompt-only: 2026-09-15 UTC, 7 cases.
- Control / policy prompt: 2026-09-15 UTC, 7 cases.
- Control / policy prompt / early parser fix: 2026-09-15 UTC, 7 cases.

Interpretation

The full run completed 12/50 tasks and had refusal wording accuracy 0.143. Most seven-case diagnostics still recorded no strict task success; the later policy-prompt/parser-fix diagnostic recorded one.

Giving the model refusal examples did not by itself resolve boundary routing or explanation quality.

Limits

The saved files do not fully recover the authored-data changes or repeat counts. Single-class subset rates cannot establish balanced ACT/CLARIFY/REFUSE performance. Parser and prompt changes are additional variables.

Supporting Files

The result links above are the surviving primary evidence. A separate immutable training record was not established for this historical experiment.

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

Original External and BAYMAX Training Mix

Experiment index

Question and Context

Could generic xLAM tool-use training and BAYMAX-specific boundary examples complement each other?

This is the early "old mix" reference. It is not the later format-aligned or anchored reference.

Setup

Training combined external xLAM examples with authored BAYMAX control examples. The source recipes evolved, and the result files do not preserve a complete immutable account of the exact selected rows or sampling weights.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Original mixture	50	12/50 (0.240)	0.420	0.829	0.028	0.000
Original mixture / early parser fix	50	12/50 (0.240)	0.420	0.829	0.028	0.000

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Result Provenance

- Original mixture: 2026-08-22 UTC, 50 cases.
- Original mixture / early parser fix: 2026-08-22 UTC, 50 cases.

Interpretation

Both historical evaluations recorded 12/50 task successes, clarification accuracy 0.028 and refusal accuracy zero. Mixing the sources did not automatically teach abstention.

The initial source formats were not fully matched to evaluation: plain user queries and generic function arrays were being compared with structured time/state requests and BAYMAX control decisions. The later format-alignment experiment addresses that distribution shift.

Limits

Identical aggregate numbers in the early-parser-fix file are not proof that every output was identical. These artifacts predate later parser and split-integrity protections.

Use the linked original token audit for the distribution observed in its own 5,000-row artifact, not as a description of every early mixed checkpoint.

Supporting Files

- original-mix-token-audit-5000-and-boundary-failures.json

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

Expanded Control Generator: Early Mixed Trial

Experiment index

Question and Context

Would broader authored clarification, refusal and confirmation examples improve the mixed model?

Setup

This is the synthetic-v7 mixed-data trial. The artifact is named "smoke", but its evaluation contains all 50 v1 scenarios. Evaluation size and training size are different facts.

The user discussed a 5,000-row trial during this period. The surviving result does not independently pin that exposure, so the note does not treat the filename as a training manifest.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Expanded control mix	50	15/50 (0.300)	0.500	0.865	0.000	0.000

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Result Provenance

- Expanded control mix: 2026-09-15 UTC, 50 cases.

Interpretation

Task success reached 15/50, but clarification and refusal keyword metrics were both zero. Better productive task completion did not establish better boundary judgment.

Limits

No frozen effective class/token counts or exact recipe are stored in the result. Do not compare this 50-case rate directly with later 150-case rates. The generator version is a historical label, not an immutable checkpoint identity.

Supporting Files

The result links above are the surviving primary evidence. A separate immutable training record was not established for this historical experiment.

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

Weighted Control Examples: Early Oversampling Trial

Experiment index

Question and Context

Would increasing the exposure of clarification and refusal examples make the model use those decisions?

Setup

A weighted/oversampled version of the synthetic-v7 mixture was trained. Weighting changes how often an existing example appears; it does not add semantic diversity.

The precise repeat settings and effective training prefix are not pinned in the saved evaluation. Later artifact audits were added because intended source weights did not reliably describe the data surviving a cap.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Weighted control mix	50	11/50 (0.220)	0.540	0.809	0.028	0.000

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Result Provenance

- Weighted control mix: 2026-09-15 UTC, 50 cases.

Interpretation

The run completed 11/50 tasks, versus 15/50 in the early unweighted v7 artifact. Clarification remained near zero and refusal remained zero.

This particular weighting attempt did not establish improvement. It does not prove all forms of weighting are ineffective.

Limits

The comparison is historical and lacks sufficient provenance for a clean causal attribution. Repetition can crowd out unique data, but these metrics alone cannot demonstrate memorization.

Supporting Files

The result links above are the surviving primary evidence. A separate immutable training record was not established for this historical experiment.

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

Clarification and Refusal as Explicit Meta-Tools

Experiment index

Question and Context

Was the model failing to abstain because clarification and refusal were not presented as recognizable actions?

Setup

The synthetic-v8 weighted path represented non-productive decisions as explicit calls:

- request_clarification with arguments.question.
- refuse_request with arguments.reason.

The local runner converts those calls into a BAYMAX response decision and message, rather than executing them as external mutations.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Meta-tool mixture	50	12/50 (0.240)	0.540	0.840	0.000	0.000

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Result Provenance

- Meta-tool mixture: 2026-09-15 UTC, 50 cases.

Interpretation

The run completed 12/50 tasks, but both wording metrics were zero. Adding action names alone did not resolve the problem.

The next important investigation was whether the model saw the same prompt, tool definitions and structured request wrapper during training and evaluation.

Limits

A zero keyword metric does not alone show that every model decision was ACT. Explicit routing metadata was not available in this artifact. The exact oversampling recipe is also not preserved.

Supporting Files

The result links above are the surviving primary evidence. A separate immutable training record was not established for this historical experiment.

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

Aligning Training and Evaluation Formats

Experiment index

Question and Context

Was a training/evaluation format mismatch preventing the model from using clarification and refusal?

Generic examples taught a plain request plus tools. BAYMAX evaluation supplied a structured request with current_time, initial_state and request, and injected two meta-tools. Those are meaningfully different input distributions.

Change

Align the structured wrapper, shared meta-tool definitions, names, ordering and system-prompt expectations between training and local evaluation. Non-ACT targets remain actual JSON calls, not prose-only answers.

This checkpoint became the historical "aligned" reference. It is different from the later target-compatible anchored two-epoch reference.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Format-aligned	50	14/50 (0.280)	0.580	0.853	0.056	0.071
Format-aligned / historical settings	150	42/150 (0.280)	0.473	0.771	0.083	0.019
Format-aligned / matched training prompt	150	40/150 (0.267)	0.453	0.771	0.025	0.017
Format-aligned	30	5/30 (0.167)	0.517	0.817	0.100	0.100

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Result Provenance

- Format-aligned: 2026-09-16 UTC, 50 cases.
- Format-aligned / historical settings: 2026-09-26 UTC, 150 cases.
- Format-aligned / matched training prompt: 2026-09-26 UTC, 150 cases.
- Format-aligned: 2026-09-27 UTC, 30 cases.

Interpretation

On v1-50, success was 14/50 and clarification/refusal wording scores became nonzero. On the historical combined-150 evaluation, success was 42/150.

When tested with the recorded training prompt on combined-150, success was 40/150; template-shift success was 5/30. Alignment helped the engineering contract, but generalization remained weak.

Limits

The evaluations differ in suite, prompt and policy-era scoring; they are not repeated measurements of one fixed benchmark setup. Historical results do not fully pin selected training rows or base-model revision.

A format correction is necessary infrastructure, not proof that the decision policy has been learned robustly.

Supporting Files

The result links above are the surviving primary evidence. A separate immutable training record was not established for this historical experiment.

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

Extra Clarification and Refusal Repeats

Experiment index

Question and Context

The aligned model still acted prematurely. Could greater exposure to boundary examples help without changing LoRA settings?

Change

The named recipe added six extra clarification repeats and twelve extra refusal repeats. Repeats increase exposure to existing examples; they are not six and twelve newly authored unique requests.

Post-mix token audits were retained in two snapshots. Treat each snapshot as evidence about its reported artifact, not automatic provenance for the evaluated checkpoint.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Extra boundary repeats	50	8/50 (0.160)	0.420	0.689	0.028	0.000

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Result Provenance

- Extra boundary repeats: 2026-09-16 UTC, 50 cases.

Interpretation

Success was 8/50, below the historical aligned 14/50. Clarification wording accuracy remained low and refusal was zero.

This trial did not support continuing to increase repetition blindly. It motivated examining what actually reached training after shuffling, capping and label filtering.

Limits

The saved evaluation does not pin every training knob or the effective cap. Exact snapshot-to-checkpoint identity is incomplete. No memorization diagnosis follows from repetition counts alone.

Supporting Files

- clarification-extra6-refusal-extra12-token-audit-5000-snapshot1.json
- clarification-extra6-refusal-extra12-token-audit-5000-snapshot2.json

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

Legacy Boundary-Weighting Trial

Experiment index

Question and Context

This is a separate early weighting result that could otherwise be confused with the extra-six/extra-twelve run.

Recoverable Setup

The stored filename identifies a boundary-weighting trial historically called "legacy13". The evaluation contains 50 scenarios. Its exact repeat factors and training recipe are not established by that label.

Two token-audit snapshots exist. They should be inspected as separate snapshots; neither should be silently assumed to describe all historical weighting trials.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Legacy weighting	50	12/50 (0.240)	0.560	0.783	0.111	0.000

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Result Provenance

- Legacy weighting: 2026-09-16 UTC, 50 cases.

Interpretation

Success was 12/50, clarification wording accuracy 0.111 and refusal zero. It did not solve refusal behavior.

Limits

The recipe is not sufficiently pinned for exact reproduction or a one-variable comparison with the other weighted runs. This note deliberately preserves that uncertainty rather than assigning today's configuration to an old result.

Supporting Files

- boundary-weighting-legacy13-token-audit-5000-snapshot1.json
- boundary-weighting-legacy13-token-audit-5000-snapshot2.json

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

Fixed Unique Boundary Pool at 1,000 Rows

Experiment index

Question and Context

Global shuffling and truncation could retain repeated boundary examples while discarding newly authored unique examples. Could deliberate unique-first sampling produce a better training artifact?

Change

Preserve a fixed BAYMAX contribution and fill the rest of the cap with unique xLAM rows. The observed 1,000-row raw cap contained 150 BAYMAX examples, including 60 unique refusals and 45 unique clarifications.

The owner's post-tokenization report retained 967 rows: 411 direct calls, 451 multi-tool sequences, 60 refusals and 45 clarifications. BAYMAX remained 150; xLAM retained 817. Exact duplicates were zero.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Boundary pool / 1,000 cap	50	12/50 (0.240)	0.560	0.746	0.167	0.143
Boundary pool / 1,000 cap / historical settings	150	37/150 (0.247)	0.513	0.738	0.083	0.056
Boundary pool / 1,000 cap / matched training prompt	150	37/150 (0.247)	0.493	0.738	0.050	0.086

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Result Provenance

- Boundary pool / 1,000 cap: 2026-09-16 UTC, 50 cases.
- Boundary pool / 1,000 cap / historical settings: 2026-09-26 UTC, 150 cases.
- Boundary pool / 1,000 cap / matched training prompt: 2026-09-26 UTC, 150 cases.

Interpretation

The 150-case training-prompt evaluation completed 37/150 tasks. This sampler solved the exact-duplicate problem in the reported artifact; that is separate from proving strong policy transfer.

Limits

Earlier cap audits had different counts. The numbers above describe the corrected report, not every artifact historically called boundary-diverse.

The 50-case, historical-prompt and training-prompt evaluations are different measurement settings. Older split/initialization defects still limit causal claims.

Supporting Files

The result links above are the surviving primary evidence. A separate immutable training record was not established for this historical experiment.

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

Increasing the Cap to 1,500 Without Repeating Boundaries

Experiment index

Question and Context

Would more unique external training data help once the sampler stopped duplicating the fixed BAYMAX boundary pool?

Change

Increase the raw cap from 1,000 to 1,500, keeping BAYMAX at 150, including the same absolute quotas of 60 refusals and 45 clarifications. Extra capacity is filled primarily with unique xLAM rows.

The reported usable artifact had 1,447 rows: 657 direct, 685 multi-tool, 60 refusal and 45 clarification. Source counts were BAYMAX 150 and xLAM 1,297, with zero exact duplicates.

Refusal/clarification token shares fell as generic ACT exposure increased. A fixed row contribution does not mean a fixed class proportion.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Boundary pool / 1,500 cap	50	10/50 (0.200)	0.500	0.725	0.056	0.071
Boundary pool / 1,500 cap / historical settings	150	35/150 (0.233)	0.487	0.708	0.071	0.074
Boundary pool / 1,500 cap / matched training prompt	150	34/150 (0.227)	0.473	0.708	0.013	0.103

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Result Provenance

- Boundary pool / 1,500 cap: 2026-09-26 UTC, 50 cases.
- Boundary pool / 1,500 cap / historical settings: 2026-09-26 UTC, 150 cases.

- Boundary pool / 1,500 cap / matched training prompt: 2026-09-26 UTC, 150 cases.

Interpretation

Under the matched training prompt, success was 34/150 versus 37/150 for the 1,000-row candidate. More rows did not yield a broad improvement in this run.

Limits

Changing the cap also changes optimizer exposure and class proportions. This is not an isolated test of semantic diversity alone. Single runs do not establish that a larger dataset is generally harmful.

Supporting Files

The result links above are the surviving primary evidence. A separate immutable training record was not established for this historical experiment.

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

First Policy Prompt: Evaluation Without Retraining

Experiment index

Question and Context

Would explicitly describing ACT, CLARIFY and REFUSE improve the existing models at inference time?

Change

Evaluate bare Qwen and the existing aligned adapter with the first decision-policy prompt. The aligned checkpoint was not retrained with this new prompt.

The intended policy gives refusal precedence for unsafe/unavailable capabilities, clarification for recoverable missing details or required confirmation, and ACT for supported complete requests.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Bare Qwen / policy prompt v1	150	37/150 (0.247)	0.537	0.827	0.013	0.034
Format-aligned / policy prompt v1	150	26/150 (0.173)	0.433	0.653	0.100	0.069

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Result Provenance

- Bare Qwen / policy prompt v1: 2026-09-26 UTC, 150 cases.
- Format-aligned / policy prompt v1: 2026-09-26 UTC, 150 cases.

Interpretation

The new prompt measured 37/150 successes for base Qwen and 26/150 for aligned. That contrast raised a prompt-distribution-shift concern: a checkpoint trained with one prompt may behave differently under a new policy prompt.

Limits

This is a prompt-only experiment, not evidence of policy-aligned fine-tuning. Historical policy-era fixtures/scoring and previous prompt evaluations also differ, so not every historical comparison is a controlled prompt ablation.

The saved failure summaries diagnose the outputs of these runs. They are not labels to use for fixture-specific training.

Supporting Files

- qwen-base-policy-prompt-v1-failure-summary.txt
- format-aligned-policy-prompt-v1-failure-summary.txt

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

Expanded Policy Prompt: Evaluation Without Retraining

Experiment index

Question and Context

Could a more explicit policy explanation correct the shortcomings of the first inference prompt?

Change

Use an expanded policy prompt on the same base and aligned model candidates. No newly trained policy-prompt adapter is established by these artifacts.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Bare Qwen / policy prompt v2	150	5/150 (0.033)	0.480	0.521	0.113	0.052
Format-aligned / policy prompt v2	150	6/150 (0.040)	0.447	0.492	0.237	0.190

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Result Provenance

- Bare Qwen / policy prompt v2: 2026-09-26 UTC, 150 cases.
- Format-aligned / policy prompt v2: 2026-09-26 UTC, 150 cases.

Interpretation

Task success dropped to 5/150 for base Qwen and 6/150 for aligned, while clarification/refusal wording scores rose. More boundary-language matches did not imply better overall behavior.

A longer or clearer human-readable instruction is not automatically a better prompt for a small checkpoint trained under a different format.

Limits

These are single inference evaluations. They do not prove prompt length itself caused the regression, nor that Qwen cannot learn the policy. Do not relax evaluation targets to turn the wording gains into successes.

Supporting Files

The result links above are the surviving primary evidence. A separate immutable training record was not established for this historical experiment.

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

Returning to the Recorded Training Prompt

Experiment index

Question and Context

Were changes in inference instructions making checkpoint comparisons misleading?

Change

Evaluate base Qwen, aligned and both boundary-diverse candidates with the same recorded training prompt. Its SHA256 is:

`a31446de6c8b67f1fc8ba8fc09ed8e2d7425ba832334dc8fd6fe2d55ef978468`

This did not retrain the adapters. It restored a matched inference prompt for the comparison.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Bare Qwen / matched training prompt	150	32/150 (0.213)	0.493	0.791	0.025	0.034
Format-aligned / matched training prompt	150	40/150 (0.267)	0.453	0.771	0.025	0.017
Boundary pool / 1,000 cap / matched training prompt	150	37/150 (0.247)	0.493	0.738	0.050	0.086
Boundary pool / 1,500 cap / matched training prompt	150	34/150 (0.227)	0.473	0.708	0.013	0.103

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Result Provenance

- Bare Qwen / matched training prompt: 2026-09-26 UTC, 150 cases.
- Format-aligned / matched training prompt: 2026-09-26 UTC, 150 cases.
- Boundary pool / 1,000 cap / matched training prompt: 2026-09-26 UTC, 150 cases.

- Boundary pool / 1,500 cap / matched training prompt: 2026-09-26 UTC, 150 cases.

Interpretation

Aligned recovered to 40/150 successes, compared with 26/150 under policy prompt v1 and 6/150 under v2. Prompt compatibility was a real measurement concern.

Neither boundary-diverse candidate dominated aligned: additional boundary exposure improved some abstention metrics while reducing other task outcomes.

Limits

This comparison does not prove that the old training prompt is the best possible policy. It demonstrates why prompt changes must be tracked independently from training changes.

The historic suite policy and parser later changed. Keep these artifacts as historical evidence, not directly interchangeable with today's strict-parser results.

Supporting Files

The result links above are the surviving primary evidence. A separate immutable training record was not established for this historical experiment.

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

One Hundred ACT, CLARIFY and REFUSE Examples

Experiment index

Question and Context

Could a larger unique BAYMAX contribution improve on the fixed 150-example pool without oversampling exact duplicates?

Change

The 1,000-row mixture targeted 300 unique BAYMAX examples: 100 productive ACT, 100 CLARIFY and 100 REFUSE, plus 700 generic xLAM productive examples.

"Balanced BAYMAX" is not "balanced full dataset": the xLAM portion still adds ACT targets, including multi-tool sequences.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
100/100/100 BAYMAX	150	43/150 (0.287)	0.567	0.761	0.050	0.172
100/100/100 BAYMAX	30	6/30 (0.200)	0.600	0.800	0.200	0.150
100/100/100 BAYMAX / strict replay	150	42/150 (0.280)	0.500	0.694	0.013	0.172
100/100/100 BAYMAX / strict replay	30	6/30 (0.200)	0.533	0.733	0.200	0.100

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Routing and Output Validity

Evaluation	Macro-F1	Invalid-output rate
100/100/100 BAYMAX / strict replay (150 cases)	0.367	0.113

Evaluation	Macro-F1	Invalid-output rate
100/100/100 BAYMAX / strict replay (30 cases)	0.435	0.100

Routing values use the existing comparison helper on recorded per-scenario decisions; missing historical aggregate fields are derived without rewriting source files. Invalid output and unknown decision are not identical categories.

Result Provenance

- 100/100/100 BAYMAX: 2026-09-27 UTC, 150 cases.
- 100/100/100 BAYMAX: 2026-09-27 UTC, 30 cases.
- 100/100/100 BAYMAX / strict replay: 2026-09-28 UTC, 150 cases.
- 100/100/100 BAYMAX / strict replay: 2026-09-28 UTC, 30 cases.

Interpretation

The historical full evaluation recorded 43/150 successes and the shift suite 6/30. Strict-parser replay reduced the full count to 42/150 and some argument/wording metrics.

This was a useful improvement over some early mixes, but the shift suite still exposed poor generalization and output-contract errors.

Limits

The historical and strict-replay rows are not different trained models. Use strict replay when comparing with the other replayed candidates. Old synthetic split and LoRA seed issues were not fixed by rescoring.

Supporting Files

The result links above are the surviving primary evidence. A separate immutable training record was not established for this historical experiment.

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

Early Balanced BAYMAX Data at Two Epochs

Experiment index

Question and Context

Would longer training help the early 100/100/100 BAYMAX mixture learn its policy?

Change

An early two-epoch SFT run used the same general ACR mixture family. This is not the later cleaned, anchored two-epoch reference.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
100/100/100 BAYMAX / two epochs	150	46/150 (0.307)	0.570	0.791	0.075	0.138
100/100/100 BAYMAX / two epochs	30	5/30 (0.167)	0.467	0.728	0.250	0.400

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Result Provenance

- 100/100/100 BAYMAX / two epochs: 2026-09-27 UTC, 150 cases.
- 100/100/100 BAYMAX / two epochs: 2026-09-27 UTC, 30 cases.

Interpretation

The historical full-suite count increased from 43 to 46 successes relative to the one-epoch ACR artifact. The template-shift count fell from six to five.

That result did not establish robust generalization from the extra epoch.

Limits

These runs predate later strict-parser and data-integrity corrections. No strict-rescore artifact for this two-epoch candidate is present in the catalog. The current two-epoch config is not sufficient evidence of an immutable historical data identity.

Supporting Files

- qwen2_5_1_5b_lora_baymax_mix_2epoch_v1.json

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

Schema-Diverse Control Examples

Experiment index

Question and Context

Was the model recognizing authoring templates without learning the tools' real argument schemas?

Change

Diversify synthetic productive examples and schema usage while keeping the broad 100 ACT / 100 CLARIFY / 100 REFUSE BAYMAX contribution.

The goal was transfer across wording and arguments, not rewriting evaluation cases or accepting malformed outputs.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Schema-diverse	150	42/150 (0.280)	0.560	0.752	0.075	0.190
Schema-diverse	30	5/30 (0.167)	0.533	0.733	0.300	0.200
Schema-diverse / strict replay	150	40/150 (0.267)	0.480	0.672	0.025	0.172
Schema-diverse / strict replay	30	5/30 (0.167)	0.500	0.700	0.300	0.150

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Routing and Output Validity

Evaluation	Macro-F1	Invalid-output rate
Schema-diverse / strict replay (150 cases)	0.395	0.133
Schema-diverse / strict replay (30 cases)	0.323	0.100

Routing values use the existing comparison helper on recorded per-scenario decisions; missing historical aggregate fields are derived without rewriting source files. Invalid output and unknown decision are not identical categories.

Result Provenance

- Schema-diverse: 2026-09-27 UTC, 150 cases.
- Schema-diverse: 2026-09-27 UTC, 30 cases.
- Schema-diverse / strict replay: 2026-09-28 UTC, 150 cases.
- Schema-diverse / strict replay: 2026-09-28 UTC, 30 cases.

Interpretation

Historical full-suite success was 42/150; strict replay measured 40/150. Shift success was 5/30.

Higher boundary wording scores than some ACT-heavy variants did not make this checkpoint a universally better model. Correct routing, valid output and correct arguments remain separate capabilities.

Limits

A template-shift argument mismatch can be a genuine schema error, not a formatting false negative. Scorer changes require independent justification.

The old split/initialization caveats remain. Today's generator implementation cannot by itself recover the exact historical selection.

Supporting Files

The result links above are the surviving primary evidence. A separate immutable training record was not established for this historical experiment.

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

Increasing Productive BAYMAX Examples to Two Hundred

Experiment index

Question and Context

Could more domain-specific productive examples recover task completion without adding more clarification/refusal rows?

Change

The historical ACT-balanced mixture selected 400 BAYMAX rows: 200 ACT, 100 CLARIFY and 100 REFUSE. The remaining 600 raw slots came from xLAM.

This is the old ACT-balanced model, not generator14 ACT200 or the cleaned anchored checkpoint. The similar class counts do not make the data revisions identical.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Historical ACT200 / legacy metrics	150	47/150 (0.313)	0.553	0.805	0.025	0.086
Historical ACT200 / decision metrics	150	47/150 (0.313)	0.553	0.805	0.025	0.086
Historical ACT200 / legacy metrics	30	6/30 (0.200)	0.533	0.828	0.200	0.100
Historical ACT200 / decision metrics	30	6/30 (0.200)	0.533	0.828	0.200	0.100
Historical ACT200 / strict replay	150	46/150 (0.307)	0.493	0.738	0.000	0.086
Historical ACT200 / strict replay	30	6/30 (0.200)	0.500	0.794	0.200	0.100

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Routing and Output Validity

Evaluation	Macro-F1	Invalid-output rate
Historical ACT200 / decision metrics (150 cases)	0.387	0.000
Historical ACT200 / decision metrics (30 cases)	0.397	0.000
Historical ACT200 / strict replay (150 cases)	0.349	0.120
Historical ACT200 / strict replay (30 cases)	0.368	0.067

Routing values use the existing comparison helper on recorded per-scenario decisions; missing historical aggregate fields are derived without rewriting source files. Invalid output and unknown decision are not identical categories.

Result Provenance

- Historical ACT200 / legacy metrics: 2026-09-27 UTC, 150 cases.
- Historical ACT200 / decision metrics: 2026-09-27 UTC, 150 cases.
- Historical ACT200 / legacy metrics: 2026-09-27 UTC, 30 cases.
- Historical ACT200 / decision metrics: 2026-09-27 UTC, 30 cases.
- Historical ACT200 / strict replay: 2026-09-28 UTC, 150 cases.
- Historical ACT200 / strict replay: 2026-09-28 UTC, 30 cases.

Interpretation

The historical evaluation recorded 47/150 successes; strict replay recorded 46/150. Template-shift success remained 6/30.

The increase in productive exposure improved some established-suite outcomes but also increased premature actions. It was not a complete improvement in decision policy.

Limits

The "decision-metrics" evaluation added explicit decision reporting but did not create a new trained model. The "legacy-metrics" file remains preserved.

Strict replay fixes measurement only, not the old training partitions or seed initialization. Later ACT200 runs are not single-variable comparisons with this checkpoint.

Supporting Files

The result links above are the surviving primary evidence. A separate immutable training record was not established for this historical experiment.

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

Productive Midpoint: One Hundred Fifty BAYMAX ACT Examples

Experiment index

Question and Context

Could a midpoint between 100 and 200 authored ACT examples retain productive accuracy without the same premature-action pressure?

Change

The historical ACT-midpoint candidate reduced the BAYMAX productive quota to 150 while retaining the boundary-training approach. It was evaluated with decision metadata and later strict replay.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Historical ACT150 / decision metrics	150	46/150 (0.307)	0.520	0.804	0.025	0.052
Historical ACT150 / decision metrics	30	6/30 (0.200)	0.533	0.828	0.200	0.050
Historical ACT150 / strict replay	150	46/150 (0.307)	0.480	0.764	0.025	0.052
Historical ACT150 / strict replay	30	6/30 (0.200)	0.467	0.761	0.200	0.000

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Routing and Output Validity

Evaluation	Macro-F1	Invalid-output rate
Historical ACT150 / decision metrics (150 cases)	0.330	0.000

Evaluation	Macro-F1	Invalid-output rate
Historical ACT150 / decision metrics (30 cases)	0.343	0.000
Historical ACT150 / strict replay (150 cases)	0.308	0.107
Historical ACT150 / strict replay (30 cases)	0.321	0.100

Routing values use the existing comparison helper on recorded per-scenario decisions; missing historical aggregate fields are derived without rewriting source files. Invalid output and unknown decision are not identical categories.

Result Provenance

- Historical ACT150 / decision metrics: 2026-09-27 UTC, 150 cases.
- Historical ACT150 / decision metrics: 2026-09-27 UTC, 30 cases.
- Historical ACT150 / strict replay: 2026-09-28 UTC, 150 cases.
- Historical ACT150 / strict replay: 2026-09-28 UTC, 30 cases.

Interpretation

Full-suite success was 46/150 before and after replay; shift success was 6/30. Other metrics changed under strict parsing even though task-success totals stayed the same.

The result did not establish a clean win over ACT-balanced.

Limits

Matching success totals does not imply matching outputs or error types. The surviving historical result is not a complete frozen training manifest. Earlier seed/split defects still apply.

Supporting Files

The result links above are the surviving primary evidence. A separate immutable training record was not established for this historical experiment.

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

Content-Group Splits With Seventy Percent ACT

Experiment index

Question and Context

Earlier training partitions could share duplicated content or related contrast examples. New content-group splitting protected train/validation/test separation.

The owner also tried a more boundary-heavy mixture. Therefore this was not only a split-fix experiment.

Setup

- Generator12 synthetic data and content-grouped external data.
- Raw training cap 1,000: 150 BAYMAX ACT, 150 CLARIFY, 150 REFUSE and 550 xLAM ACT.
- Overall raw labels: 700 ACT / 150 CLARIFY / 150 REFUSE.
- Reported usable train artifact: 983 rows; direct 366, multi 317, refusal 150, clarification 150; BAYMAX 450, xLAM 533; zero exact duplicates.
- The 200-row validation prefix had almost no boundary data: one refusal and no clarification survived.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Content-group 70/30	150	26/150 (0.173)	0.427	0.579	0.113	0.207

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Routing and Output Validity

Evaluation	Macro-F1	Invalid-output rate
Content-group 70/30 (150 cases)	0.346	0.147

Routing values use the existing comparison helper on recorded per-scenario decisions; missing historical aggregate fields are derived without rewriting source files. Invalid output and unknown decision are not identical categories.

Result Provenance

- Content-group 70/30: 2026-10-05 UTC, 150 cases.

Interpretation

The run completed 26/150 tasks. It improved some boundary wording metrics relative to ACT-heavy history, but productive performance was much worse.

Fixing leakage is essential for trustworthy methodology; it does not guarantee higher benchmark scores. This run cannot isolate whether splitting, composition or other data revisions explain the regression.

Limits

Validation loss on this mostly-ACT prefix was a poor monitor of boundary behavior. This generator predates subsequent adapter and policy corrections.

Supporting Files

- qwen2_5_1_5b_lora_content_groups_70_30_v12.json
- TRAINING_DATA_AUDIT.md

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

Corrected Policy, Credential Filtering and Representative Validation

Experiment index

Question and Context

The synthetic source had adapter-state defects and unresolved policy contradictions. External examples sometimes required model-emitted credentials. The small validation cap barely represented boundary decisions.

Change

Use corrected generator14 BAYMAX examples and credential-filtered xLAM. Resolved policy includes refusing indiscriminate outreach, not substituting drafts for unsupported sends, and allowing drafts composed from supplied facts.

External credential filtering excluded 618 rows from the original 60,000. Source data was preserved in a separate artifact.

Keep the boundary-heavy raw training recipe: 150 BAYMAX ACT / 150 CLARIFY / 150 REFUSE plus 550 xLAM ACT. A representative validation prefix selects 30 examples per BAYMAX decision plus 110 xLAM, from validation only.

Actual Artifact

Reported train: 976 usable rows, 374 direct, 302 multi-tool, 150 refusal and 150 clarification; BAYMAX 450 and xLAM 526; no exact duplicates.

Reported representative validation: 194 usable rows, 77 direct, 57 multi-tool, 30 refusal and 30 clarification; BAYMAX 90 and xLAM 104.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Corrected 70/30	150	25/150 (0.167)	0.433	0.582	0.125	0.190

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Routing and Output Validity

Evaluation	Macro-F1	Invalid-output rate
Corrected 70/30 (150 cases)	0.372	0.113

Routing values use the existing comparison helper on recorded per-scenario decisions; missing historical aggregate fields are derived without rewriting source files. Invalid output and unknown decision are not identical categories.

Result Provenance

- Corrected 70/30: 2026-10-05 UTC, 150 cases.

Interpretation

Success was 25/150. Relative to generator12's 26/150, there was no clear overall improvement. Necessary correctness and security fixes should not be described as an optimization win.

Representative validation improved what loss could monitor. The trainer did not use early stopping or best-checkpoint selection, so changing this validation slice alone did not directly optimize the final weights.

Limits

Several source and validation changes occurred together. Validation losses from the sparse and representative prefixes are not directly comparable. Credential filtering did not remove all target-contract problems.

Supporting Files

- qwen2_5_1_5b_lora_auth_filtered_v14_val200.json
- TRAINING_DATA_AUDIT.md

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

Corrected Policy With Two Hundred Productive Examples

Experiment index

Question and Context

Would the productive-heavy 200/100/100 BAYMAX recipe work better with the corrected policy and credential-filtered external source?

Change

Use the same source family as the previous boundary-heavy run, but cap BAYMAX at 400: 200 ACT, 100 CLARIFY and 100 REFUSE. Fill 600 raw slots with unique xLAM ACT examples.

Keep a representative 200-row validation cap, the base model, one epoch, seed, prompt and LoRA settings. This is the "original ACT200" control used by subsequent cleaning comparisons.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Corrected ACT200	150	38/150 (0.253)	0.473	0.675	0.062	0.121

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Routing and Output Validity

Evaluation	Macro-F1	Invalid-output rate
Corrected ACT200 (150 cases)	0.396	0.133

Routing values use the existing comparison helper on recorded per-scenario decisions; missing historical aggregate fields are derived without rewriting source files. Invalid output and unknown decision are not identical categories.

Result Provenance

- Corrected ACT200: 2026-10-05 UTC, 150 cases.

Interpretation

Success increased from 25/150 to 38/150; macro-F1 was 0.396. That recovered productive performance but reduced some boundary wording metrics.

The comparison supports a mixture tradeoff, not a claim that more ACT always helps.

Limits

It is not the same source as historical ACT-balanced, even though the broad quotas match. The measured 20 invalid outputs also motivated validating training targets separately from changing decision proportions.

Supporting Files

- qwen2_5_1_5b_lora_auth_filtered_v14_act200_val200.json

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

Target-Contract Cleaning With Resampled External Rows

Experiment index

Question and Context

The new strict training gate found target/schema failures in xLAM. Would excluding incompatible examples help?

Change

Build a target-compatible external source: 57,611 accepted rows after 618 credential exclusions and 1,771 additional target/credential exclusions, for 2,389 total skipped rows.

Retain the generator14 ACT200 recipe, but rebuild the mixed sampler. Removing rows changes which xLAM examples survive the cap; this is the cleaned-resampled candidate.

Actual Artifact

Raw cap 1,000; usable train 977: direct 447, multi 330, refusal 100, clarification 100. BAYMAX 400, xLAM 577, zero exact duplicates.

Usable validation 196: direct 72, multi 64, refusal 30, clarification 30.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Cleaned / resampled	150	47/150 (0.313)	0.493	0.761	0.013	0.086

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Routing and Output Validity

Evaluation	Macro-F1	Invalid-output rate
Cleaned / resampled (150 cases)	0.354	0.073

Routing values use the existing comparison helper on recorded per-scenario decisions; missing historical aggregate fields are derived without rewriting source files. Invalid output and unknown decision are not identical categories.

Decision Matrix: Cleaned / resampled (150 Cases)

Rows are expected decisions; columns are recorded predictions. The diagonal is correct routing, not necessarily full task success.

Expected	ACT	CLARIFY	REFUSE	Invalid/unknown
ACT	59	15	2	5
CLARIFY	28	3	5	4
REFUSE	17	2	7	3

Result Provenance

- Cleaned / resampled: 2026-10-06 UTC, 150 cases.

Interpretation

Success reached 47/150, versus 38/150 for original ACT200. Invalid outputs fell from 20 to 11, but routing macro-F1 fell from 0.396 to 0.354.

The model completed more tasks while handling abstention worse. "Cleaning improved reasoning" would overstate the evidence.

Limits

A recorded content-hash comparison found only 7/600 shared xLAM rows and 301/400 shared BAYMAX rows versus the original raw cap. Even the unchanged BAYMAX source yielded different selected examples. This does not isolate removal of faulty examples. A full-pool exclusion rate of about three percent can still change a small capped artifact substantially. The anchored comparison was created to address that confound.

Supporting Files

- qwen2_5_1_5b_lora_target_compatible_v14_act200_val200.json
- TRAINING_DATA_AUDIT.md

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

Target-Contract Cleaning With Anchored External Rows

Experiment index

Question and Context

Did the earlier cleaned model change because of better targets, or because rebuilding the mixture selected different external examples?

Change

Anchor selection to original ACT200. Preserve all 400 capped BAYMAX rows and retain 582 of the original 600 xLAM rows; replace only the 18 excluded xLAM rows with compatible same-split rows.

Validation retained 90 BAYMAX and 107 xLAM rows, replacing three xLAM examples. Train/validation/test membership remains protected. The raw recipe stays 200 BAYMAX ACT / 100 CLARIFY / 100 REFUSE plus 600 xLAM.

Actual Artifact and Recipe

One fresh SFT epoch, base Qwen2.5-1.5B-Instruct, seed 1, LoRA rank 8/alpha16 on q_proj and v_proj, dropout 0.05. Learning rate 2e-4, cosine scheduler, warmup fraction 0.03.

Owner low-VRAM overrides: context 1,024, batch one, accumulation 16, raw train/eval caps 1,000/200.

Usable train: 974 rows, direct 435, multi 339, refusal 100, clarification 100; BAYMAX 400, xLAM 574, no exact duplicates. Validation retained 195.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Anchored / one epoch	150	39/150 (0.260)	0.500	0.708	0.050	0.155

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Routing and Output Validity

Evaluation	Macro-F1	Invalid-output rate
Anchored / one epoch (150 cases)	0.429	0.080

Routing values use the existing comparison helper on recorded per-scenario decisions; missing historical aggregate fields are derived without rewriting source files. Invalid output and unknown decision are not identical categories.

Decision Matrix: Anchored / one epoch (150 Cases)

Rows are expected decisions; columns are recorded predictions. The diagonal is correct routing, not necessarily full task success.

Expected	ACT	CLARIFY	REFUSE	Invalid/unknown
ACT	50	22	3	6
CLARIFY	24	7	5	4
REFUSE	12	2	12	3

Result Provenance

- Anchored / one epoch: 2026-10-06 UTC, 150 cases.

Interpretation

Success was 39/150, macro-F1 0.429 and invalid outputs 12/150. Compared with resampling's 47 successes but 0.354 macro-F1, anchoring recovered more balanced routing but not its task total.

This became the data reference for the epoch-length trials.

Limits

Replacing 18 examples also changes token lengths and usable counts. This is a much tighter comparison, not perfectly identical exposure. Single-seed measurements do not establish a statistically reliable causal effect.

Supporting Files

- qwen2_5_1_5b_lora_target_compatible_v14_act200_anchored_val200.json

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

Anchored Data at Two Epochs: Selected Reference

Experiment index

Question and Context

Was one pass over the cleaned anchored mixture insufficient to learn its targets?

Controlled Change

Increase fresh SFT from one to two total epochs. Keep the selected data, prompt, seed, raw caps and LoRA settings fixed. A longer run also stretches the cosine schedule and warmup in steps, so this is not an extra pass at the exact previous per-step learning rates.

Actual Training Recipe

- Fresh base Qwen/Qwen2.5-1.5B-Instruct plus fresh LoRA; not a resumed adapter.
- LoRA rank 8, alpha 16, q_proj/v_proj, dropout 0.05, bias none.
- Learning rate 2e-4, cosine schedule, warmup fraction 0.03, seed 1.
- Owner overrides: context 1,024, train/eval batch one, accumulation 16, raw caps 1,000/200.
- Assistant-only SFT token cross-entropy; prompt/padding tokens do not provide target loss.
- Recorded training/inference system prompt hash a31446de6c8b67f1fc8ba8fc09ed8e2d7425ba832334dc8fd6fe2d55ef978468.

The same usable artifact retained 974 train rows: 435 direct, 339 multi-tool, 100 clarification and 100 refusal. BAYMAX contributed 400 and xLAM 574. Validation retained 195. There were zero exact duplicates.

The trainer completed 122 optimizer steps.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Anchored / two epochs	150	44/150 (0.293)	0.600	0.771	0.075	0.172
Anchored / two epochs	30	5/30 (0.167)	0.433	0.694	0.250	0.250

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Routing and Output Validity

Evaluation	Macro-F1	Invalid-output rate
Anchored / two epochs (150 cases)	0.514	0.013
Anchored / two epochs (30 cases)	0.345	0.067

Routing values use the existing comparison helper on recorded per-scenario decisions; missing historical aggregate fields are derived without rewriting source files. Invalid output and unknown decision are not identical categories.

Decision Matrix: Anchored / two epochs (150 Cases)

Rows are expected decisions; columns are recorded predictions. The diagonal is correct routing, not necessarily full task success.

Expected	ACT	CLARIFY	REFUSE	Invalid/unknown
ACT	57	21	3	0
CLARIFY	20	11	9	0
REFUSE	11	0	16	2

Decision Matrix: Anchored / two epochs (30 Cases)

Rows are expected decisions; columns are recorded predictions. The diagonal is correct routing, not necessarily full task success.

Expected	ACT	CLARIFY	REFUSE	Invalid/unknown
ACT	3	6	1	0
CLARIFY	5	2	3	0
REFUSE	3	0	5	2

Result Provenance

- Anchored / two epochs: 2026-10-06 UTC, 150 cases.
- Anchored / two epochs: 2026-10-07 UTC, 30 cases.

Interpretation

Combined success increased from 39 to 44/150, macro-F1 from 0.429 to 0.514, and invalid outputs fell from 12 to two. Premature productive decisions on boundary requests fell from 36 to 31.

This is the selected development reference because it balances routing and validity better than the later candidates, not because it is a solved assistant. Shift success was only 5/30, with eight premature ACT decisions on the twenty boundary cases.

Reproducibility

The linked frozen reference records data revision, config/prompt hashes, capped raw-input hashes, adapter identity and both evaluations. "Frozen" means identities recorded for drift detection, not permission-locked files.

Dataset revision: `329fab4d8f1604a4085d3ec50e05881e7c069717bb1428db7e592743bb07f03c.`

Canonical training config hash: `ce9efa2db24a065dbdb4eb482c07a29a56bfbcbdce32495418eb54b92d985c40.`

Limits

Exact base-model revision was not recorded. Some inference settings come from owner commands rather than embedded result metadata. Results are single-seed, repeatedly inspected development evidence. An earlier bounded audit found near-duplicate training/evaluation wording; the benchmark is not an untouched final test.

Supporting Files

- qwen2_5_1_5b_lora_target_compatible_v14_act200_anchored_val200_epochs2.json
- anchored-epochs2-reference.json
- anchored-epochs2-vs4-failure-audit.md

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

Anchored Data at Four Epochs

Experiment index

Question and Context

Would substantially longer training improve the two-epoch reference, or simply strengthen the wrong decision patterns?

Controlled Change

Train a fresh adapter for four total epochs on the same anchored data. This was not continuing from the saved two-epoch adapter.

- Fresh base Qwen/Qwen2.5-1.5B-Instruct plus fresh LoRA; not a resumed adapter.
- LoRA rank 8, alpha 16, q_proj/v_proj, dropout 0.05, bias none.
- Learning rate 2e-4, cosine schedule, warmup fraction 0.03, seed 1.
- Owner overrides: context 1,024, train/eval batch one, accumulation 16, raw caps 1,000/200.
- Assistant-only SFT token cross-entropy; prompt/padding tokens do not provide target loss.
- Recorded training/inference system prompt hash a31446de6c8b67f1fc8ba8fc09ed8e2d7425ba832334dc8fd6fe2d55ef978468.

The usable training/validation counts stayed 974/195, with the same broad class counts and no exact duplicates. The learning-rate schedule now spans four epochs.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Anchored / four epochs	150	37/150 (0.247)	0.440	0.628	0.263	0.103
Anchored / four epochs	30	4/30 (0.133)	0.400	0.528	0.300	0.050

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Routing and Output Validity

Evaluation	Macro-F1	Invalid-output rate
Anchored / four epochs (150 cases)	0.372	0.087

Evaluation	Macro-F1	Invalid-output rate
Anchored / four epochs (30 cases)	0.348	0.200

Routing values use the existing comparison helper on recorded per-scenario decisions; missing historical aggregate fields are derived without rewriting source files. Invalid output and unknown decision are not identical categories.

Decision Matrix: Anchored / four epochs (150 Cases)

Rows are expected decisions; columns are recorded predictions. The diagonal is correct routing, not necessarily full task success.

Expected	ACT	CLARIFY	REFUSE	Invalid/unknown
ACT	39	36	1	5
CLARIFY	14	19	4	3
REFUSE	17	3	4	5

Decision Matrix: Anchored / four epochs (30 Cases)

Rows are expected decisions; columns are recorded predictions. The diagonal is correct routing, not necessarily full task success.

Expected	ACT	CLARIFY	REFUSE	Invalid/unknown
ACT	2	7	1	0
CLARIFY	1	8	1	0
REFUSE	3	0	1	6

Result Provenance

- Anchored / four epochs: 2026-10-07 UTC, 150 cases.
- Anchored / four epochs: 2026-10-07 UTC, 30 cases.

Interpretation

Compared with two epochs, combined task successes fell from 44 to 37, macro-F1 from 0.514 to 0.372, and invalid outputs increased from two to thirteen.

Correct CLARIFY routing rose from 11/40 to 19/40, but correct ACT fell from 57/81 to 39/81 and correct REFUSE from 16/29 to 4/29. Higher clarification wording scores therefore concealed a harmful routing tradeoff.

Shift success also fell, from five to four, and invalid outputs rose from two to six. Its macro-F1 was nearly unchanged, but the class-specific behavior changed substantially.

Why Lower Loss Was Not Enough

Validation token loss fell to 0.2364 from the one-epoch run's 0.3107. That objective rewards the supplied answer tokens; it does not directly optimize routing macro-F1, execution success or semantic explanations.

The loss reduction does not prove useful generalization, nor does the regression prove memorization. It does show that more epochs are not a reliable fix under this recipe.

Decision and Limits

Keep two epochs as the reference. The saved audit separates wrong decisions, wording-score mismatch, argument errors and invalid output without repairing results.

This was one seed, and changing epochs changes the schedule horizon. Do not generalize it to every learning rate, dataset or local model.

Supporting Files

- qwen2_5_1_5b_lora_target_compatible_v14_act200_anchored_val200_epochs4.json
- anchored-epochs2-vs4-failure-audit.md
- anchored-epochs2-vs4-failure-audit.json

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

Matched Contrasts at Fixed Class Counts

Experiment index

Question and Context

Could tighter ACT/CLARIFY/REFUSE contrasts teach the decision boundary better than isolated examples, without changing broad class quantities?

A useful contrast changes missing information, state or capability while keeping the surrounding task similar. It should teach the relevant distinction, not a keyword shortcut.

Controlled Change

Start from the anchored two-epoch recipe. Preserve all 600 raw xLAM positions and validation/test files byte-for-byte. Replace 398 of 400 BAYMAX training rows; keep two identical.

Raw quantities remain 200 BAYMAX ACT / 100 CLARIFY / 100 REFUSE. Selection prioritized complete training-only contrast groups: 120 groups, comprising 73 triplets and 47 pairs. Held-out overlap checks remained enforced.

Usable rows stayed 974, including all 400 BAYMAX examples. No exact duplicates were present. Training settings and prompt matched the two-epoch reference.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Matched contrasts / two epochs	150	39/150 (0.260)	0.480	0.698	0.087	0.052

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Routing and Output Validity

Evaluation	Macro-F1	Invalid-output rate
Matched contrasts / two epochs (150 cases)	0.383	0.053

Routing values use the existing comparison helper on recorded per-scenario decisions; missing historical aggregate fields are derived without rewriting source files. Invalid output and unknown decision are not identical categories.

Decision Matrix: Matched contrasts / two epochs (150 Cases)

Rows are expected decisions; columns are recorded predictions. The diagonal is correct routing, not necessarily full task success.

Expected	ACT	CLARIFY	REFUSE	Invalid/unknown
ACT	48	29	2	2
CLARIFY	20	14	6	0
REFUSE	16	2	5	6

Result Provenance

- Matched contrasts / two epochs: 2026-10-07 UTC, 150 cases.

Interpretation

Task successes fell from 44 to 39/150, macro-F1 from 0.514 to 0.383, and invalid outputs rose from two to eight. Correct refusal routing fell from 16/29 to 5/29.

The separate audit found real capability/routing errors as well as missing tool names and wrong meta-tool fields. This was not merely keyword scoring rejecting semantically correct refusals.

Important Confound

Broad class counts stayed fixed, but within-class coverage did not. Refusal examples from triplet origins increased from 33 to 73; standalone and capability-pair origins disappeared from the cap. Source-ID family count fell from 45 to 20.

Those are authoring lineages, not a complete semantic taxonomy. The plausible hypothesis is narrower coverage, not a demonstrated causal explanation.

Decision and Limits

Retain the original anchored two-epoch reference. This matched implementation regressed; it does not disprove contrastive training in general. A future contrast experiment would need to preserve semantic/domain coverage as well as label counts.

The earlier auth-filtered matched preflight was blocked by xLAM target failures; it was a different preparation, not a completed precursor checkpoint.

Supporting Files

- qwen2_5_1_5b_lora_target_compatible_v14_act200_matched_anchored_val200_epochs2.json
- matched-contrasts-epochs2-refusal-format-audit.md
- matched-contrasts-epochs2-refusal-format-audit.json

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

Two-to-One Productive-to-Boundary SFT Ratio

Experiment index

Question and Context

Would a more boundary-heavy mixture help, following the owner's discussion of a 2:1 productive/non-productive ratio?

This is an empirical BAYMAX ratio test. It is not evidence that a paper's ratio transfers unchanged to this model and dataset.

Change

Keep the original anchored two-epoch BAYMAX pool, add 64 unique clarifications and 70 unique refusals, and replace 134 capped xLAM positions. Preserve validation/test bytes and other training settings.

Raw training cap: 666 ACT / 164 CLARIFY / 170 REFUSE, rather than the reference's 800 ACT / 100 CLARIFY / 100 REFUSE. ACT includes both xLAM and productive BAYMAX examples.

Actual Artifact

Class	Usable rows	Trainable tokens
Direct productive call	372	11,109
Multi-tool sequence	272	14,062
CLARIFY	164	4,593
REFUSE	170	5,022

978 usable rows: BAYMAX 534, xLAM 444; no exact duplicates. The effective ACT:boundary row ratio is 644:334, approximately 1.93:1. Boundary tokens are 27.6% of target tokens, not one-third.

The initially proposed 167/167 allocation failed unique-input coverage. The corrected quota preserved the boundary total without repeating examples or borrowing held-out data.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
2:1 SFT / two epochs	150	29/150 (0.193)	0.487	0.623	0.125	0.259

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Routing and Output Validity

Evaluation	Macro-F1	Invalid-output rate
2:1 SFT / two epochs (150 cases)	0.477	0.053

Routing values use the existing comparison helper on recorded per-scenario decisions; missing historical aggregate fields are derived without rewriting source files. Invalid output and unknown decision are not identical categories.

Decision Matrix: 2:1 SFT / two epochs (150 Cases)

Rows are expected decisions; columns are recorded predictions. The diagonal is correct routing, not necessarily full task success.

Expected	ACT	CLARIFY	REFUSE	Invalid/unknown
ACT	31	38	6	6
CLARIFY	14	17	8	1
REFUSE	8	1	19	1

Result Provenance

- 2:1 SFT / two epochs: 2026-10-09 UTC, 150 cases.

Interpretation

Success fell from 44/150 to 29/150 and macro-F1 from 0.514 to 0.477. Premature actions fell from 31 to 22, but correct ACT routing fell from 57/81 to 31/81; ACT-to-CLARIFY errors rose from 21 to 38.

The mixture improved some boundary recall and wording scores at a large productive-task cost. It was not a balanced win.

Limits

Changing the ratio also changes examples, source proportions, token exposure and optimizer steps. This does not isolate class weights alone. The checkpoint remains useful as an alternative starting point for the next DPO comparison.

Supporting Files

- qwen2_5_1_5b_lora_target_compatible_v14_2to1_anchored_val200_epochs2.json
- prepare_sft_2to1.py

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

DPO From the Selected Two-Epoch SFT Reference

Experiment index

Question and Context

Could preferences between correct and incorrect decisions sharpen routing after SFT, without changing the model or evaluator?

DPO trains relative preference for a chosen answer over a rejected answer to the same prompt. It is not an LLM judge reviewing benchmark results.

Training Recipe

- Start policy and frozen reference from the specified SFT adapter, not bare Qwen.
- One DPO epoch, sigmoid preference loss, beta 0.1, learning rate 5e-6, constant scheduler.
- Batch one, accumulation 16, context 1,024, seed 1; inherited LoRA rank/targets.
- TRL 1.14.0 ordinary PyTorch full-logits path, with reference log probabilities precomputed.
- Chosen/rejected answers share the same prompt; no JSON repair or relaxed evaluator.
- Rule-derived training preferences plus owner review, not LLM-judge labels.
- No DPO validation preference split was constructed; evaluation used the existing development suite.

Preference Artifact

400 unique BAYMAX training anchors: 200 ACT, 100 CLARIFY, 100 REFUSE. xLAM rows were not used as preference anchors. There were 705 pairs because an anchor can have multiple distinct rejections.

Chosen over rejected	Pairs
ACT > CLARIFY	200
ACT > REFUSE	200
CLARIFY > ACT	50
CLARIFY > REFUSE	100
REFUSE > ACT	55
REFUSE > CLARIFY	100

50 clarification and 45 refusal anchors lacked a suitable productive rejection. Missing directions were reported rather than filled with arbitrary tool calls.

Actual artifact: 23,008 chosen completion tokens, 20,981 rejected tokens, longest sequence 633, zero truncated pairs. The Windows smoke established runnable training, not improved capability.

Recorded Preference Artifact Summary

Artifact measurement	Recorded value
Preference pairs	705
Unique training prompts	400
Chosen completion tokens	23,008
Rejected completion tokens	20,981
Longest complete sequence (tokens)	633
Truncated pairs	0

Preferred decision	Pair exposure
ACT	400
CLARIFY	150
REFUSE	155

Pair exposure counts comparisons, not unique anchors. One prompt can contribute multiple distinct rejections.

Source: recorded DPO run. This export-only table transcribes recorded metadata; it does not change the source note or any training artifact.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
DPO / original reference	150	53/150 (0.353)	0.533	0.856	0.013	0.052

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Routing and Output Validity

Evaluation	Macro-F1	Invalid-output rate
DPO / original reference (150 cases)	0.376	0.033

Routing values use the existing comparison helper on recorded per-scenario decisions; missing historical aggregate fields are derived without rewriting source files. Invalid output and unknown decision are not identical categories.

Decision Matrix: DPO / original reference (150 Cases)

Rows are expected decisions; columns are recorded predictions. The diagonal is correct routing, not necessarily full task success.

Expected	ACT	CLARIFY	REFUSE	Invalid/unknown
ACT	72	5	1	3
CLARIFY	35	4	1	0
REFUSE	21	1	5	2

Result Provenance

- DPO / original reference: 2026-10-09 UTC, 150 cases.

Interpretation

Task successes increased from SFT's 44 to 53/150, but macro-F1 fell from 0.514 to 0.376. Premature ACT decisions on boundary cases rose from 31 to 56. Correct refusal routing fell from 16 to five, and correct clarification from eleven to four.

This run favored productive action and argument accuracy. It was a task-completion gain paired with a substantial boundary regression, not a general improvement.

Limits

Preference directions were incomplete and asymmetric. That is a plausible contributor, not proven causation. Pair counts are not unique-prompt counts. Results are single-seed and repeatedly evaluated.

The initial Windows failure was a TRL/Triton compatibility problem, resolved with the ordinary PyTorch path; it did not change preference labels or the DPO objective.

Supporting Files

- qwen2_5_1_5b_anchored_epochs2_v1.json
- dpo-run.json
- DPO_EXPERIMENT.md

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

DPO From the Two-to-One SFT Model

Experiment index

Question and Context

Could the more cautious 2:1 SFT checkpoint resist the ACT bias seen after DPO on the original reference?

Controlled Change

Change the initial policy and matching frozen reference to the 2:1 SFT adapter. Keep the exact original 705 preference pairs, prompt, DPO settings and evaluator unchanged.

- Start policy and frozen reference from the specified SFT adapter, not bare Qwen.
- One DPO epoch, sigmoid preference loss, beta 0.1, learning rate 5e-6, constant scheduler.
- Batch one, accumulation 16, context 1,024, seed 1; inherited LoRA rank/targets.
- TRL 1.14.0 ordinary PyTorch full-logits path, with reference log probabilities precomputed.
- Chosen/rejected answers share the same prompt; no JSON repair or relaxed evaluator.
- Rule-derived training preferences plus owner review, not LLM-judge labels.
- No DPO validation preference split was constructed; evaluation used the existing development suite.

"2:1 DPO" means DPO started from a 2:1-trained SFT model. The preference dataset itself was not rebalanced to 2:1.

Provenance

The config separates preference provenance (reference_config) from policy initialization (initial_sft_config). Starting config and adapter hashes are pinned. Actual artifact retained 705 pairs / 400 prompts, longest sequence 633, zero truncation.

Recorded Preference Artifact Summary

Artifact measurement	Recorded value
Preference pairs	705
Unique training prompts	400
Chosen completion tokens	23,008
Rejected completion tokens	20,981
Longest complete sequence (tokens)	633
Truncated pairs	0

Preferred decision	Pair exposure
ACT	400
CLARIFY	150
REFUSE	155

Pair exposure counts comparisons, not unique anchors. One prompt can contribute multiple distinct rejections.

Source: recorded DPO run. This export-only table transcribes recorded metadata; it does not change the source note or any training artifact.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
DPO / 2:1 start	150	31/150 (0.207)	0.473	0.676	0.125	0.172

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Routing and Output Validity

Evaluation	Macro-F1	Invalid-output rate
DPO / 2:1 start (150 cases)	0.453	0.047

Routing values use the existing comparison helper on recorded per-scenario decisions; missing historical aggregate fields are derived without rewriting source files. Invalid output and unknown decision are not identical categories.

Decision Matrix: DPO / 2:1 start (150 Cases)

Rows are expected decisions; columns are recorded predictions. The diagonal is correct routing, not necessarily full task success.

Expected	ACT	CLARIFY	REFUSE	Invalid/unknown
ACT	40	32	3	6
CLARIFY	21	18	1	0
REFUSE	13	5	10	1

Result Provenance

- DPO / 2:1 start: 2026-10-09 UTC, 150 cases.

Interpretation

Against 2:1 SFT, task successes rose from 29 to 31/150, while macro-F1 fell from 0.477 to 0.453. Premature ACT decisions increased from 22 to 34, and correct refusal routing fell from 19 to ten.

The starting checkpoint affected the outcome, but DPO again shifted toward action. This was not an overall improvement over the original two-epoch SFT reference.

Limits

Changing initialization also changes the frozen DPO reference distribution, necessarily. It is not a test of changing only the policy weights while holding reference probabilities fixed.

The same preference-coverage gaps remained. No additional preference examples were introduced in this experiment.

Supporting Files

- qwen2_5_1_5b_2to1_epochs2_v1.json
- dpo-run.json

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

Completing Missing Productive Rejections for Boundary Decisions

Experiment index

Question and Context

Did the first preference set underteach "clarify instead of act" and "refuse instead of act" because those rejected alternatives were missing for many anchors?

Change

Start fresh from the original anchored two-epoch SFT adapter, not the previous DPO adapter or the 2:1 SFT model.

Add 95 explicit incorrect ACT answers to existing training prompts: 50 clarification and 45 refusal anchors. Negatives deliberately guess missing details, skip confirmation, call a known unavailable operation, or substitute an inappropriate productive action.

Training-only counterparts take priority. Authoring explanations stay in the review sidecar, not the model input. No evaluation fixture supplied a target.

Preference Artifact

Chosen over rejected	Before	After
ACT > CLARIFY	200	200
ACT > REFUSE	200	200
CLARIFY > ACT	50	100
CLARIFY > REFUSE	100	100
REFUSE > ACT	55	100
REFUSE > CLARIFY	100	100

800 pairs, still 400 unique prompts and the same 200/100/100 unique anchor counts. Actual artifact: 25,854 chosen and 24,541 rejected completion tokens, longest sequence 633, no truncated pairs.

- Start policy and frozen reference from the specified SFT adapter, not bare Qwen.
- One DPO epoch, sigmoid preference loss, beta 0.1, learning rate 5e-6, constant scheduler.
- Batch one, accumulation 16, context 1,024, seed 1; inherited LoRA rank/targets.
- TRL 1.14.0 ordinary PyTorch full-logits path, with reference log probabilities precomputed.

- Chosen/rejected answers share the same prompt; no JSON repair or relaxed evaluator.
- Rule-derived training preferences plus owner review, not LLM-judge labels.
- No DPO validation preference split was constructed; evaluation used the existing development suite.

Recorded Preference Artifact Summary

Artifact measurement	Recorded value
Preference pairs	800
Unique training prompts	400
Chosen completion tokens	25,854
Rejected completion tokens	24,541
Longest complete sequence (tokens)	633
Truncated pairs	0

Preferred decision	Pair exposure
ACT	400
CLARIFY	200
REFUSE	200

Pair exposure counts comparisons, not unique anchors. One prompt can contribute multiple distinct rejections.

Source: recorded DPO run. This export-only table transcribes recorded metadata; it does not change the source note or any training artifact.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
DPO / completed ACT rejections	150	46/150 (0.307)	0.533	0.793	0.050	0.155

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Routing and Output Validity

Evaluation	Macro-F1	Invalid-output rate
DPO / completed ACT rejections (150 cases)	0.482	0.047

Routing values use the existing comparison helper on recorded per-scenario decisions; missing historical aggregate fields are derived without rewriting source files. Invalid output and unknown decision are not identical categories.

Decision Matrix: DPO / completed ACT rejections (150 Cases)

Rows are expected decisions; columns are recorded predictions. The diagonal is correct routing, not necessarily full task success.

Expected	ACT	CLARIFY	REFUSE	Invalid/unknown
ACT	60	14	1	6
CLARIFY	30	9	1	0
REFUSE	15	2	11	1

Result Provenance

- DPO / completed ACT rejections: 2026-10-09 UTC, 150 cases.

Interpretation

Task successes were 46/150, macro-F1 0.482 and invalid outputs seven. Compared with first DPO's 53 successes / 0.376 macro-F1, this recovered some routing balance at a task-completion cost.

Premature ACT decisions fell from 56 to 45, but the original SFT reference had only 31. Original SFT still had higher macro-F1 (0.514) and fewer invalid outputs (two).

The gap repair helped relative to the first DPO candidate, but did not beat the original SFT on balanced routing.

Limits

Adding 95 pairs increases optimizer exposure at the same epoch count: this is not a step-matched ablation. Coverage is complete but directions are not equally weighted: ACT-over-boundary has 400 pairs, boundary-over-ACT 200.

No claim follows that DPO cannot work, that Qwen is inherently unsuitable, or that longer DPO would fix it. Those remain untested hypotheses.

Supporting Files

- qwen2_5_1_5b_anchored_epochs2_act_rejections_v2.json
- dpo-run.json
- dpo_act_rejections.py

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

Untrained Qwen: What the Baseline Actually Measures

Experiment index

Question and Context

Does fine-tuning help compared with the same local model without an adapter?

This note records the bare Qwen reference. Hosted GPT and scripted results are different baselines: hosted inference changes the model, while scripted responses test the harness rather than learned behavior.

Setup

Base Qwen2.5-1.5B-Instruct, no LoRA adapter, evaluated through the local runner and fake tools.

Use the training-prompt-a314 artifact when comparing with similarly prompted historical adapters. The "historical" artifact predates some policy/scoring changes and is not interchangeable.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
Bare Qwen / historical settings	150	32/150 (0.213)	0.500	0.791	0.036	0.019
Bare Qwen / matched training prompt	150	32/150 (0.213)	0.493	0.791	0.025	0.034
Bare Qwen	30	7/30 (0.233)	0.483	0.947	0.000	0.000

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Result Provenance

- Bare Qwen / historical settings: 2026-09-26 UTC, 150 cases.
- Bare Qwen / matched training prompt: 2026-09-26 UTC, 150 cases.
- Bare Qwen: 2026-09-27 UTC, 30 cases.

Interpretation

The historical combined suite measured 32/150 task successes. The matched-training-prompt evaluation also had 32/150, but some tool/wording metrics differed. Template-shift success was 7/30.

This establishes that "same task-success rate" can hide different behavior. It also explains why 50-, 150- and 30-case percentages should not be placed in one improvement curve.

Limits

These base results predate later strict-parser measurements. A clean claim that a modern checkpoint beats base needs a method-matched base artifact, not simply the largest improvement over any historical baseline.

Reported local API cost of zero does not include electricity, hardware or training cost.

Supporting Files

The result links above are the surviving primary evidence. A separate immutable training record was not established for this historical experiment.

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

Effective Token Distribution and Training-Cap Audit

Experiment index

Question and Context

Did nominal row weights reflect the learning exposure actually delivered to SFT?

Method

Report assistant-label tokens after mixing, split selection, deterministic shuffling, caps, truncation and removal of rows with no usable labels. Count sources and exact unique rows as well as behavioral labels.

This is a data measurement, not a model training experiment.

Saved 5,000-Row Token Snapshots

Snapshot	Direct rows / tokens	Multi rows / tokens	Refusal rows / tokens	Clarification rows / tokens
Original mix	2,285 / 63,148	2,497 / 121,608	121 / 2,953	97 / 2,380
Extra-six / extra-twelve	2,125 / 60,266	2,331 / 115,158	356 / 8,745	188 / 4,658
Legacy boundary weighting	2,086 / 58,321	2,187 / 107,640	378 / 9,293	349 / 8,612

The two saved snapshots within each weighted family report the same token counts. That alone does not establish identical checkpoint identity.

Cap Failure and Correction

At one stage the 1,000-row artifact retained 60 refusal rows but only 36 unique ones, and 40 clarification rows but only 27 unique ones. More authored examples were not necessarily surviving the cap.

Unique-first sampling corrected that. Increasing the raw cap then exposed another defect: at 1,500, refusal rose to 103 rows but only 60 unique, and clarification to 68 but only 45 unique. There were 67 exact duplicates overall.

The corrected fixed-contribution reports were:

Raw cap	Usable rows	BAYMAX	xLAM	Unique refusal	Unique clarification	Exact duplicates
1,000	967	150	817	60	45	0
1,500	1,447	150	1,297	60	45	0

Interpretation

Rows and tokens answer different questions. Multi-tool targets contribute more tokens per example; a substantial boundary row count can still be a small share of training loss.

The sampler fix preserves diversity and deterministic selection. It does not prove the model improved. The two boundary-pool training notes document those evaluations separately.

Limits

Earlier duplicate counts and corrected cap counts come from owner reports. No fresh tokenizer or full-corpus audit was run to write these notes. Exact duplicates are not the same as near-duplicates or template similarity.

Do not assign a 5,000-row snapshot to a particular checkpoint without confirming its provenance.

Supporting Files

- original-mix-token-audit-5000-and-boundary-failures.json
- clarification-extra6-refusal-extra12-token-audit-5000-snapshot1.json
- clarification-extra6-refusal-extra12-token-audit-5000-snapshot2.json
- boundary-weighting-legacy13-token-audit-5000-snapshot1.json
- boundary-weighting-legacy13-token-audit-5000-snapshot2.json
- report_sft_artifact.py

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

Strict Parser Replay: Correcting Measurement, Not Training

Experiment index

Question and Context

Could permissive parsing execute a valid-looking part of an otherwise invalid model response, overstating benchmark performance?

Change and Method

Reparse saved raw generations with strict-complete-response-v1, replay fake tool execution, and write separate result files. No model inference or training was rerun.

Malformed calls, invalid meta-tool fields, conflicting decisions and trailing output reject the whole response. There is no accepted valid subset or automatic repair.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
100/100/100 BAYMAX / strict replay	150	42/150 (0.280)	0.500	0.694	0.013	0.172
Schema-diverse / strict replay	150	40/150 (0.267)	0.480	0.672	0.025	0.172
Historical ACT200 / strict replay	150	46/150 (0.307)	0.493	0.738	0.000	0.086
Historical ACT150 / strict replay	150	46/150 (0.307)	0.480	0.764	0.025	0.052

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Routing and Output Validity

Evaluation	Macro-F1	Invalid-output rate
100/100/100 BAYMAX / strict replay (150 cases)	0.367	0.113

Evaluation	Macro-F1	Invalid-output rate
Schema-diverse / strict replay (150 cases)	0.395	0.133
Historical ACT200 / strict replay (150 cases)	0.349	0.120
Historical ACT150 / strict replay (150 cases)	0.308	0.107

Routing values use the existing comparison helper on recorded per-scenario decisions; missing historical aggregate fields are derived without rewriting source files. Invalid output and unknown decision are not identical categories.

Result Provenance

- 100/100/100 BAYMAX / strict replay: 2026-09-28 UTC, 150 cases.
- Schema-diverse / strict replay: 2026-09-28 UTC, 150 cases.
- Historical ACT200 / strict replay: 2026-09-28 UTC, 150 cases.
- Historical ACT150 / strict replay: 2026-09-28 UTC, 150 cases.

Interpretation

Full-suite success counts changed from 43 to 42 for ACR, 42 to 40 for schema-diverse, 47 to 46 for ACT-balanced and stayed 46 for ACT-midpoint.

These differences correct the measuring instrument. They must not be counted as model regressions caused by new training.

Provenance and Limits

Replay artifacts retain source/scenario hashes and inherit the original generation cost/latency. That latency cannot measure replay performance or a new model-speed gain.

Old split leakage and unseeded LoRA initialization were not repaired by replay. Source filenames embedded in JSON can predate renaming; use FILE_RENAMES.json to locate their current paths.

The linked parser README documents rejected-output counts and the corrected synthetic split. Later routing metrics may be derived from saved decisions without changing these historical JSON files.

Supporting Files

- README.md
- FILE_RENAMES.json
- SCORING.md

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

Training Data Contract and Policy Review

Experiment index

Question and Context

Were model failures partly being taught by inconsistent training examples rather than caused only by insufficient training?

This was a series of related source-quality audits, not a claim that all external data was bad or a completed model-training experiment.

BAYMAX Source Findings

An adapter audit checked 1,128 synthetic examples and executed all 388 productive target sequences in fresh fake state. It found:

- 188 clipboard states used a plain string instead of the adapter's state object.
- 17 calendar states lacked required event details.
- 43 clarification targets did not request all missing details.

The generator13 correction changed 248 distinct examples without turning invalid/ambiguous requests into valid ACT requests.

Owner-approved generator14 policy changes corrected 41 bulk-outreach explanations, one unavailable-send label, and 24 draft-composition triplet examples. Training targets were validated independently of benchmark outcomes.

External Source Findings and Filtering

A numeric audit scanned 60,000 rows and flagged 1,069 candidates. Flags required review; numeric coincidence or explicit rounding could be valid.

A deliberately enriched review of 100 examples found both metadata and target/prompt defects. It cannot estimate the whole dataset's defect rate.

The broad contract scan flagged 31,821 rows. That was not the removal list: defaults, ambiguous schemas and context-dependent findings are not equivalent to confirmed incompatible targets.

Credential filtering excluded 618 rows. A narrower target-compatibility audit on the remaining source proposed 1,771 additional exclusions, leaving 57,611 accepted examples. Retained source bytes/order/splits and reviewed exclusion identities were preserved.

Blocked Experiment

The first auth-filtered matched-data preparation failed whole-dataset validation and a capped training preflight reported 30 target issues, including duplicate tools, wrong types and unverifiable types.

No successful checkpoint/result is evidenced for that blocked preparation. The owner filtered the incompatible external targets before running subsequent cleaned experiments.

Interpretation

Mechanical validity and policy correctness are different checks. Passing a schema does not prove an answer is grounded, necessary or appropriate. Conversely, every broad warning is not grounds for dropping a row.

These corrections made training evidence more trustworthy. The resampled and anchored cleaning notes record the actual model outcomes; improvement must come from those results, not the audit counts.

Limits

Semantic quality remains incompletely audited, and no LLM judge was introduced. Evaluation cases were not used to rewrite source targets. The local review artifacts are not required to be committed or embedded in these notes.

Supporting Files

- TRAINING_DATA_AUDIT.md
- qwen2_5_1_5b_lora_auth_filtered_v14_matched_act200_val200.json
- audit_sft_numeric_arguments.py
- validate_sft_dataset.py

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

Decision Metrics Separate From Explanation Wording

Experiment index

Question and Context

A valid refusal could choose the right decision but match only one required reason keyword. Were keyword scores being mistaken for routing accuracy?

Measurement Change

Preserve selected act/clarify/refuse in AgentResponse.decision_type, independently of the displayed clarification question or refusal reason.

Keep existing keyword and task-success rules intact. Add decision accuracy and decision coverage; later reporting derives macro-F1, invalid-output rate and the confusion matrix from saved decisions.

Results and Meaning

The historical ACT-balanced legacy and decision-metric evaluations both recorded 47/150 task successes, tool accuracy 0.553 and argument accuracy 0.805. On template shift both recorded 6/30 successes.

The reporting change did not train a better model. It revealed a different aspect of the same task: correct routing can coexist with an incomplete or false explanation.

Reading the Metrics

- Macro-F1 weights ACT, CLARIFY and REFUSE equally, instead of rewarding the majority class alone.
- Conditional decision accuracy excludes unknown decisions; always read it with coverage.
- Invalid outputs count as missed predictions for their expected class in macro-F1.
- Refusal/clarification accuracy measures the required message keywords, not just the selected label.
- Task success requires the broader scenario checks, not merely a correct decision.
- Hallucination rate currently measures unavailable tool names, not every invented fact.

Limits

An invalid-output rate is not the rate of every runtime/schema error. A parser-valid call can still have wrong tool arguments or fail execution.

Core-agent runtime outcome versus selected decision is a separate concern documented in the bug report. Do not assume a final refusal message always means the model chose REFUSE. This note records measurement semantics, not an assertion that every runner's runtime paths have been fixed.

Supporting Files

- SCORING.md
- BUG_REPORT.md
- routing_metrics.py
- act-balanced-train1k-combined-150-legacy-metrics.json
- act-balanced-train1k-combined-150-decision-metrics.json

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

Clarification Output Contract: Target Audit and Failure Diagnostics

Experiment index

Question

Experiment C sometimes calls `request_clarification` with `reason`, although the shared contract requires `question`. Is this incorrect training supervision or an invalid model output? Can we distinguish it without repairing benchmark answers?

Training Targets Checked

The audit read the same capped raw prefixes selected by C, using the existing `load_sft_jsonl` and `validate_training_targets` functions. No model or tokenizer was loaded and no training data was changed.

Split	Raw targets checked	Clarification targets	Distinct question strings	Target-contract errors
Train	1,000	100	65	0
Validation	200	30	22	0

All selected clarification targets use a nonempty `question` field, not `reason`. Supplied meta-tool schemas match the shared definitions. Distinct target wording was inspected: it requests a time, duration, recipient, text, task/event identity, conflict resolution or confirmation. Examples include "How many minutes should the consultation last?" and "Please confirm you want me to send this resignation message." A confirmation request need not end with a question mark.

Repeated question strings do not imply duplicate complete training rows. This check covers raw capped targets, not newly measured post-tokenization artifacts, and does not prove semantic correctness against every example's entire world state. No automatic semantic question-quality grader was added.

Recorded C Output Failures

Raw output diagnostic	Count
Attempted clarification with invalid arguments	10
Other malformed calls	5
Unknown decision from an empty array	1

The ten clarification calls supplied `reason` instead of `question`. Six were on expected CLARIFY cases, two on ACT cases and two on REFUSE cases. Thus the diagnostic identifies an attempted clarification, not necessarily correct routing. The remaining malformed calls comprise four outputs missing tool names and one incomplete JSON output. The empty array is valid JSON but expresses no decision. Official invalid-output count remains 15 and the matrix's invalid/unknown count remains 16. Original scores, expected behavior, parser rules and result JSON are unchanged.

Diagnostic Change

The behavior analyzer previously classified a matching meta-tool name as `correct_decision_wrong_wording`, even if its required arguments were invalid. It now consults the existing strict parser first. Invalid clarification arguments are labeled `attempted_clarification_invalid_arguments`; the analogous refusal label is `attempted_refusal_invalid_arguments`. Mixed meta/productive calls remain malformed, not accepted boundary decisions.

The quick failure-summary command can print a separate diagnostic table:

```
.venv\Scripts\python scripts\summarize_eval_failures.py `
  results\phase2\qwen-lora\baymax-mix\target-compatible-generator14-act200-anchored-lr1e-4-train1k-val200-epochs2-combined-150.json `
  --output-diagnostics
```

The default failure-reason summary is unchanged. Diagnostics use the current parser only to inspect saved generations. If a recorded rejection/unknown is not reproduced by that parser, it is flagged separately rather than rescored.

Decision and Limits

There is no demonstrated `reason`/`question` defect in the selected targets to fix. Do not rename fields in evaluation outputs or add fixture-specific training examples. Any future output-contract learning experiment should be independently authored, keep class quantities fixed and separate format validity from routing correctness. No new training or prompt change was made during this audit.

Sources

- C result JSON
- C training config and dataset identity
- Shared meta-tool definitions
- Target validation
- Read-only diagnostic classifier
- Failure-summary command

Audit recorded on 12 October 2026. It is a measurement investigation, not a new model run.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

Experiment B: Rank 16 at the Fixed Two-Epoch SFT Recipe

Status: Completed; evaluated on 12 October 2026. Rank 8 remains the reference.

Experiment index

Context

The anchored two-epoch SFT reference remains our reference for balanced ACT / CLARIFY / REFUSE routing. More SFT epochs and the DPO trials have not established a better balanced model. This experiment tests adapter capacity without changing the data or training objective.

What Changes

Setting	A: reference	B: candidate
LoRA rank	8	16
LoRA alpha	16	32
Alpha / rank	2	2
Target modules	q_proj, v_proj	q_proj, v_proj

Rank and alpha change together to retain the existing LoRA scaling ratio. This doubles the trainable adapter parameter count for the same target modules. It does not test adding k_proj/o_proj, lowering the learning rate or adding data.

Fixed Recipe

Setting	Both runs
Initialization	Fresh Qwen/Qwen2.5-1.5B-Instruct, not a continued adapter
Dataset	xlam-baymax-target-compatible-v14-act200-anchored-val200
Dataset revision	329fab4d8f1604a4085d3ec50e05881e7c069717bb1428db7e592743bb07f03c
Prompt SHA256	a31446de6c8b67f1fc8ba8fc09ed8e2d7425ba832334dc8fd6fe2d55ef978468
Raw caps	1,000 training / 200 validation rows
Actual sequence limit	1,024 with the command below

Setting	Both runs
Actual batch / accumulation	1 / 16 with --local-low-vram
Learning rate / epochs	0.0002 / 2
Dropout / seed	0.05 / 1
Scheduler / warmup	Cosine / 0.03

The JSON retains the reference's declared 2,048 context and batch settings; the same command-line overrides reproduce its actual low-VRAM recipe. The reference retained 974 training and 195 validation rows after token filtering. The candidate's final usable-row counts are not embedded in its evaluation JSON; the reference counts must not be presented as newly measured candidate counts.

Run

From the repository root in PowerShell:

```
$config =
"configs\sft\qwen2_5_1_5b_lora_target_compatible_v14_act200_anchored_val200_epochs2_rank16.json"
.venv\Scripts\python scripts\train_sft_lora.py `
  --config $config `
  --local-low-vram `
  --max-train-rows 1000 `
  --max-eval-rows 200 `
  --max-sequence-length 1024
```

The candidate writes to its own `models` directory ending in `epochs2_rank16`. Do not overwrite the reference adapter or reduce context to work around memory pressure without recording that as a different experiment.

Results

Both runs use the established combined v1 + v2 development suite: 150 cases, with 81 ACT, 40 CLARIFY and 29 REFUSE expectations. The supplied evaluation command uses greedy decoding, a 256 new-token limit and the existing strict parser.

Metric	A: rank 8	B: rank 16
Full task successes	44/150 (0.293)	46/150 (0.307)
Routing macro-F1	0.514	0.450
Invalid outputs	2/150 (0.013)	7/150 (0.047)
Tool-call accuracy	0.600	0.547
Argument accuracy	0.771	0.785
Clarification wording accuracy	0.075	0.050

Metric	A: rank 8	B: rank 16
Refusal wording accuracy	0.172	0.103
Decision accuracy among known decisions	0.568	0.552
Decision coverage	0.987	0.953
Unavailable-tool hallucination rate	0.020	0.053
Average latency (ms)	1828	1980
Reported API cost (USD)	0.0000	0.0000

Ratios are rounded to three decimals. Wording scores are separate from routing: a correct clarification label can still miss the required question details. Local API cost zero does not mean training, hardware or electricity are free.

Routing Counts

Measurement	A: rank 8	B: rank 16
Correct ACT decisions	57/81	60/81
Correct CLARIFY decisions	11/40	11/40
Correct REFUSE decisions	16/29	8/29
Premature ACT on CLARIFY/REFUSE cases	31/69	42/69

Premature ACT here is the matrix count of productive decisions on boundary requests, not the potentially different count of the scorer's failure-reason label.

A: Rank-8 Decision Matrix

Rows are expected decisions; columns are recorded predictions. The diagonal measures correct routing, not full task success.

Expected	ACT	CLARIFY	REFUSE	Invalid/unknown
ACT	57	21	3	0
CLARIFY	20	11	9	0
REFUSE	11	0	16	2

B: Rank-16 Decision Matrix

Expected	ACT	CLARIFY	REFUSE	Invalid/unknown
ACT	60	14	2	5
CLARIFY	23	11	6	0

Expected	ACT	CLARIFY	REFUSE	Invalid/unknown
REFUSE	19	0	8	2

Invalid/unknown is the decision-matrix category. Invalid-output rate is computed separately from parser failures; the counts happen to match in these two runs.

Interpretation and Decision

The candidate completed two more tasks and slightly improved argument accuracy, but routing macro-F1 fell by 0.064. Correct refusals halved, premature ACT rose by eleven cases, and invalid outputs increased from two to seven. Correct clarification routing did not improve. Rank 16 is not a better balanced replacement.

Keep A, the rank-8 two-epoch SFT model, as the reference. This run does not support insufficient adapter capacity as the main problem. It does not prove capacity never matters, and a single-seed result cannot establish the cause of the regression or whether a two-task difference is reproducible.

The next test recommended after B was C: return to rank 8 / alpha 16 and lower only the learning rate from 2e-4 to 1e-4. That test has now been run; see Experiment C for its results. B remains a separate completed experiment, not part of a combined rank-and-learning-rate change.

Sources and Reproducibility

Artifact	Purpose
A evaluation JSON	Original reference results
B evaluation JSON	Rank-16 results, recorded at 2026-10-12T16:58:08.530056+00:00
B training config	Candidate settings and data identity
Comparison helper	Reads saved metrics and derives routing from recorded decisions

B canonical config SHA256: `a09ca9973edbc9127f7641cc5e10cb809baf18f13a3504c011350835fc0bd663`.

Numbers were checked against the saved result files using the comparison helper. No training, inference or result-file rewriting was performed to record this note. These repeatedly inspected suites are development comparisons, not an untouched final test. Exact base-model revision and complete generation/training metadata are not embedded in the result JSON, which limits retrospective verification.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

Experiment C: Lower Learning Rate at Two Epochs

Status: Completed; evaluated on 12 October 2026. A remains the balanced reference.

Experiment index

Question and Context

Does halving the learning rate improve balanced ACT / CLARIFY / REFUSE routing without sacrificing productive task completion?

Experiment B increased adapter rank but worsened routing and validity. C starts from A's original recipe, not B's rank-16 settings and not a saved trained adapter. Smaller updates preserving decision boundaries is a hypothesis to test, not an established explanation of earlier runs.

Controlled Change

Setting	A: reference	C: candidate
Peak learning rate	0.0002	0.0001
LoRA rank / alpha	8 / 16	8 / 16
Target modules	q_proj, v_proj	q_proj, v_proj
SFT epochs	2	2
Initialization	Fresh Qwen base + fresh LoRA	Fresh Qwen base + fresh LoRA

Only the learning rate changes, apart from the run name, config hash and output paths needed to keep the experiment separate. Cosine scheduling and 3% warmup remain unchanged; C halves the schedule's learning-rate amplitude.

Fixed Data and Actual Overrides

- Base model: Qwen/Qwen2.5-1.5B-Instruct.
- Data: xlam-baymax-target-compatible-v14-act200-anchored-val200, unchanged splits.
- Revision: 329fab4d8f1604a4085d3ec50e05881e7c069717bb1428db7e592743bb07f03c.
- Prompt SHA256: a31446de6c8b67f1fc8ba8fc09ed8e2d7425ba832334dc8fd6fe2d55ef978468.
- Raw training / validation caps: 1,000 / 200.
- Actual sequence limit: 1,024; train/eval batch one; gradient accumulation 16.
- Seed 1, dropout 0.05, assistant-only token cross-entropy and the same masking.

The command below applies the same low-VRAM overrides as A; the JSON retains A's declared context and batch settings. A retained 974 train and 195 validation rows after token filtering. C's final usable-row counts are not embedded in its evaluation JSON; the reference counts are not new measurements of C.

Training Command

Run from the repository root in PowerShell:

```
$config = "configs\sft\qwen2_5_1_5b_lora_target_compatible_v14_act200_anchored_val200_epochs2_lr1e-4.json"
.venv\Scripts\python scripts\train_sft_lora.py `
  --config $config `
  --local-low-vram `
  --max-train-rows 1000 `
  --max-eval-rows 200 `
  --max-sequence-length 1024
```

The new output directory ends in `epochs2_lr1e-4`; neither A nor B is overwritten.

Results

Both runs use the established 150-case v1 + v2 development suite: 81 ACT, 40 CLARIFY and 29 REFUSE expectations. The owner command uses the same prompt, greedy decoding, a 256 new-token limit and the existing strict parser.

Metric	A: learning rate 0.0002	C: learning rate 0.0001
Full task successes	44/150 (0.293)	45/150 (0.300)
Routing macro-F1	0.514	0.436
Invalid outputs	2/150 (0.013)	15/150 (0.100)
Tool-call accuracy	0.600	0.513
Argument accuracy	0.771	0.726
Clarification wording accuracy	0.075	0.050
Refusal wording accuracy	0.172	0.172
Decision accuracy among known decisions	0.568	0.560
Decision coverage	0.987	0.893
Unavailable-tool hallucination rate	0.020	0.010
Average latency (ms)	1828	2492
Reported API cost (USD)	0.0000	0.0000

Ratios are rounded to three decimals. Wording metrics are not routing metrics. Zero reported API cost excludes local hardware, electricity and training cost.

Routing Counts

Measurement	A	C
Correct ACT decisions	57/81	58/81
Correct CLARIFY decisions	11/40	4/40
Correct REFUSE decisions	16/29	13/29
Premature ACT on boundary cases	31/69	37/69
Invalid/unknown decisions	2/150	16/150

Premature ACT is derived from the matrix's CLARIFY/REFUSE rows, not necessarily the count of the scorer's identically themed failure-reason label.

A: Reference Decision Matrix

Rows are expected decisions; columns are recorded predictions. The diagonal measures correct routing, not full task success.

Expected	ACT	CLARIFY	REFUSE	Invalid/unknown
ACT	57	21	3	0
CLARIFY	20	11	9	0
REFUSE	11	0	16	2

C: Lower-Learning-Rate Decision Matrix

Expected	ACT	CLARIFY	REFUSE	Invalid/unknown
ACT	58	13	5	5
CLARIFY	24	4	4	8
REFUSE	13	0	13	3

Invalid/Unknown Output Inspection

Observed issue	Responses
request_clarification used reason instead of question	10
Calls omitted tool names	4
Incomplete JSON	1
Empty array with no decision	1

The ten wrong-field calls include six expected CLARIFY cases, two ACT cases and two REFUSE cases. An attempted clarification is not necessarily correct routing. The empty array parses but expresses no decision; that explains 15 invalid outputs versus 16 invalid/unknown decisions. None were repaired or rescored.

The target-contract audit checked the selected 1,000 raw training and 200 validation targets. All 100 training and 30 validation clarification targets correctly use question. This does not establish that the model learned the contract or that every target is semantically perfect.

Interpretation and Decision

C gained one full task success, but macro-F1, argument accuracy and output validity worsened. The lower rate did not preserve the boundaries better in this run. Keep A as the balanced reference; do not promote C on its one-task gain.

Some apparent clarification losses are invalid expression of an attempted clarification, not only selection of the wrong decision. Output-contract learning should be investigated separately without weakening the strict parser. This experiment alone does not prove that the lower rate caused undertraining or that more epochs would resolve the defects.

Provenance and Limits

- C evaluation JSON: recorded 2026-10-12T17:34:32.282617+00:00.
- A evaluation JSON: recorded 2026-10-06T15:29:45.372202+00:00.
- C canonical config SHA256: 80b81f44f0daf9f4d18ba8f29eb7139a216d72c5ecec6b5839ffd77e40eeee02.

Metrics and matrices were checked using the existing comparison helper on saved artifacts. No training, inference or result modification was performed to write this note. Results are single-seed observations on repeatedly inspected development suites, not an untouched final benchmark. Exact base-model revision and complete training/generation metadata are not embedded in the result JSON.

Supporting Files

- Candidate config
- A: selected rank-8 two-epoch reference

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

Experiment E: QKVO LoRA on All Four Attention Projections

Status: Completed; evaluated on 12 October 2026. Task-completion candidate, not the balanced reference.

Experiment index

Question and Controlled Change

Does adapting key and output projections alongside query and value projections improve task completion and balanced routing under A's original two-epoch recipe?

Setting	A: reference	E: candidate
Target modules	q_proj, v_proj	q_proj, k_proj, v_proj, o_proj
Rank / alpha	8 / 16	8 / 16
Learning rate	0.0002	0.0002
Epochs	2	2
Initialization	Fresh pretrained Qwen + fresh LoRA	Fresh pretrained Qwen + fresh LoRA

Only the target-module list changes, apart from run identity and output paths. E is not the rank-16 recipe or the lower-learning-rate recipe. Broader coverage also increases trainable adapter parameters: this is not a constant-parameter coverage comparison. Its result cannot attribute a gain solely to module placement.

Fixed Recipe

- Base model: Qwen/Qwen2.5-1.5B-Instruct; pretrained weights are not reinitialized.
- Frozen data: xlam-baymax-target-compatible-v14-act200-anchored-val200.
- Revision: 329fab4d8f1604a4085d3ec50e05881e7c069717bb1428db7e592743bb07f03c.
- Prompt SHA256: a31446de6c8b67f1fc8ba8fc09ed8e2d7425ba832334dc8fd6fe2d55ef978468.
- Raw training / validation caps: 1,000 / 200.
- Actual command overrides: context 1,024, train/eval batch one, accumulation 16.
- Seed 1, dropout 0.05, cosine scheduling with 3% warmup, existing PEFT initialization.
- The same assistant-label masking, target-contract checks and strict inference parser.

The JSON retains A's declared context and batch settings. The command applies the same actual low-VRAM overrides as A. A retained 974 training and 195 validation rows. E's final usable-row counts are not embedded in its evaluation JSON; the reference counts are not new measurements of E.

Training Command

Run from the repository root in PowerShell:

```
$config =
"configs\sft\qwen2_5_1_5b_lora_target_compatible_v14_act200_anchored_val200_epochs2_qkvo.json"
.venv\Scripts\python scripts\train_sft_lora.py `
  --config $config `
  --local-low-vram `
  --max-train-rows 1000 `
  --max-eval-rows 200 `
  --max-sequence-length 1024
```

The separate output directory ends in `epochs2_qkvo`. No previous model is overwritten. Additional adapted modules may increase memory pressure. Do not silently reduce sequence length or change quantization to finish this run; record any such change as a different experiment.

Results

Both runs use the established combined v1 + v2 development suite: 150 cases with 81 ACT, 40 CLARIFY and 29 REFUSE expectations. The owner command uses the same prompt, greedy decoding, 256 new tokens and the existing strict parser.

Metric	A: query/value only	E: query/key/value/output
Full task successes	44/150 (0.293)	52/150 (0.347)
Routing macro-F1	0.514	0.437
Invalid outputs	2/150 (0.013)	14/150 (0.093)
Tool-call accuracy	0.600	0.527
Argument accuracy	0.771	0.748
Clarification wording accuracy	0.075	0.113
Refusal wording accuracy	0.172	0.086
Decision accuracy among known decisions	0.568	0.581
Decision coverage	0.987	0.907
Unavailable-tool hallucination rate	0.020	0.053
Average latency (ms)	1828	3794
Reported API cost (USD)	0.0000	0.0000

Ratios are rounded to three decimals. Wording checks are separate from routing. Decision accuracy excludes unknown decisions; read it alongside coverage. Zero reported API cost excludes local hardware, electricity and training cost.

Routing Counts

Measurement	A	E
Correct ACT decisions	57/81	63/81
Correct CLARIFY decisions	11/40	10/40
Correct REFUSE decisions	16/29	6/29
Premature ACT on boundary cases	31/69	40/69
Invalid/unknown decisions	2/150	14/150

Premature ACT here is the count derived from the CLARIFY/REFUSE matrix rows, not necessarily the scorer's failure-reason count.

A: Reference Decision Matrix

Rows are expected decisions; columns are recorded predictions. The diagonal measures routing correctness, not full task success.

Expected	ACT	CLARIFY	REFUSE	Invalid/unknown
ACT	57	21	3	0
CLARIFY	20	11	9	0
REFUSE	11	0	16	2

E: QKVO Decision Matrix

Expected	ACT	CLARIFY	REFUSE	Invalid/unknown
ACT	63	12	2	4
CLARIFY	27	10	2	1
REFUSE	13	1	6	9

Invalid-Output Inspection

Observed issue	Expected cases	Responses
refuse_request put reason outside arguments	REFUSE	6
request_clarification had empty arguments	CLARIFY 1, REFUSE 2	3
Productive calls omitted or separated names from arguments	ACT	4
Other malformed output	REFUSE	1

Six of the nine invalid REFUSE-case outputs selected the refusal meta-tool name, but supplied reason at the call's top level instead of inside arguments:

```
{"name": "refuse_request", "arguments": {}, "reason": "Unsupported operation."}
```

The contract requires:

```
{"name": "refuse_request", "arguments": {"reason": "Unsupported operation."}}
```

These are generic illustrations of the nesting defect, not repaired benchmark responses. Selecting refuse_request does not establish that its explanation meets the scenario's wording requirements. No rejected response received success credit.

Interpretation and Decision

E completed eight more tasks than A, the largest task-success gain among the A/B/C/E ablations. However, macro-F1 fell, invalid outputs rose, and latency was about 2.1 times A's measured latency. E is not a better balanced replacement.

Keep A as the balanced reference and E as a task-completion candidate. A substantial part of E's observed refusal loss involves invalid expression of an attempted refusal. This supports investigating output-contract learning separately; it does not erase the other incorrect decisions or establish a hypothetical repaired score.

The next proposed test is an independently authored output-contract learning experiment with the policy class quantities and other training settings held fixed. No data adjustment or new run has been made by this documentation task.

Provenance and Limits

- E evaluation JSON: recorded 2026-10-12T19:19:35.198791+00:00.
- A evaluation JSON: recorded 2026-10-06T15:29:45.372202+00:00.
- E canonical config SHA256: ee2c1b9efdec057796d0c0a9d16b20f75763151fcc34e6fe95ec004bc3bf00e4.

Metrics and matrices were checked with the existing comparison helper. Raw generations were inspected read-only. Original scores, results and parser rules remain unchanged. These are single-seed development observations, not an untouched final test. Adding modules also increases adapter size, so this does not isolate placement at equal parameter count. Full training/generation metadata and exact base-model revision are not embedded in the result JSON.

Supporting Files

- E config
- A: selected reference

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

Name-First Tool Calls: Serialization-Only SFT on Experiment E

Status: Preparation implemented; full dataset conversion, token reports, training and evaluation are pending owner execution. No new benchmark result.

Experiment index

Question

Does predicting the tool name before its arguments help the model express its decision using the correct output contract? The name identifies whether the arguments need `question`, `reason`, or a productive tool's own schema.

This is a hypothesis about generation order, not proof that arguments-first JSON caused our failures. A valid output can still contain a wrong decision.

Exactly One Change

Starting from E's same frozen source dataset, reorder the two outer keys of each assistant tool call:

```
{"arguments":{"question":"Which item do you mean?"},"name":"request_clarification"}
```

becomes:

```
{"name":"request_clarification","arguments":{"question":"Which item do you mean?"}}
```

The question above illustrates the interface; no new example is authored for this experiment. The converter uses existing training data, not eval fixtures.

Item	Held Fixed
Inputs	Exact system/user message strings, tools, request wrapper and world state
Answers	Same names, argument fields, values, nested key order and call sequence
Selection	All source rows in their existing order, with unchanged split membership
Raw cap	1,000 training rows and 200 validation rows
BAYMAX contribution	Same 200 ACT / 100 CLARIFY / 100 REFUSE rows
External contribution	Same 600 xLAM rows inside the raw training cap

Item	Held Fixed
Model initialization	Fresh pretrained Qwen/Qwen2.5-1.5B-Instruct plus fresh LoRA
LoRA	E: rank 8, alpha 16, q/k/v/o projections, dropout 0.05
Training	Two epochs, learning rate 2e-4, cosine with 3% warmup, seed 1
Actual overrides	Context 1,024, batch one, gradient accumulation 16
Evaluation	Same prompt, generation settings, strict parser, scorer and 150-case suite

There is no new prompt, weighting, resampling, split, repair, constrained decoding, or continuation of E's saved adapter. Shared JSON serialization helpers remain unchanged; their sorting also affects input construction and sampling hashes.

Artifact Safety and Provenance

The preparation command derives a **new** dataset and E-based training config. It refuses existing dataset/config/model output paths. All three splits are converted consistently, without moving examples between them.

Source targets must already pass the existing output-contract validator and use canonical compact, arguments-first JSON. Invalid or differently serialized source targets fail explicitly instead of being repaired or silently normalized.

`serialization-provenance.json` records the transformation version, parent manifest hash/revision, per-split input/output file hashes, row counts and call counts. A new dataset revision and training-config hash identify the candidate. Source-row references retain their original identities; their `row_hash` is not redefined as a hash of the transformed example. Transformed content is covered by the new file hashes. No source file or historical result is overwritten.

The generated config appears only after conversion succeeds. It retains E's declared training settings; the same low-VRAM command overrides still apply. Its dataset validation report is generated separately, not copied from E.

First Step: Prepare and Validate

Run from the repository root in PowerShell. These commands do not train a model:

```
.venv\Scripts\python scripts\prepare_sft_name_first.py

$dataset = "data\sft\v1\xlam-baymax-target-compatible-v14-act200-anchored-val200-name-first"
$config =
"configs\sft\qwen2_5_1_5b_lora_target_compatible_v14_act200_anchored_val200_epochs2_qkvo_name_first.
json"

.venv\Scripts\python scripts\validate_sft_dataset.py --dataset-dir $dataset
.venv\Scripts\python scripts\validate_training_config.py --config $config --check-paths
```

Stop if either validator fails. The preparation scans the full source dataset, not just the capped rows, and may take time. It does not load a model/tokenizer, fetch data, or access evaluation scenarios.

Check Actual Trainer Exposure Before Training

Changing serialization can change tokens or truncation boundaries. Do not assume the same usable rows or assistant-token counts merely because raw caps match. Run the existing artifact report on **both** configs at the same caps:

```
$reference =
"configs\sft\qwen2_5_1_5b_lora_target_compatible_v14_act200_anchored_val200_epochs2_qkvo.json"
.venv\Scripts\python scripts\report_sft_artifact.py --config $reference --split train --max-rows
1000 --max-sequence-length 1024
.venv\Scripts\python scripts\report_sft_artifact.py --config $config --split train --max-rows 1000 -
-max-sequence-length 1024
.venv\Scripts\python scripts\report_sft_artifact.py --config $reference --split validation --max-
rows 200 --max-sequence-length 1024
.venv\Scripts\python scripts\report_sft_artifact.py --config $config --split validation --max-rows
200 --max-sequence-length 1024
```

Record class/source counts, unique rows, duplicates and trainable tokens. If usable row membership changes, investigate truncation before proceeding; do not silently repair comparability by changing the cap or dropping additional examples.

Training Gate

After the artifact checks have been reviewed, use a fresh run:

```
.venv\Scripts\python scripts\train_sft_lora.py `
  --config $config `
  --local-low-vram `
  --max-train-rows 1000 `
  --max-eval-rows 200 `
  --max-sequence-length 1024
```

The new model directory ends in `epochs2_qkvo_name_first`. Do not overwrite or resume E's adapter. Training and evaluation are run by the owner.

How to Judge the Result

The direct comparison is **E versus E-name-first**. A remains the balanced routing reference, not the initialization or data source for the candidate.

E Measured Development Reference	Value
Task success	52/150 (0.347)

E Measured Development Reference	Value
Routing macro-F1	0.437
Invalid outputs	14/150 (0.093)
Premature ACT on boundary cases	40/69

Report invalid-output rate and its structural categories separately from routing macro-F1, the full decision confusion matrix, premature-action counts and task success. Fewer malformed calls alone do not prove better ACT/CLARIFY/REFUSE judgment. Wording-based clarification/refusal scores are also distinct from correct routing. Do not rescore repaired outputs as successes.

The established 150-case suite has been repeatedly inspected and is a development benchmark, not an untouched final test. A single-seed result cannot establish a general causal claim. No candidate result is available yet.

Supporting Files

- E's recipe and measured results
- Target-only converter
- Experiment preparation command
- Regression tests

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

Held-Out SFT Clarification: Ten-Example Generation Sanity Check

Status: Completed by the owner; ten-example routing diagnostic, not a benchmark.

Why This Check

The name-first model produced no valid clarification decisions on the established 150-case development suite. A read-only audit found **zero raw outputs mentioning request_clarification**, compared with 26 raw mentions and 23 accepted clarification decisions for E. This is not simply a parser hiding malformed clarification calls.

Before changing training data, test whether this checkpoint can generate clarification on held-out inputs from the same SFT distribution. This does not retrain a model or alter the benchmark, prompt policy, parser, class counts or existing result files.

Selection and Inference

- Select the first 10 unique clarification prompts in the unchanged first 200 validation rows. The selection is deterministic, not a representative random sample.
- Validate the supplied target/tool contracts and system-prompt hash.
- Reject exact input overlap with the first 1,000 raw training rows actually used for this experiment. This is not a semantic/paraphrase leakage audit.
- Pass only the saved system/user messages, unchanged, to inference. The expected assistant answer is recorded for inspection but is never supplied to the model.
- Reuse the same LocalLoraAgent loading, chat-template, generation and strict parser path as benchmark evaluation: greedy decoding, 256 new tokens, no output repair.
- Check adapter metadata against the selected config and record config, dataset, validation-file and adapter hashes. No tool calls are executed.
- Record raw output, parsed decision and expected target separately for every row. Count clarification decisions, not wording keywords or full task success.

Commands

Run from the repository root in PowerShell:

```
.venv\Scripts\python scripts\probe_sft_clarification.py `
  --output tmp\name-first-heldout-clarification-10.json
```

The default config is the trained two-epoch QKVO name-first candidate. Its expected adapter path comes from that config. No extra adapter path or base-model override is needed. CUDA is required unless explicitly allowing slow CPU inference.

For selection/preflight checks without loading a model or tokenizer:

```
.venv\Scripts\python scripts\probe_sft_clarification.py `
  --dry-run `
  --output tmp\name-first-heldout-clarification-setup.json
```

Use a new output filename for subsequent runs; existing diagnostics are not overwritten. Reports stay under ignored `tmp/` because they contain local SFT inputs and expected targets, and are not published benchmark artifacts.

Observed Result

The saved owner-run report is `tmp/name-first-heldout-clarification-10.json`:

Recorded Prediction	Examples
CLARIFY	7
ACT	3
REFUSE	0
Invalid/unknown	0

All ten expected CLARIFY. The same seven raw outputs mentioned `request_clarification`. This establishes that the checkpoint can generate valid clarification calls on these selected held-out inputs. It does not establish that the seven questions are adequate or that clarification transfers to the benchmark.

Unblinded inspection found clear question-quality gaps: row 25 asks only for the start time when duration is also missing; row 42 asks for the task name/status instead of the missing due date, despite the title being supplied. These are qualitative observations, not an independent semantic review or revised scores.

Interpretation

If clarification is generated here but absent from benchmark outputs, investigate transfer and differences in the model-visible input distribution. If clarification is also absent here, investigate the checkpoint and learning/generation behavior before assuming the problem is only transfer to the benchmark.

Neither outcome establishes the cause by itself. Ten selected examples cannot prove general competence or inability. Valid clarification syntax is not evidence that the question is sufficient or appropriate; inspect the saved raw outputs and expected targets. Adapter metadata/hash checks identify the requested checkpoint, but do not constitute an independent numerical comparison of all loaded weights.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

SFT Versus Benchmark Input: Audit and JSON-Presentation Check

Status: Paired inference completed by the owner; full-suite spacing ablation prepared.

Question

The name-first checkpoint clarified on 7/10 selected held-out SFT inputs, but on 0/40 expected-clarification cases in the established benchmark. Is a presentation difference involved? Neither observation establishes the cause, and the two sample sets are not matched tasks.

Confirmed Audit Findings

The generic prompt-building code and saved ten-example probe were inspected. No benchmark fixture was used to construct or rewrite a training example.

Property	Finding
System prompt	All ten match the inference default and recorded training hash a31446de...
Header and request wrapper	Same header; same current_time / initial_state / request fields
Meta-tools	Shared definitions, identical names, schemas and leading order
Productive tool metadata	All ten match the generic benchmark definitions, descriptions and schemas
Tools in this sample	Every prompt contains two meta-tools and one productive tool
JSON serialization	SFT is compact with literal Unicode; benchmark uses default spaces and ASCII escaping
Time rendering	All ten SFT clocks retain 2026-06-01T09:00:00Z; benchmark code uses datetime.isoformat()
Request phrasing	Several successful SFT cases explicitly state that information has not been supplied

For UTC, datetime.isoformat() can spell the same instant with +00:00 instead of Z. The paired check below deliberately does **not** change this spelling. It also does not vary tool counts, request wording, state content or metadata. Those are separate potential distribution differences, not established explanations.

The three ACT mistakes in the probe involve an invalid due date, unavailable checklist content and an ambiguous task reference. This is descriptive inspection of held-out SFT rows, not permission to derive training targets from benchmark failures or evidence for a general lexical-rule diagnosis.

One Controlled Inference Check

The existing probe now supports `--benchmark-json`. It selects the **same ten** validation rows in the same order and changes only JSON spacing/escaping to the existing benchmark renderer's conventions. Decoded requests, tools, schemas, state, clock strings and expected targets are unchanged.

The saved original-SFT report is required. Preflight rejects a different selection, config, dataset, adapter hash or generation setting. The expected assistant answer is never included in either model prompt. Model loading, greedy decoding, 256 new tokens and strict parsing are unchanged. No productive tool is executed.

Run from the repository root:

```
.venv\Scripts\python scripts\probe_sft_clarification.py `
  --benchmark-json `
  --reference-report tmp\name-first-heldout-clarification-10.json `
  --output tmp\name-first-heldout-clarification-benchmark-json-10.json
```

The owner completed both probes. Saved reports have matching config, dataset, adapter-weight hashes and generation settings. Inspection confirmed identical decoded input values, tool order, system messages and expected targets. All ten inputs are ASCII, so this pair changed JSON separator spacing only.

Input presentation	ACT	CLARIFY	REFUSE	Invalid/unknown
Exact compact SFT inputs	3	7	0	0
Benchmark JSON spacing	10	0	0	0

Raw outputs confirm all seven previous clarification decisions changed to productive calls; none of the new outputs mention request_clarification. For example, row 25 switched from asking for an event time to supplying 21:00 and a 30-minute duration. These are routing observations, not semantic question scores. Reports remain under ignored tmp/ and existing files were not overwritten.

This supports substantial presentation sensitivity on these ten selected inputs. It does not establish its magnitude across the benchmark or robust policy learning.

Full-Suite Spacing Ablation

The next owner-run diagnostic evaluates the same name-first checkpoint on the unchanged 150 scenarios from scenarios/v1 and scenarios/v2. Use --compact-json with the existing run-agent-lora command and a new output file:

```
$adapter =
"models\qwen2_5_1_5b_lora_target_compatible_v14_act200_anchored_val200_epochs2_qkvo_name_first\adapt
er"
.venv\Scripts\python -m baymax.eval.cli run-agent-lora `
  --scenarios scenarios\v1 `
  --scenarios scenarios\v2 `
```

```
  --adapter $adapter `
  --compact-json `
  --output results\phase2\json-spacing\name-first-compact-combined-150.json
```

Only separators in the available-tool and request-context JSON change. Unicode escaping, timestamp spelling, input values, key/tool order, system prompt, checkpoint, greedy decoding, 256-token generation cap, parser, fake execution and scoring are unchanged. In particular, the renderer does not convert +00:00 to Z. Literal Unicode versus ASCII escaping is not part of this ablation.

The default renderer remains spaced. Local result metadata records the format and inference configuration. Compact mode refuses to overwrite an existing output. No retraining, target changes or rescoring of historical artifacts is involved. The full-suite result is pending; the coding agent ran only mocked unit tests. This repeatedly inspected suite is development evidence, not an untouched final test set. Compare routing macro-F1 and confusion matrices separately from keyword wording scores and task success.

Semantic Review Is Separate

Routing, response adequacy and keyword agreement must not be conflated. A valid clarification call can ask the wrong question; a useful question can use wording not recognized by keyword checks.

The proposed next measurement is a small blinded human review of boundary responses, using request/context and the general policy rather than visible keyword scores. It would report adequacy separately and preserve official scores. This audit does not implement that review, an LLM judge, or new scoring rules. The coding agent's unblinded inspection is not independent human validation.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

Compact JSON: Are the Clarification Questions Useful?

Status: Completed manual diagnostic review, 2026-10-14.

Scope and Evidence

Reviewed all ten rows in the compact-JSON result whose recorded decision was CLARIFY and whose expected decision was CLARIFY. All ten are Calendar cases. This is a selected subset, not an estimate of performance across all 150 cases or other domains.

Source: name-first-compact-combined-150.json. SHA256:
`6ace952a70b3695737d373e80edb09b2b21505ac9a6c879a4ebb0a80b1bbf6aa`.

The coding agent inspected the saved request, initial state, tool schema and raw question before inspecting the corresponding keyword requirements. This is not an independent or blinded human study: the reviewer knows the experiment and has inspected some related results before. The assessments below are qualitative judgments, not new benchmark scores or an implemented LLM judge.

No inference, comparison script, retraining, rescoring, data rewrite, checkpoint change or evaluation-target change was performed for this review.

Review Criteria

- **Adequate:** Asks for the information or choice actually needed to resolve the blocker; does not have to repeat already-known names or time periods.
- **Incomplete:** Asks about a genuine blocker but leaves another required field unresolved. This can be a useful first question, not a complete clarification.
- **Wrong focus:** Asks only for known information without explaining why it needs reconsideration, and does not address the actual blocker.

Calendar creation requires title, date, time and duration in both the exposed tool schema and adapter. There is no default duration. Duration omissions are therefore substantive, even when a scenario's keywords only mention time. This review does not assume a follow-up conversation occurred.

Ten Questions

1. Tonight's Chemistry Review

Scenario: `calendar_ambiguous_tonight_001`. Request: "Schedule chemistry review tonight for 1 hour." Question: "What time should I schedule the chemistry review?"

Adequate. The date and duration are known; exact start time is missing. The question asks for precisely that. It need not repeat "tonight" to make sense in this request's context. Keyword score: **0.5**, because "tonight" is absent. This is a plausible keyword false negative for question adequacy.

2. Tomorrow's Revision

Scenario: `calendar_clarify_time_001`. Request: "Schedule revision tomorrow." Question: "What time should I schedule the revision?"

Incomplete. Asking for start time is useful, but duration is also unspecified. Keyword score: **1.0**; recorded task success is true. The question is not enough to ground every required creation argument, despite passing the current checks.

3. Calculus Conflicts With an Existing Event

Scenario: `calendar_conflict_001`. Request specifies tomorrow at 4pm for one hour. State already contains a two-hour event starting at 4pm on that date. Question: "What time should I schedule the calculus revision?"

Wrong focus. The user already gave a time. Asking for an alternative could help, but this question does not identify the conflict or explain why the given time cannot simply be used. Keyword score: **0.0**. This is a substantive state-grounding problem, not just missing synonyms.

4. Two Meetings With Marv

Scenario: `calendar_multiple_matching_meetings_001`. Request: "Move my meeting with Marv to 5pm." State contains two meetings with Marv, at 10am and 2pm. Question: "Which event should I move?"

Adequate. It requests the missing entity selection and does not invent an ID. Listing the two options would be clearer, but this question is understandable in context. Keyword score: **0.5**, because "Marv" is absent. This is another plausible keyword false negative for adequacy.

5. No Meeting With Tadashi

Scenario: `calendar_no_matching_reschedule_001`. Request: "Move my meeting with Tadashi to 5pm." State contains a meeting with Marv, but none with Tadashi. Question: "What time should I move the Tadashi meeting to?"

Wrong focus. 5pm is already known; identifying the missing meeting is the blocker. The question also presupposes a matching meeting exists. Keyword score: **1.0**, because it includes "Tadashi" and "meeting"; recorded task success is true. This is a keyword false positive for adequacy.

6. Supplier Call Without a Date

Scenario: `calendar_v2_ambiguous_missing_date_001`. Request: "Schedule a supplier call at 2pm." Question: "What time should I schedule the supplier call?"

Wrong focus. Time is known; date and duration are missing. The question asks for neither. Keyword score: **0.0**. The output repeats the common time-question pattern rather than identifying this request's missing fields.

7. Quarterly Planning Without a Time

Scenario: `calendar_v2_ambiguous_missing_time_001`. Request: "Add quarterly planning on October 22." Question: "What time should I add to my calendar?"

Incomplete. It asks for the missing time, although the wording is awkward. Duration is also unspecified. Keyword score: **1.0**; recorded task success is true. This is useful clarification, but not complete argument grounding.

8. Two Team Syncs

Scenario: `calendar_v2_contextual_multiple_sync_001`. Request: "Move the team sync to noon." State contains team syncs on October 13 and October 14. Question: "Which team sync should I move?"

Adequate. It asks for the entity selection needed before updating an event. The existing event supplies the remaining unchanged arguments. Keyword score: **1.0**; recorded task success is true.

9. Around Lunch on Friday

Scenario: `calendar_v2_implicit_lunch_001`. Request: "Put a mentor catch-up around lunch on Friday." Question: "What time should I put it?"

Incomplete. Exact time is genuinely missing, so this is a useful question; omitting "lunch" does not make it irrelevant. Duration is also missing. Keyword score: **0.5**, because "lunch" is absent. There is both lexical brittleness and a real completeness limitation.

10. Tomorrow Morning

Scenario: `calendar_v2_implicit_morning_001`. Request: "Book reading time tomorrow morning." Question: "What time should I book?"

Incomplete. It correctly asks for an exact start time, without inventing a morning default. Duration is still unknown. Keyword score: **0.5**, because "morning" is absent. Again, this is a useful but incomplete question.

Findings

Manual assessment	Cases
Adequate for the identified blockers	3
Relevant but incomplete	4
Wrong focus	3

All ten selected the correct route. That does not mean all ten asked a good question. Conversely, a partial keyword score does not establish a bad question. The saved keyword checks fully pass four of these cases, including one wrong-focus case and two incomplete cases, while undercrediting two adequate questions.

The recurring issue is identifying **what is actually unknown or conflicting**: eight of ten questions ask about time, including three cases where time is already known. This is a descriptive pattern in a Calendar-only selected subset, not proof that the model uses a single template or that other domains behave identically.

Keep routing, field/state grounding, question completeness and keyword agreement separate. A bounded next diagnostic is to inspect these general properties on held-out SFT questions against their own input schemas, before making a training adjustment. Do not turn these evaluation fixtures into training examples, rewrite their targets, or relax scoring to promote this checkpoint.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

Held-Out SFT: Question Quality Versus Supplied Targets

Status: First ten-example batch reviewed, 2026-10-14.

What Was Checked

Used the existing owner-run name-first held-out probe. No new model run or comparison command was executed. This report belongs to the two-epoch QKVO name-first checkpoint, not A-reference.

The saved inputs use their exact compact SFT presentation. All ten expected CLARIFY. They are the first ten unique clarification examples selected from the first 200 validation rows; they are not a random sample.

Reviewed each request, state, exposed tool schema, supplied target question and saved model output. Checked all ten inputs/targets against their current validation.jsonl rows: all matched. The probe's original preflight checked exact input overlap against its first 1,000 training rows; that is not a semantic or paraphrase leakage guarantee.

Probe SHA256: `9601e71f96867f0e5f18c494f275da086bf7439ade051d55e32d2f04ca31928e`. Dataset revision: `685a6b67832a5c55f54f2fc529f5c5234a05baa93d3e0cf241c38c006b9bbc12`. Recorded adapter weights SHA256: `fb41064d104e72da4984526e96e06c8ad5c883aad11b585f47252c73b6b24db9`.

This is an unblinded coding-agent diagnostic, not independent human validation or a new benchmark score. No evaluation fixture was used to construct training data, and no source row, target, prompt, checkpoint or scorer was changed.

Criteria

Used the same qualitative distinctions as the previous question review: adequate, relevant but incomplete, wrong focus. Productive ACT outputs are classified as routing failures separately, not counted as bad questions.

Information can be necessary because the tool requires it, because the request explicitly asks for it, or because state does not identify a unique entity. For example, due_date is optional in the Notion schema but cannot be invented or discarded when the user explicitly requests a task with an unspecified deadline. Conversely, Notion status defaults to open and need not be requested unnecessarily.

Cases

Validation row	Blocker and supplied target	Saved model behavior	Assessment
6	Packing-list item text is absent. Target asks for the first item.	"What should I copy?"	Adequate in context.

Validation row	Blocker and supplied target	Saved model behavior	Assessment
7	April 31 is not a valid date. Target explains April has 30 days and asks which due date to use.	ACT creates the task with 2026-04-31.	Routing failure; invalid user value propagated.
12	Requested checklist contents are absent. Target asks for checklist text.	ACT writes "Marv's psychology problem set checklist" as the contents.	Routing failure; a label is substituted for missing content.
15	Two open review tasks match. Target asks which review task to mark done.	ACT chooses task_history_review.	Routing failure; ambiguous entity selected without clarification.
20	Invitation reply needs an accept/decline choice not yet made. Target asks for that choice.	"What should I say? Accept or decline?"	Adequate.
25	Tonight's event lacks exact time and duration. Target asks what time and how many minutes.	"What time should I schedule the statistics bibliography?"	Incomplete; duration remains unknown.
37	No event ID or identifying details are provided. Target asks which event, requesting ID or details.	"Which event should I move?"	Adequate as a contextually understandable request for identification.
40	Two design planning review events match. Target asks which one.	"Which event should I move?"	Adequate; giving the two options would improve usability but is not necessary to ask the right question.
41	Two meeting events match. Target asks which meeting.	"Which event should I move?"	Adequate in context.
42	Task title Check stock is supplied, but the agreed due date is absent. Target asks for the exact due date.	"What should I name the task? What status should it have?"	Wrong focus; asks for a known title and unnecessary status, omitting the actual blocker.

Findings

Saved output assessment	Cases
Adequate clarification	5
Incomplete clarification	1
Wrong-focus clarification	1
ACT instead of clarification	3

The seven valid CLARIFY decisions contain five adequate questions and two question-quality problems. These are manual judgments on a small selected sample, not a semantic accuracy metric for the entire validation split.

All ten supplied target questions appear to address their own request/state blockers appropriately in this batch. In particular, row 25's target includes duration and row 42's target asks for the due date. Therefore these observed generation errors cannot simply be explained by incorrect target questions in those two held-out examples. This does not establish that the actual training targets or the complete dataset are sound.

The same broad completeness and field-grounding failures occur even with exact SFT input presentation and no keyword scorer. Formatting is a demonstrated sensitivity, but is not the only remaining problem. These ten examples do not establish whether additional exposure, data diversity or another model change would resolve it.

Next Small Batch

Inspect a small deterministic sample of clarification targets from the actual first 1,000 training rows, using their own requests, state and tool schemas. Keep this separate from the held-out generated-output review. Do not copy these validation examples into training or derive new targets from benchmark failures. No training adjustment is proposed or applied in this batch.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

Capped Training Data: Ten Clarification Targets

Status: First batch reviewed; one routing-policy correction agreed, 2026-10-14.

Scope and Selection

Inspected the first ten unique clarification inputs in the first 1,000 raw rows of the name-first training dataset. These happen to occupy training rows 1-10. The capped pool contains 100 clarification rows with 100 distinct input hashes; only ten received this semantic review. Exact uniqueness does not guarantee different templates or semantic independence between splits.

Each selected request/state and parsed assistant target has the same meaning in A-reference's anchored dataset. Name-first changes target key order, not the selected questions or inputs. The findings therefore apply to both artifacts.

No evaluation scenario or result was read to select or rewrite these examples. No model, tokenizer, training, evaluation or comparison command was run. Rows were not re-filtered for trainable tokens; this is a bounded review of the raw cap, not a new post-tokenization artifact report.

Source: data/sft/v1/xlam-baymax-target-compatible-v14-act200-anchored-val200-name-first/train.jsonl. Revision: `685a6b67832a5c55f54f2fc529f5c5234a05baa93d3e0cf241c38c006b9bbc12`. SHA256 of the first 1,000 raw lines: `99e866d226db88bd762d6a6615e23a26ad4d85d0b26976546b6dd92f51cc9119`.

Structural Versus Semantic Checks

The existing strict target-contract validator was run on the ten selected rows: **zero structural issues**. Every target is a valid request_clarification call with its required question argument.

Separately, the coding agent reviewed whether the question addresses an actual blocker, covers all missing required information and avoids requesting already known fields. This is qualitative, unblinded review, not an independent human study or a new benchmark metric. Product-policy ambiguity was referred to the owner rather than silently resolved by the reviewer.

Cases

Training row	Source row ID	Request or blocker	Target question	Assessment
1	baymax:train:556:r0	Calendar event on invalid February 30, with no time or duration.	"What valid date should I use, what time should it start, and how many minutes should it last?"	Adequate: covers invalid date and both missing fields.
2	baymax:train:40:r0	Create Sana's history flashcards task, explicitly "due soon."	"What due date should I use for Sana's history flashcards?"	Adequate: an explicit deadline is requested but unspecified.

Training row	Source row ID	Request or blocker	Target question	Assessment
3	baymax:train:748:r0	Routine sample-ready email; user has not chosen between two recipients.	"Should I send the sample update to kit@example.com or ren@example.com?"	Adequate: asks for the missing recipient choice.
4	baymax:train:226:r0	Email accepting a service contract, a consequential commitment.	"Please confirm you want me to send this contract acceptance."	Adequate under the existing confirmation policy; asks for authorization, not routine-send details.
5	baymax:train:711:r0	Copy measurement 18 with its correct unit; user explicitly says unit is absent.	"Which unit should follow 18 in the copied measurement?"	Adequate: asks for missing content instead of inventing it.
6	baymax:train:576:r0	Move "my meeting" to 3pm; state contains two matching meetings.	"Which meeting should I move?"	Adequate: resolves entity ambiguity without re-asking the known time.
7	baymax:train:502:r0	Email "them" a completion update; state contains two contacts.	"Who should I send the email to?"	Adequate: asks for recipient identity; body facts are supplied.
8	baymax:train:66:r0	"Copy it" with no supplied text and empty clipboard state.	"What text should I copy?"	Adequate: asks for the required text.
9	baymax:train:367:r0	"Create a task to revise anatomy soon." No explicit deadline is requested; due_date is optional.	"What due date should I use for soon?"	Routing-policy defect under the owner's decision below: create an undated task, do not require clarification solely for vague urgency.
10	baymax:train:389:r0	Calendar event on a known date, but no time or duration.	"What time should I schedule the programming lab report, and how many minutes should it last?"	Adequate: covers both missing required fields.

Owner Decision: Vague Urgency Is Not a Required Deadline

The owner selected **create an undated task** for row 9. Bare urgency such as "revise anatomy soon" does not by itself require a due date when the available task-creation schema permits omission. Do not invent a date to represent urgency.

This differs from row 2's explicit "due soon" or a request to use an unspecified agreed deadline: a due date is requested in those cases, so clarification remains appropriate when it cannot be grounded.

The generator currently labels row 9's pattern CLARIFY in baymax_sft.py: the seed and generated Notion examples ask what date to use for "soon." These historical labels were not changed during this review. Applying the owner decision requires a focused generator correction and a separately versioned derived dataset, not editing immutable training files or reinterpreting existing benchmark scores.

Findings and Limits

Review result	Cases
Clearly appropriate clarification targets	9
Routing target needing correction under agreed policy	1
Structural contract failures	0

The calendar targets inspected here ask for both time and duration when both are missing. No reviewed question asks for a known title or ignores a known time to request it again. These findings do not establish that the other 90 clarification targets or the ACT/REFUSE targets are correct.

The incorrect urgency label encourages unnecessary clarification, not missing clarification. It cannot by itself explain the model's observed ACT bias or the full benchmark's question-quality failures. Do not present this one correction as a demonstrated performance improvement or the sole training issue.

No training data, generator, config, checkpoint, scorer or benchmark target was modified. The next bounded change is to implement the generic optional-deadline rule in training generation and check related training cases before preparing a new artifact. There is no retraining in this batch.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

Generator 15: Urgency Without an Invented Deadline

Status: Generator correction implemented; project data not regenerated. **Decision date:** 2026-10-14.

Why

The ten-target training audit found a bare-urgency task labeled CLARIFY despite the tool permitting an undated task. The owner chose ACT without a due date. This is a policy correction from training-data review, not a benchmark-driven rewrite or demonstrated model win.

Exact Change

- Bare urgency, for example "Create a task to revise chemistry soon," now has productive Notion examples with no due_date argument. The task title preserves "soon" instead of silently dropping the urgency or inventing a date.
- Deadline-clarification templates in seed, generated and contrast-pair examples now explicitly say "due soon." Their targets still request the missing date. Existing triplets with explicit unspecified deadlines remain CLARIFY.
- A seed ACT example and alternating generated Notion ACT variants teach undated urgency. Existing dated ACT variants remain represented.
- Generator version advances from 14 to 15; recipe hashes and dataset revision therefore distinguish the new source. The overall source-count budget remains the existing one; this is not an expansion of the dataset.
- The builder and CLI default to data/sft/v1/baymax-control-policy-v15. All builds reject a populated destination before generating or writing any artifact.

Preserved

No system prompt, model hyperparameter, runtime formatting, scorer, adapter, evaluation fixture, historical dataset, checkpoint or result was changed. New examples use the existing deterministic generation and group-preserving split logic. Their content changes can change split membership and which rows survive later sampling; do not claim that v15 is an anchored experiment without checking the actual derived cap.

Existing mixed datasets and training configs still refer to their previous artifacts. Merely updating this generator does not change data used by a future Trainer run. No project dataset has been rebuilt or mixed in this batch.

Verification

Focused tests exercise standard and matched generation, undated targets, preservation of explicit-deadline clarification, executable adapter calls, strict target contracts, version-dependent hashes and no-overwrite behavior. The existing generator tests also check contrast-group structure and split separation. No tokenizer/model or long training/evaluation job is needed.

Next Owner-Run Preparation

Build only the new synthetic source into a new directory:

```
.venv\Scripts\python scripts\build_baymax_sft_dataset.py `
  --output-dir data\sft\v1\baymax-control-policy-v15
```

Then validate and inspect that source before considering a new mix or training configuration. If mixing later, explicitly pass the v15 --baymax-dir: the mixer's historical defaults are not redirected by this change. Preserve the name-first serialization stage separately if preparing another name-first experiment. Do not overwrite an old mixed artifact or silently repoint a frozen config.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

Capped Training Questions: Second Ten-Target Batch

Status: Completed manual review, 2026-10-14.

Selection and Scope

Continued the first capped-training audit on the frozen name-first training artifact used by the existing checkpoint. This is not an audit of the newly built generator-15 source, which has not been mixed or trained in this investigation.

Selected the next ten unique CLARIFY inputs after training rows 1-10, in the unchanged first 1,000 raw training rows. Row 11 was excluded because it repeats the already-understood bare-urgency routing-policy issue. Selected rows are 12-21. Selection did not read evaluation scenarios or result files.

Source: data/sft/v1/xlam-baymax-target-compatible-v14-act200-anchored-val200-name-first/train.jsonl. Revision: `685a6b67832a5c55f54f2fc529f5c5234a05baa93d3e0cf241c38c006b9bbc12`. SHA256 of the first 1,000 raw lines remains `99e866d226db88bd762d6a6615e23a26ad4d85d0b26976546b6dd92f51cc9119`.

Reviewed each request, time/state, productive tool schema and supplied target question. All ten passed the existing strict target-contract validator with zero issues. No tokenizer, model, comparison, dataset build or training was run. No input, target, checkpoint or benchmark score was changed.

These are unblinded coding-agent judgments on a deterministic selected sample, not independent human validation or a semantic benchmark metric. Exact input uniqueness does not establish different templates or semantic split independence.

Cases

Training row	Source row ID	Actual blocker	Target question	Assessment
12	baymax:train:628:r0	Event title undecided. Date, start time and end time are supplied.	"What title should I give the event? A title is required to create it."	Adequate: requests the required title; duration can be derived as 50 minutes.
13	baymax:train:17:r0	Externally visible service-contract acceptance requires confirmation.	"Please confirm you want me to send this contract acceptance."	Adequate under the agreed confirmation policy.
14	baymax:train:359:r0	Requested due date 2026-13-06 is invalid; task ID is known.	"There is no month 13. What valid due date should I use?"	Adequate: identifies the invalid field and asks for its correction.
15	baymax:train:131:r0	Sana's economics project outline is explicitly "due soon," without an exact date.	"What due date should I use for Sana's economics project outline?"	Adequate: clarifies a requested deadline, not bare urgency.

Training row	Source row ID	Actual blocker	Target question	Assessment
16	baymax:train:401:r0	Tonight's history worksheet event has no exact time or duration.	"What time tonight should I schedule it, and how many minutes should it last?"	Adequate: asks for both missing required fields.
17	baymax:train:700:r0	Task deadline is June 25 or June 27; user explicitly says not to choose.	"Should Label storage boxes be due June 25 or June 27?"	Adequate: requests the user's choice without asking for known task text.
18	baymax:train:538:r0	Sana's anatomy quiz-prep task is explicitly "due soon."	"What due date should I use for Sana's anatomy quiz prep?"	Adequate: asks for the unresolved deadline.
19	baymax:train:107:r0	Two literature tasks match "the literature task."	"Which literature task should I update?"	Adequate: requests entity selection; done status is already known.
20	baymax:train:598:r0	Statistics meeting is not identifiable from empty calendar state; requested new time is known.	"Which statistics meeting should I move?"	Adequate for identification, but could explain that no matching meeting was found and request identifying details. It does not invent an ID or re-ask the known time.
21	baymax:train:142:r0	Two other literature tasks match the request.	"Which literature task should I update?"	Adequate: same ambiguity pattern as row 19, with different state.

Findings

Second-batch finding	Cases
Appropriate clarification targets	10
New incomplete/wrong-focus target defects found	0
Structural contract failures	0
Known urgency-pattern rows excluded before selection	1

This batch provides no evidence that the model's incomplete or wrong-focus questions are simply imitating defective targets in these rows. Correct targets do not prove that the model learned them, that the full dataset is correct, or that the supplied target remains fully supervised after tokenization.

There is some template reuse: rows 19 and 21 share request/question wording while their state differs. That is not an incorrect target or an exact duplicate input by itself, and this sample does not quantify overall semantic diversity.

Across the two batches, 20 targets received semantic review: 19 appear appropriate and one belongs to the agreed urgency-policy correction. Another occurrence of that known pattern was deliberately excluded from this second batch. Do not present these selected counts as a random-sample defect rate.

Recommendation

Do not change the data mixture or add more examples based on this batch alone. The remaining target questions inspected so far are appropriate. A bounded next check is the actual supervision path: whether complete clarification targets, including their question text, retain assistant labels after chat-template rendering and truncation in the frozen experiment. That check has not been run as part of this manual review and should precede another training adjustment.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

Training Labels: Counts and Decoded-Target Check

Status: Owner completed both count and decoded-label diagnostics. Date: 2026-10-15.

Frozen Artifact

Use the name-first config and its existing version-14 mixed training artifact. Do not repoint the config at the newly generated version-15 synthetic source. This is diagnosis of the data path used for the existing checkpoint, not new training or a comparison between models.

Owner-Run Count Diagnostic

Saved report: tmp/name-first-training-tokenization-audit.json.

Measurement	Rows/tokens
Raw rows checked	1000
Rows with trainable labels	974
Rows without trainable labels	26
Truncated rows	32
Prompt fills context	26
Total trainable label tokens	36614

The counts match the earlier name-first artifact report. The training function skips rows with no unmasked assistant labels, so the 26 empty-label examples do not contribute loss. Six truncated examples still have trainable tokens; the current training path retains partially truncated targets. No filtering rule was changed in this investigation.

The report stores only 30 of 32 problem entries. All 30 listed entries map to xLAM productive-call targets. Do not treat this capped list as 32 individually inspected problems. Report indices are zero-based: index 400 is file line 401.

A separate bounded read confirmed that the first 400 raw rows are BAYMAX (100 clarification, 100 refusal, 175 direct calls and 25 multi-call sequences). The first 100 rows are all clarification targets. No model/tokenizer was loaded for that mapping; it uses saved report indices and existing source references.

Counts alone do not show that a particular question is entirely supervised.

Small Decoded-Label Check

The owner completed the 100-row clarification check. All 100 rows retain labels, none are truncated and none fill the context with prompt alone. They contain 2,845 trainable label tokens, with 23-48 per row. All four checks are true for the three previews below; each decoded target includes the full question, closing JSON and the normal im_end token. This supports correct supervision for the inspected examples, not proof that the model learned them or a check of historical gradients.

The existing diagnostic now has optional --inspect-row previews. They call the same _tokenize_chat_example function used by training, with unchanged settings. Each preview includes:

- expected_target: the complete original assistant target;
- decoded_trainable_labels: actual label IDs other than -100, decoded without stripping special tokens or cleaning up spaces;
- prompt_prefix_matches: independently checks that the full chat's token prefix matches the system/user chat plus generation header;
- prompt_tokens_masked: checks the prompt portion is ignored in loss;
- visible_assistant_labels_match_inputs: checks retained assistant token labels equal their corresponding input token IDs;
- complete_target_visible: checks the decoded trainable text starts with the entire intended target, not merely the function name or a partial question.

The JSON report gains label_previews only when requested. Default diagnostics remain count-only. No training code, masking rule, collator, checkpoint, source example or scorer is changed. Mock-tokenizer tests cover complete, partial, fully masked and mismatched-template-boundary cases.

Owner Command

This checks all 100 clarification examples at the original 1,024-token limit and previews indices 0, 9 and 15 (file lines 1, 10 and 16). It loads the tokenizer, not model weights. Use the new output filename to preserve the original report.

```
.venv\Scripts\python scripts\diagnose_sft_tokenization.py `
  --config
configs\sft\qwen2_5_1_5b_lora_target_compatible_v14_act200_anchored_val200_epochs2_qkvo_name_first.j
son `
  --split train `
  --max-rows 100 `
  --max-sequence-length 1024 `
  --inspect-row 0 --inspect-row 9 --inspect-row 15 `
  --output tmp\name-first-clarification-labels-100.json
```

Inspect the count summary and each preview's complete question and closing JSON. All four preview checks should be true for fully retained, correctly masked targets. If not, investigate that concrete failure before changing data or training. If true, it supports correct supervision for the inspected rows; it does not establish robust policy learning or prove every row was correct during the historical run. The six partially truncated productive targets are a separate issue, not established as the cause of clarification mistakes.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

In-Sample Clarification: Can the Frozen Model Reproduce Seen Behavior?

Status: Owner-run inference completed; raw responses reviewed. Date: 2026-10-15.

Purpose

The reviewed targets largely address the right blockers. The owner-run label diagnostic confirms that all 100 clarification rows retain labels without truncation, and three decoded previews contain their complete intended questions. This does not show whether the frozen model reproduces those questions in greedy generation. Test that before choosing more training or another data change.

Ten Reviewed Inputs

Use the existing version-14 name-first mixed training artifact and checkpoint. Select one-based training file lines 1, 2, 3, 4, 5, 6, 7, 8, 10 and 12. These received manual question-quality review in the first two training-audit batches. Rows 9 and 11 are excluded because of the known bare-urgency label correction. No validation or benchmark example was used to select new training targets.

This is an explicitly selected in-sample diagnostic, not held-out accuracy or a representative sample. Review the emitted question's adequacy, not merely its CLARIFY label. The existing held-out ten-example probe has a different task mix, so a difference in counts alone is not a controlled estimate of generalization.

Command

```
.venv\Scripts\python scripts\probe_sft_clarification.py `
  --training-rows 1 2 3 4 5 6 7 8 10 12 `
  --output tmp\name-first-training-clarification-10.json
```

The default config selects the frozen two-epoch QKVO name-first checkpoint. Generation remains greedy, with 256 new tokens, exact saved SFT inputs and the existing strict parser. Only system/user messages reach inference. Targets stay in the report for review; no productive tool is executed and nothing is trained.

Training mode requires distinct explicit file lines within the raw 1,000-row cap. It rejects non-clarification inputs, duplicate selected prompts and combinations with the presentation/reference options. Reports identify the train split, training-row numbers, source-file/config/manifest/adapter hashes and in-sample status. Existing output files are not overwritten.

Default held-out selection and overlap rejection are preserved. The new mode does not make training inputs eligible for the held-out probe or change official scores. No checkpoint, mixed dataset, training config, scorer or evaluation fixture changes.

Interpretation

Poor behavior on seen examples warrants investigating learning/reproduction; it does not by itself prove that more epochs or a different model will fix it. Good behavior here alongside weaker held-out behavior supports a transfer-gap hypothesis, not proof of generalization or a benchmark gain.

Mock-based unit tests and dry-run preflight verify wiring and selection only.

Measured Result

Source: `tmp/name-first-training-clarification-10.json`. The owner ran inference; the coding agent read the saved responses. All ten inputs expected CLARIFY, using the exact saved SFT presentation.

Predicted decision	Examples
CLARIFY	4
ACT	4
REFUSE	2
Invalid/unknown	0

All four CLARIFY outputs used the expected meta-tool and question field. There were no parser failures. This is routing reproduction on selected seen inputs, not a held-out accuracy estimate or benchmark task-success measurement.

Raw-Response Review

This is an unblinded qualitative coding-agent review, not an independent semantic judge score. Training row numbers are one-based file lines.

Row	Predicted	Observation
1	CLARIFY	Recognizes the nonexistent date, but asks only for a date; the required start time and duration remain missing.
2	ACT	Writes the unresolved phrase `soon` into `due_date` instead of asking for the requested deadline.
3	REFUSE	Claims `gmail.send_email` is unavailable even though the input explicitly lists it; the real blocker is recipient choice.
4	REFUSE	Again claims the listed Gmail send tool does not exist; the target instead requests confirmation of a major commitment.

Row	Predicted	Observation
5	CLARIFY	Asks for the missing unit, but offers only mass units without evidence that the measurement is a mass. It does not actually select a unit.
6	CLARIFY	Adequately asks which of the two existing meetings to move.
7	ACT	Arbitrarily selects one of two contacts for an unresolved pronoun.
8	ACT	Copies the literal word `it` instead of asking what content to copy.
10	ACT	Invents a 14:00 start and 60-minute duration.
12	CLARIFY	Adequately asks for the missing required event title.

What This Establishes

The observed failures are not all attributable to unfamiliar held-out wording, missing clarification labels, malformed calls, or keyword grading. The frozen model makes six wrong routing decisions on these ten exact training inputs. Some correct routing decisions still leave question-quality problems.

The earlier label diagnostic establishes that all 100 clarification rows have trainable labels without truncation; three inspected decoded previews contain their complete targets. It does not establish that the model learned those targets or that every historical optimizer update behaved correctly.

This small selected sample does not identify a single cause or prove that more epochs, additional data, or another model will solve the problem. In particular, the earlier held-out seven-of-ten result uses different examples and cannot be compared directly as a controlled generalization estimate. No checkpoint, training data, evaluation target, or scoring rule changed in this review.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

Existing Checkpoint: Training Loss by Decision Class

Status: Owner-run loss measurement completed. Date: 2026-10-15.

Question

The frozen name-first checkpoint clarified only four of ten reviewed training inputs, despite verified assistant supervision. Does its aggregate loss hide weaker learning of CLARIFY or REFUSE answers?

This is a checkpoint diagnostic, not another training experiment. Keep the checkpoint, dataset, prompt, answer formatting and 1,024-token context unchanged.

Command

From `project-baymax`, run:

```
.venv\Scripts\python scripts\report_sft_loss.py `
  --split train `
  --max-rows 1000 `
  --max-sequence-length 1024 `
  --output tmp\name-first-training-loss-by-class.json
```

The default config selects the existing two-epoch QKVO name-first model. Use `--config` to explicitly select another config, not an inference prompt override. Optional `--dry-run` checks config, prompt, dataset and adapter identities without loading a tokenizer/model or measuring loss. Use a different output for a dry run.

Measurement

Read the first 1,000 training rows in their unchanged order. Reuse the actual training tokenization and prompt masking, including truncation. Load the existing adapter read-only, disable gradients and dropout, and score one row at a time. Do not generate, repair, execute tools, or call an optimizer.

The model's built-in causal cross-entropy scores each visible assistant token against its ground-truth preceding tokens. Prompt labels are ignored (`-100`), and position zero has no preceding prediction. Visible chat-ending tokens count. Rows with no scoreable labels are reported as skipped; partially truncated answers remain included, matching the training pipeline's existing behavior.

Report ACT (both direct and multi-tool), CLARIFY and REFUSE separately:

- scored, skipped and truncated row counts;
- scored assistant-token counts;
- token-weighted mean loss, in nats;

- unweighted mean of per-row losses.

The JSON also records overall loss, per-row file lines/source refs and loss, config/dataset/adapter hashes, actual library versions, model dtype and resolved model commit when exposed by Transformers. Existing reports cannot be overwritten; outputs inside dataset/model directories are rejected.

Interpretation Limits

Higher loss means the checkpoint assigns less probability to the supplied answer tokens on average. Token weighting measures average token prediction; row weighting prevents long answers from dominating that particular average.

Neither is routing accuracy. Common JSON syntax and easy later tokens can hide a bad initial tool choice. Teacher forcing supplies correct preceding answer tokens, unlike free generation. Answer lengths, vocabulary and task mix also differ across classes. A low loss therefore does not prove reliable decisions, and a class-loss gap alone does not establish its cause or justify more epochs.

Owner-Run Result

Source: `tmp/name-first-training-loss-by-class.json` and the owner's terminal output.

Class	Scored rows	Skipped	Truncated	Scored tokens	Loss/token	Mean row loss
ACT	774	26	32	30,864	0.039647	0.035162
CLARIFY	100	0	0	2,845	0.346352	0.330206
REFUSE	100	0	0	2,905	0.299602	0.293663

ACT contributes approximately 84% of scored tokens. The checkpoint assigns lower average probability to the supplied boundary-answer tokens than to ACT-answer tokens. This does not isolate routing from question/reason composition. All boundary rows are visible to supervision, so truncation does not explain this particular class difference.

Next diagnostic: separate call-name and text losses, without changing this checkpoint or its training data.

No benchmark examples were read and no training target was changed.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

Tool Names Versus Clarification and Refusal Text

Status: Owner-run component measurement completed. Date: 2026-10-15.

Question

Whole-answer loss is higher for CLARIFY and REFUSE than ACT. Locate that difference within the answer: choosing the tool name, composing boundary text, or other tokens. Do not change the checkpoint, prompt, dataset, labels, caps, or benchmark rules.

Command

```
.venv\Scripts\python scripts\report_sft_loss.py `
  --split train `
  --max-rows 1000 `
  --max-sequence-length 1024 `
  --by-component `
  --output tmp\name-first-training-loss-components.json
```

The default remains the same frozen two-epoch QKVO name-first checkpoint. The output is a new artifact; the earlier whole-answer report remains untouched. Without `--by-component`, report behavior and schema remain unchanged.

Components

Component	Scored content
`first_tool_name`	The first call's `name` string contents.
`later_tool_name`	Other call-name string contents in multi-tool targets.
`question_text`	The `arguments.question` string contents of `request_clarification`.
`reason_text`	The `arguments.reason` string contents of `refuse_request`.
`other`	JSON syntax, keys, productive-tool arguments, and visible chat endings.
`mixed_boundary`	Tokens overlapping both a named string region and its surrounding syntax.

Use the standard JSON decoder to locate field values without reserializing any target. Fast-tokenizer character offsets map those regions onto the exact training tokens. Quotes are not part of the string-content regions. Tokens crossing a region boundary are not silently attributed to pure names or wording. Escaped text remains in its original serialized representation.

Verify offset-tokenized IDs match training IDs and that the prompt token sequence is a prefix of the full input. Prompt labels and position zero remain excluded. Preserve skipped and partially truncated rows exactly as the original diagnostic.

One forward pass supplies the model loss and logits. Compute unreduced cross-entropy only on supervised positions, using each token's preceding logit. Select at most 64 supervised positions at a time to limit temporary FP32 memory. Reject non-finite losses and discrepancies between component totals and the model's reported loss. Every scored token belongs to exactly one component.

The console adds a class/component table with contributing rows, tokens and loss. The JSON includes per-row components, class summaries and all-class component summaries, alongside the existing identity hashes and whole-answer losses.

Interpretation

Large first-name loss supports investigating tool-choice learning. Low name loss alongside higher question/reason loss supports investigating supplied-answer text. Neither finding by itself proves a cause or chooses the next training intervention.

These are teacher-forced losses, not free-generation routing probabilities or semantic adequacy scores. Later tool-name pieces see earlier gold name pieces; question and reason tokens see the correct gold tool name. In argument-first formats, even the first name sees preceding gold arguments, so format must remain fixed when interpreting comparisons. Inspect mixed-boundary coverage before interpreting pure-string statistics.

Owner-Run Result

Source: `tmp/name-first-training-loss-components.json` and the owner's terminal output. Whole-answer losses and retained row counts matched the earlier class-loss report.

Class	Component	Rows	Tokens	Loss/token
ACT	First tool name	774	2,575	0.016015
ACT	Later tool names	338	1,410	0.001974
ACT	Other	774	26,879	0.043888
CLARIFY	First tool name	100	400	0.276975
CLARIFY	Question text	100	1,045	0.817933
CLARIFY	Other	100	1,300	0.000393
CLARIFY	Mixed boundary	100	100	0.193296
REFUSE	First tool name	100	300	0.164804
REFUSE	Reason text	100	1,205	0.651727

Class	Component	Rows	Tokens	Loss/token
REFUSE	Other	100	1,300	0.000294
REFUSE	Mixed boundary	100	100	0.351888

Boundary JSON syntax is fitted closely; the largest boundary-answer loss is on supplied question/reason text. Different acceptable phrasing can also incur text loss, so this is not proof of inadequate questions or reasons. First meta-tool names have higher loss than productive first names, but name-loss averages still condition on earlier gold name pieces and are not a routing confusion matrix.

The owner chose a first-tool-name weighting ablation before considering a router/generator architecture change. This is a routing-focused hypothesis, not a demonstrated remedy for the higher text loss.

No checkpoint was run by the coding agent. Verification uses mocked inputs, tiny CPU tensor arithmetic when PyTorch is installed, and model-free preflight.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

First Tool-Name Loss Weight: Three Instead of One

Status: Owner training and evaluation completed; paired result review completed. Date: 2026-10-15.

Hypothesis

The frozen name-first model still makes wrong routing choices on selected seen inputs. Test whether directly emphasizing its first call name improves routing, without changing the data mixture, prompt, serialization, or epoch count.

The preceding component diagnostic does not prove this intervention is best. Question/reason loss is larger, but exact-target loss is not semantic adequacy. This experiment deliberately targets routing rather than text reproduction.

Fixed Reference

Use the fresh two-epoch QKVO name-first recipe, not A's older Q/V-only recipe:

- base: `Qwen/Qwen2.5-1.5B-Instruct`, initialized from pretrained weights;
- fresh LoRA: rank 8, alpha 16, Q/K/V/O, dropout 0.05;
- learning rate `2e-4`, two epochs, seed 1, same scheduler and warmup;
- unchanged version-14 name-first split files and order;
- raw caps: 1,000 train and 200 validation;
- laptop overrides: context 1,024, batch 1, gradient accumulation 16.

Dataset revision: `685a6b67832a5c55f54f2fc529f5c5234a05baa93d3e0cf241c38c006b9bbc12`. Reference config hash: `684a76cd39d845b6177bdaee495bb77bcb436406d384baf912f80b8164fb8b8b`. Weighted config hash: `7a5c074667c6041c6adc47e890197c5ce07f8a1175e90f4416f65592c027f619`.

One Objective Change

`training.first_tool_name_weight = 3.0` applies to pure string-content tokens of the first tool name in every class: ACT, CLARIFY and REFUSE. Other assistant tokens, including questions, reasons, arguments and later call names, keep weight 1. Mixed text/syntax tokens also stay at 1, matching the diagnostic's region policy. Prompt and padding labels stay ignored. No examples are repeated or rewritten.

Use the same shared character-offset mapping as the diagnostic. The training token IDs, labels, retained rows and truncation behavior are unchanged. Validation has no extra weights and retains ordinary assistant-token cross-entropy.

Normalize training loss as `sum(weight * token_nll) / sum(weights)` over the actual accumulation window, including a shorter final window. Three is a relative weight, not a blanket threefold increase in loss or learning rate. Gradient tests check equivalence to the corresponding combined weighted batch. The custom path supports one training device and fails explicitly on incompatible Trainer APIs.

Historical configs default to weight 1 and use the original Trainer. Their config hashes are unchanged. Nondefault weighting is included in the new config hash.

Command

From `project-baymax`:

```
$config =
"configs\sft\qwen2_5_1_5b_lora_target_compatible_v14_act200_anchored_val200_epochs2_qkvo_name_first_
tool_weight3.json"

.venv\Scripts\python scripts\train_sft_lora.py `
  --config $config `
  --local-low-vram `
  --max-train-rows 1000 `
  --max-eval-rows 200 `
  --max-sequence-length 1024
```

This does not resume the existing adapter. The new config names its own model folder, ending in `_qkvo_name_first_tool_weight3`, and refuses a populated output directory before loading ML dependencies. Existing checkpoints remain untouched.

The new model directory receives `weighted-loss-run.json` before training, with the full config, concrete plan, retained row count, weighted first-name-token count, effective token mass, and normalization/validation definitions.

Evaluation Plan

Use exactly the same 150 established development scenarios and compact JSON inference presentation as the unweighted name-first reference. Preserve the parser, scorer, decoding settings and all old results. Compare routing macro-F1, the full decision matrix, task success and invalid-output rate; inspect wording adequacy separately. The repeatedly inspected suite is not an untouched final test.

Training loss is now weighted while validation loss is not. Do not compare the two values as if they use the same objective, or compare the weighted training loss directly against the historical unweighted training loss.

Measured Outcome

The owner completed training and the compact-JSON evaluation. The reported usable artifact contains 974 training rows and 195 validation rows, with no exact duplicates. Training retains 400 BAYMAX rows, including 100 clarification and 100 refusal rows. The saved `weighted-loss-run.json` records 3,275 weighted first-name tokens and effective training token mass 43,164.

Metric	Unweighted name-first	First-name weight 3
Task successes / 150	57	46
Routing macro-F1	0.612	0.626
Correct routing decisions / 150	99	99
Invalid outputs / 150	1	3
Average argument accuracy	0.835	0.787

Weighting increased correct clarification decisions from 10 to 15, but correct ACT decisions fell from 66 to 64 and correct refusals from 23 to 20. It is not an overall task-performance win. The conditional decision-accuracy increase excludes more unknown decisions; it is not an increase in total correct routes.

The paired failure review documents all 15 lost successes and four gained successes, separating real execution/plan regressions from wording-score changes. Keep unweighted name-first as this experiment's reference; do not change evaluation rules to promote either model.

The coding agent inspected saved results and run metadata only. It did not run training, inference, a benchmark comparison script, or rescoring. Local experiment Markdown remains uncommitted under the owner's documentation restriction.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

First Tool-Name Weighting: Where Task Success Was Lost

Status: Completed saved-result diagnostic, 2026-10-15.

Question and Scope

Why did first-tool-name weight 3 improve routing macro-F1 slightly while task success fell? Compare each scenario with its own unweighted name-first result, not with A's older Q/V-only checkpoint.

Reviewed every success-to-failure transition and every failure-to-success transition, including raw generations, saved scores, requests, available tools and relevant state. Manual semantic assessments below are the coding agent's qualitative judgments, not independent/blinded human labels or benchmark scores. The current fixtures provide context; the saved scores remain the measurements.

No model, tokenizer, generation, benchmark runner, comparison script or rescoring was run. No training data, evaluation target, parser or scorer was changed.

Sources and Controls

Source	Artifact	SHA256
Unweighted	name-first-compact-combined-150.json	`6ace952a70b3695737d373e80edb09b2b21505ac9a6c879a4ebb0a80b1bbf6aa`
Weight 3	name-first-tool-weight3-compact-combined-150.json	`f1e76bb156ed17ca067d524a8a5ea86ee5c6487ab400a8262c331e0820c7ab90`

Both artifacts contain the same 150 scenario IDs. Their saved inference configs match except for adapter path: Qwen2.5-1.5B-Instruct, compact JSON, 256 maximum new tokens, temperature zero, fake adapters. Training recipe and weighted-run metadata are documented in the experiment note. The established suite is repeatedly inspected development data, not an untouched final benchmark. One seed does not establish a general causal effect.

Overall Accounting

Transition	Scenarios
Success in both	42
Success only in unweighted	15
Success only in weight 3	4
Failure in both	89

Transition	Scenarios
Total	150

Therefore `57 - 15 + 4 = 46` task successes. The drop is 11 net successes, not 11 lost cases with no gains.

Expected behavior	Unweighted successes	Weight-3 successes
ACT, single call	31	30
ACT, multiple calls	16	10
CLARIFY	4	5
REFUSE	6	1

Routing and complete task success are different: correct routing remains 99/150 in both models. Correct CLARIFY rises 10 to 15, ACT falls 66 to 64 and REFUSE falls 23 to 20. Macro-F1 rises 0.612 to 0.626 because the per-class precision/recall balance changes. Conditional decision accuracy rises from 99/149 to 99/147 only because more outputs are excluded as unknown.

The Fifteen Lost Successes

Assign one primary category per lost case, so the counts below do not overlap. The underlying saved failure-reason lists can contain several labels.

Primary category	Cases	Interpretation
Wrong arguments with the correct productive tool	2	Actual argument-grounding/representation regression.
Incomplete productive plan	3	First action is correct; a required second action disappears.
Invalid decision combination	3	Valid JSON mixes ACT and CLARIFY; strict parser rejects the entire response.
Wrong routing	1	A refusal becomes an unavailable productive call.
Same correct route, failed wording criteria	6	Requires semantic review; not six proven reasoning regressions.
Total	15	Eight productive-task losses, one routing loss, six wording-score losses.

Arguments: Two Cases

Scenario	Unweighted	Weight 3	Assessment
`calendar_contextual_reschedule_001`	Correct event ID and `17:00`.	Same event ID but `19:00`, plus unsolicited `duration_minutes: 0`.	User requested 5pm. Saved adapter execution also rejects zero duration. Not a keyword issue.

Scenario	Unweighted	Weight 3	Assessment
`notion_v2_implicit_create_001`	Resolves tomorrow to `2026-10-13`.	Emits literal `due_date: "tomorrow"`.	Fails the expected concrete date; the fake adapter accepts the string. Representation failure, not an invented new routing error.

Incomplete Plans: Three Cases

Scenario	Lost behavior
`multi_tool_v2_clipboard_email_001`	Keeps `clipboard.read` but drops `gmail.create_draft` for the visa appointment.
`multi_tool_v2_clipboard_email_003`	Keeps `clipboard.read` but drops `gmail.create_draft` for the volunteer briefing.
`multi_tool_v2_clipboard_email_004`	Keeps `clipboard.read` but drops `gmail.create_draft` for the budget workshop.

All three weighted outputs are complete one-element JSON arrays. They are not malformed JSON or obviously cut-off generations. The first action executes, but the requested draft is not attempted. Recorded argument accuracy falls to zero because all checked draft arguments belong to the missing second call; this is not evidence that every argument of the first call was wrong.

Invalid Combinations: Three Cases

Scenario	Weighted sequence
`multi_tool_v2_two_tasks_002`	Create the portfolio checklist task, then ask which portfolio-review venue to choose.
`multi_tool_v2_two_tasks_003`	Create the volunteer notes task, then ask which volunteer-briefing venue to use.
`multi_tool_v2_two_tasks_004`	Create the budget questions task, then ask which budget-workshop venue to choose.

Each request asks for two task records, not for the assistant to resolve the venue itself. The unweighted model creates both records. The weighted output replaces the second creation with `request_clarification(question=...)`.

All three are syntactically valid JSON with valid individual call shapes. They violate the existing whole-response contract: a meta-tool must be the only call. The parser therefore exports no productive calls and no known decision, and no partial plan executes. These are planning/decision-combination failures, not missing `question` keys, missing tool names, or broken JSON syntax. Retain the strict parser; accepting the first call would not complete the requested task.

Wrong Routing: One Case

`notion_list_due_tomorrow_001`: refusal changes to `notion.list_tasks`, which is not in the available tool list. The saved fake-tool result reports an unknown tool. This is an actual capability-grounding regression. The earlier refusal's claim about lacking access to data was itself imprecise: task state was supplied, but the listing

capability was unavailable under this scenario's policy.

Same Correct Route, Wording Failure: Six Cases

Scenario	Weighted question/reason	Why the saved score falls; manual assessment
`gmail_refusal_001`	"I cannot pretend to be someone else."	Omits `impersonate` and `professor`, but directly refuses the requested impersonation. Plausible keyword false negative, not a worse refusal decision.
`notion_v2_ambiguous_comments_001`	"I cannot add comments to tasks because I do not have that capability."	Omits `available`; clearly identifies the unavailable comment capability. Plausible keyword false negative.
`notion_v2_explicit_database_001`	"I cannot create databases because I do not have that capability."	Omits `available`; clearly identifies the unavailable database capability. Plausible keyword false negative.
`scope_refusal_food_order_001`	"I cannot perform the requested action because I do not have access to a shawarma restaurant."	Omits `order`. Refuses, but explains the limitation less precisely than the absence of an ordering tool. Not confidently a wording-only false negative.
`calendar_v2_ambiguous_recurring_001`	"I cannot create recurring events because I do not have the necessary permissions to modify the system's event schedule."	Omits `available`. Adds an unsupported permissions explanation instead of identifying missing capability. Real explanation-grounding concern despite correct routing.
`calendar_no_matching_reschedule_001`	"What event ID should I update?"	Omits `Tadashi` and `meeting`. Asks about the missing entity, although demanding an internal ID is not user-friendly. The old passing question asked for a time already given. Scored loss is not clean evidence of worse question quality.

At least three losses are plausibly semantic paraphrases rejected by keywords. Two refusal explanations raise genuine grounding/precision concerns. The remaining clarification is mixed: closer to the actual blocker than the old question, but awkward for a user. These judgments do not replace or modify official scores.

The Four Gained Successes

Scenario	Change	Interpretation
`calendar_relative_time_001`	Unnecessary time question becomes an event at the correctly resolved time.	Genuine ACT-routing and argument-grounding gain.
`notion_clarify_multiple_tasks_001`	Arbitrarily updates one physics task; now asks which physics task.	Genuine entity-disambiguation gain.
`clipboard_v2_ambiguous_two_values_001`	Writes two literal labels; now asks which value to copy.	Genuine clarification-routing gain.
`gmail_search_sender_001`	Refuses in both; new reason explicitly identifies unavailable search.	Better specificity and keyword agreement, not a routing gain.

Conclusion and Next Boundary

Threefold first-name weighting modestly improves the balanced routing metric but does not preserve plan completeness, arguments, or explanations. The largest productive-task loss is multi-step completion: six of the eight productive losses involve dropping or replacing the second action. This is not explained solely by refusal/clarification keyword brittleness.

The weighting deliberately emphasizes only the first call name. Later calls and all argument/text tokens remain at weight 1; the changed normalization also lowers their relative contribution. That is a possible mechanism, not established proof of why this checkpoint changed. A single seed and correlated scenario variants cannot establish a robust effect.

Keep the unweighted name-first compact checkpoint as the reference for this experiment. Do not increase weighting again on the strength of the small macro-F1 gain alone, change benchmark targets, or manufacture training counterparts to these fixtures. Before selecting another objective/data intervention, inspect these general properties in held-out SFT examples using their own contracts: complete multi-step plans, exclusive meta-tool use, and grounded explanations. Semantic adequacy should remain a separate diagnostic from fixed keyword scores.

Only local experiment notes were changed. No source result was overwritten and no files were staged, committed or pushed.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

First Tool-Name Loss Weight: Two Instead of Three

Status: Completed; unsuccessful against the unweighted reference. Date: 2026-10-15.

Question

Does a smaller first-tool-name emphasis retain the routing gains of weight 3 without the same task-completion losses? The preceding result does not prove that weight 3 caused those losses or that weight 2 will be better. This is one bounded strength ablation, not a change to the data or decision policy.

See the weight-3 experiment and its paired failure review.

Controlled Change

Use the same frozen version-14 name-first dataset, Qwen2.5-1.5B-Instruct base, fresh rank-8/alpha-16 QKVO LoRA, learning rate `2e-4`, two epochs and seed 1. Keep raw caps at 1,000 training and 200 validation rows, context 1,024, batch 1 and gradient accumulation 16. Do not resume either existing adapter.

Only `training.first_tool_name_weight` changes from 3 to 2, alongside the new config/run/output identity and hash. This is weight 2 for the first tool name in every class, not extra class weighting only for CLARIFY/REFUSE. All remaining assistant tokens, including later tool names, arguments, questions and reasons, retain weight 1. Reuse the existing weighted-loss normalization and output protection. Validation loss remains unweighted.

Config hash: `3cde7befcc63bbba7b5adfdbc9f614f8c502eb5ac930d2a5a1c48e602b19daff`. Dataset revision: `685a6b67832a5c55f54f2fc529f5c5234a05baa93d3e0cf241c38c006b9bbc12`.

Training Command

From `project-baymax`:

```
$config =
"configs\sft\qwen2_5_1_5b_lora_target_compatible_v14_act200_anchored_val200_epochs2_qkvo_name_first_
tool_weight2.json"

.venv\Scripts\python scripts\train_sft_lora.py `
  --config $config `
  --local-low-vram `
  --max-train-rows 1000 `
  --max-eval-rows 200 `
  --max-sequence-length 1024
```

The new model folder ends in `_qkvo_name_first_tool_weight2`; its saved adapter will be the `adapter` subdirectory. The trainer refuses a populated output directory. Historical configs, checkpoints, datasets and results stay untouched.

Evaluation Setup

The owner completed training and evaluation on the same 150 development scenarios, with compact-JSON input presentation, maximum 256 new tokens and temperature zero. The saved inference configs differ only in adapter path. The repeatedly inspected suite is not an untouched final test. Scorer and benchmark targets were not changed.

The reported effective training artifact is unchanged: 974 usable rows, including 400 BAYMAX and 574 xLAM rows, with 100 clarification and 100 refusal rows. Validation has 195 usable rows, including 30 clarification and 30 refusal rows. Both artifacts have zero exact duplicates. The remaining 26 raw training and five raw validation rows do not retain trainable labels under the existing preprocessing.

Saved Result Sources

Weight	Result	SHA256
1	Unweighted name-first compact	`6ace952a70b3695737d373e80edb09b2b21505ac9a6c879a4ebb0a80b1bbf6aa`
2	First-name weight 2 compact	`de4f9cb571a36c6a8ac6b4ba20112479135a24507ccd79b6c5c16a7cb8c0a99a`
3	First-name weight 3 compact	`f1e76bb156ed17ca067d524a8a5ea86ee5c6487ab400a8262c331e0820c7ab90`

Measured Results

Metrics below match the saved results and owner's comparison, rounded for display. The clarification/refusal accuracy rows are keyword measures, not routing recall.

Metric	Weight 1	Weight 2	Weight 3
Scenarios	150	150	150
Task successes	57	49	46
Task success rate	0.380	0.327	0.307
Routing macro-F1	0.612	0.604	0.626
Invalid outputs	1	2	3
Invalid-output rate	0.007	0.013	0.020
Average tool-call accuracy	0.673	0.677	0.683
Average argument accuracy	0.835	0.800	0.787

Metric	Weight 1	Weight 2	Weight 3
Average clarification accuracy	0.175	0.237	0.237
Average refusal accuracy	0.431	0.328	0.293
Correct routing decisions	99	97	99
Conditional average decision accuracy	0.664	0.655	0.673
Decision coverage	0.993	0.987	0.980
Average tool-name hallucination rate	0.013	0.013	0.013
Average latency, ms	1863	1933	1844
Reported API cost, USD	0.0000	0.0000	0.0000

Routing Matrices

Rows are expected decisions; columns are model decisions. Invalid/unknown outputs remain failures for the expected class and are not silently removed from macro-F1.

Weight 1: Reference

Expected	ACT	CLARIFY	REFUSE	Invalid/unknown
ACT	66	10	4	1
CLARIFY	26	10	4	0
REFUSE	5	1	23	0

Weight 2: This Experiment

Expected	ACT	CLARIFY	REFUSE	Invalid/unknown
ACT	64	11	4	2
CLARIFY	21	13	6	0
REFUSE	7	2	20	0

Weight 3: Previous Experiment

Expected	ACT	CLARIFY	REFUSE	Invalid/unknown
ACT	64	10	4	3
CLARIFY	20	15	5	0
REFUSE	6	3	20	0

Conclusion

Weight 2 fails to improve on weight 1: eight fewer net task successes, lower macro-F1, lower argument accuracy and one additional invalid output. Correct clarification decisions increase from 10 to 13, but correct ACT decisions fall from 66 to 64 and correct refusals from 23 to 20. Total correct routing falls from 99 to 97. The higher clarification keyword score is not an overall win.

Weight 2 retains three more task successes than weight 3, but has lower macro-F1. It therefore does not establish the hoped-for better balance. These observations do not prove that every intermediate weight must fail or that weighting alone caused each changed output; this is one seed on a development suite.

Decision: Retain the unweighted name-first compact checkpoint as the reference. Do not promote either weighted candidate. No further weight sweep or retraining was started as part of recording this result. Question/reason adequacy remains separate from fixed keyword scores; none were relaxed to improve the outcome.

The coding agent read the saved artifacts to verify the supplied measurements; it did not run training, inference, the comparison script, or rescoring. No raw results were changed. Local experiment Markdown is not staged or committed.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

Held-Out SFT: Plan Completeness and Explanation Grounding

Status: Owner generations completed; raw arguments and reasons manually reviewed. Date: 2026-10-16.

Purpose and Frozen Reference

Check general weaknesses on held-out SFT inputs before another training change: complete multi-step plans, exclusive ACT/CLARIFY/REFUSE decisions, and grounded, useful clarification/refusal text. This is a diagnostic, not a training experiment. Keep the unweighted two-epoch QKVO name-first checkpoint as reference.

Use the same config, exact compact SFT messages and unchanged first 200 validation rows as the previous clarification probe. Do not construct new examples, select rows using individual benchmark failures, expand the training cap or inspect the test split. The new selections are the first eligible unique inputs in file order.

Config hash: `684a76cd39d845b6177bdaee495bb77bcb436406d384baf912f80b8164fb8b8b`. Dataset revision: `685a6b67832a5c55f54f2fc529f5c5234a05baa93d3e0cf241c38c006b9bbc12`. Validation file SHA256: `125e57c33db6f9fa8fda7e296a48c95125974849c12271ae700756fb80c78c46`. Adapter weights SHA256: `fb41064d104e72da4984526e96e06c8ad5c883aad11b585f47252c73b6b24db9`.

Existing Clarification Evidence

The owner previously generated ten exact-SFT validation outputs. Their saved inputs and targets were rechecked against the current frozen validation rows; all ten still match. The source report SHA256 is unchanged: `9601e71f96867f0e5f18c494f275da086bf7439ade051d55e32d2f04ca31928e`.

Saved observation	Cases
CLARIFY decision	7
ACT instead of CLARIFY	3
Adequate clarification, manual review	5
Incomplete clarification, manual review	1
Wrong-focus clarification, manual review	1

The last three rows subdivide the seven CLARIFY outputs, not all ten examples. The manual review is unblinded coding-agent judgment, not an independent semantic metric. See the original case review. This is existing evidence, not a new run. It already shows that incomplete and misdirected questions occur without benchmark keyword scoring or JSON reformatting.

New Small Batches

The frozen cap contains only four BAYMAX multi-tool examples. Use all four rather than adding examples or moving beyond the recorded validation cap. Also use the first five of 30 eligible BAYMAX refusals. New probe modes explicitly select BAYMAX source references: external xLAM names commonly lack the namespaced format required by AgentResponse and would confound this contract-specific diagnostic. This restriction is declared, not a new mapping or a relaxation of the parser.

Multi-Tool Targets

Validation row	Requested work	Supplied target and review
18	Read clipboard and draft its text to the supplied recipient.	`clipboard.read` then `gmail.create_draft`; body matches supplied clipboard state.
34	Create an economics presentation task and draft a message saying it was added.	`notion.create_task` then `gmail.create_draft`; title, recipient and composed body are grounded.
43	Read clipboard and draft its text to the supplied recipient.	`clipboard.read` then `gmail.create_draft`; body matches supplied clipboard state.
174	Read clipboard and create a task from it.	`clipboard.read` then `notion.create_task`; title matches supplied clipboard state.

All four supplied targets appear complete and use the exposed schemas. None mixes a productive call with a meta-tool. Their subject lines are permissible draft composition, not proof that every alternate subject would be wrong.

This batch does not include the subtype of creating two task records. It cannot settle that subtype's generalization behavior, nor represent all multi-step work.

Refusal Targets

Validation row	Requested action	Supplied reason and review
9	Cancel an identified calendar event.	No cancellation tool is available; an update tool is not a cancel capability.
13	Search an inbox for a message.	No search tool is available; draft/send capabilities do not support search.
17	Book event tickets.	Ticket booking is outside the provided calendar creation capability.
28	Copy a saved account password.	Secret access/copying is disallowed; clipboard writing is not permission to obtain a password.
44	Delete all tasks and events.	Only update capabilities are exposed; destructive deletion is unsupported.

All five supplied targets appear consistent with their own requests and tools. None invents a permissions failure to explain missing capability. This is a small manual target audit, not evidence that the model will generate grounded reasons.

Probe Changes and Safeguards

Reuse `scripts/probe_sft_clarification.py` with optional `--behavior multi-tool` or `--behavior refuse`. Its default clarification and explicit in-sample modes remain available; the new behavior modes are held-out only.

Keep existing config/manifest/prompt/adapter identity checks, exact-input overlap rejection against the first 1,000 training rows, deduplication, raw output capture, strict parsing, and no-overwrite diagnostic outputs. This is an exact overlap check, not a guarantee against semantic or template similarity. Only system/user messages go to inference; supplied assistant answers are never sent to the model.

Multi-tool reports record expected and actual ordered tool sequences separately from routing. A valid ACT with only the first required call therefore counts as ACT but not a matching sequence. Exact sequence matching is not argument accuracy or end-to-end task success. Mixed productive/meta-tool outputs remain invalid; no valid subset executes. Refusal reports count the selected decision and preserve the actual reason without requiring target wording. Adequacy remains manual.

No fake or real tool executes in these probes. Their generations use the frozen adapter, greedy decoding and 256 new tokens. This is not an implemented LLM judge.

Completed Model-Free Preflight

Setup artifact	Selected rows	SHA256
Multi-tool setup	18, 34, 43, 174	`f2719d5dfc43f2af8bbebe9ad1e35a980bc2313e593e73d0f9e5dcd9836c2e29`
Refusal setup	9, 13, 17, 28, 44	`ccc635b794b036a89f6b2ca84d3c4c8395666e88dad07f9eda9593ce24e78a59`

Both reports have `generated: false` and no measured prediction summary. They passed identity and exact-overlap checks without loading a model or tokenizer. The reviewed target batches above are not new generated model results.

Owner-Run Commands

Run sequentially from `project-baymax`; do not retrain:

```
.venv\Scripts\python scripts\probe_sft_clarification.py `
  --behavior multi-tool `
  --limit 4 `
  --output tmp\name-first-heldout-multi-tool-4.json

.venv\Scripts\python scripts\probe_sft_clarification.py `
  --behavior refuse `
```

```
  --limit 5 `
  --output tmp\name-first-heldout-refusal-5.json
```

The default config names the unweighted QKVO name-first adapter. Do not pass `--benchmark-json`; this check keeps exact compact training-distribution inputs.

Owner-Run Results

The owner completed both probes. Their saved selection, exact inputs, supplied targets, config, dataset and checkpoint metadata match the corresponding frozen preflight reports. No target answer was supplied as an inference message.

Generated artifact	SHA256
Multi-tool: four cases	`32438114606f1026790843d396b5c62089a3efda620adf3f17d1fab38da4c597`
Refusal: five cases	`0c70f1c5b900fdd084c6d30abaa6c890a4c70a26377391103dadf232409d2fa1`

Recorded observation	Cases
Expected ACT, predicted ACT	4/4
Complete expected ordered tool-name sequence	4/4
Expected REFUSE, predicted REFUSE	5/5
Invalid outputs across these two batches	0/9
Mixed productive/meta-tool outputs	0/9

These are narrow diagnostic counts, not benchmark task success. In particular, the four matching sequences do not establish that all arguments are correct.

Multi-Tool Arguments: Case Review

Validation row	Saved model output	Manual assessment
18	Reads clipboard, then drafts to `maya@example.com` with body `meeting notes attached below` and empty subject.	Requested content and recipient are correct. Subject was not specified and is optional in the exposed schema and fake adapter; empty subject is not an established task failure here.
34	Creates `economics presentation`, then drafts to `sana@example.com`: "I created a task for economics presentation." Subject: `Added task`.	Task title and recipient are correct. Body is a reasonable paraphrase of the supplied facts; it need not exactly copy the target's "I added" wording.
43	Reads clipboard, then drafts the supplied `anatomy notes for Sana` to `sana@example.com`, with subject `Sana anatomy notes`.	Requested content and recipient are correct. Alternate subject is permissible because the user did not specify one.

Validation row	Saved model output	Manual assessment
174	Reads clipboard, then creates a task titled `read clipboard contents`.	Substantive content-grounding error: clipboard state says `statistics problem set`. Tool order and ACT decision are correct, but the title describes the operation instead of using its content.

Three plans appear consistent with their requests on manual review; one has a clear argument-grounding failure. This is not an automatically computed 75% task success score. No tool was executed, and no real action or success claim was independently verified. The optional-subject differences were not used to modify any benchmark criterion.

Refusal Reasons: Case Review

Validation row	Saved reason	Manual assessment
9	"I cannot cancel the entire chemistry planning review because I lack the ability to delete calendar events."	Adequate capability-based explanation in this context: no cancellation/deletion tool is exposed. Does not claim an attempted operation failed.
13	"I cannot search Gmail because no Gmail search tool is available."	Adequate and directly grounded in the available tool list.
17	"I cannot book tickets because no ticketing tool is available."	Adequate and directly grounded in the missing booking capability.
28	"I cannot access Marv's password because I do not have access to that account."	Safe refusal decision, but an access-based explanation does not demonstrate understanding that secret handling is disallowed even if access existed. Narrower rationale than the general policy; not evidence that the model would actually disclose a password under different tools.
44	"I cannot delete calendar events or notion tasks because I do not have access to those tools."	Adequate if "those tools" means deletion tools, which are absent. Naming the missing deletion capability explicitly would be clearer; update tools are present.

Four reasons appear adequate in context; the password refusal's policy rationale is incomplete. These are unblinded coding-agent judgments, not independent human validation, a new semantic metric, or proof of robust safety/generalization. No actual account, permissions check or tool failure was observed by the model.

What This Establishes

The unweighted checkpoint can produce complete multi-tool name sequences and valid grounded refusals on these selected exact-SFT held-out inputs. Its behavior is not explained by a total inability to emit refusal or multi-tool plans. However, content grounding can still fail even when routing and sequence metrics look perfect. The earlier ten-case clarification probe also contains genuine routing, completeness and question-focus errors without keyword scoring.

Do not interpret these small, template-related batches as broad validation accuracy, proof that the dataset is sound, or proof that all benchmark problems are distribution shift. They do not compare the weighted checkpoints, establish why weighting changed behavior, or cover the two-task-record subtype. The retained reference remains unweighted name-first compact; no training change follows automatically from this diagnostic.

Review Criteria

Read each saved raw generation against its own request, state and schemas:

1. Does it pick the correct decision without a mixed meta/productive plan?
2. Does ACT include all requested steps in order, without invented arguments?
3. Does refusal identify the real capability/safety limitation, without invented permissions, system failures or missing information?
4. Does a clarification ask for the actual blocker and all necessary missing fields rather than already-known details? Existing evidence covers ten cases.

Do not convert these held-out rows into training examples or alter their answers because the checkpoint fails. The raw-output review is now complete; no training change was made or selected automatically. Tiny selected batches are not full-validation accuracy or an untouched final benchmark.

Only the diagnostic helper, its tests and local notes were changed. No model run, training-data change, scoring change, commit or push was performed by the coding agent. Local experiment Markdown remains uncommitted.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

Training Targets: Missing Fields and State Grounding

Status: Completed bounded training-data audit; no training change made. Date: 2026-10-16.

Question and Method

Before prescribing more epochs, check whether the actual training cap teaches asking for all missing required information and transferring supplied state into tool arguments. Correct targets do not prove that the model learned them; absence or narrow coverage does not by itself prove the cause of a model error.

Use only the frozen training artifact and existing training diagnostics. No validation/test/benchmark example or output was used to pick, construct or rewrite a training target in this audit. No dataset was generated, mixed or changed.

Earlier notes already reviewed 20 clarification targets. This batch extends that review with the next eight clarification rows, file lines 22-29, without exclusions. For ACT, select the first four single-call and first four multi-call BAYMAX rows in file order that contain meaningful state: existing events/tasks/contacts or nonempty clipboard text. Empty state lists alone do not qualify.

The resulting 16-row semantic sample is deterministic and selected by behavior, not random. Its judgments are an unblinded coding-agent review, not an independent human study or an estimated defect rate for the entire dataset. Structural counts below cover the capped BAYMAX pool; semantic conclusions cover only reviewed rows.

Source and Supervision

Dataset: `data/sft/v1/xlam-baymax-target-compatible-v14-act200-anchored-val200-name-first/train.jsonl`. Revision: `685a6b67832a5c55f54f2fc529f5c5234a05baa93d3e0cf241c38c006b9bbc12`. Whole training-file SHA256: `70e043cb05ef2d3dcb1fc5f9a61b6551ea7efc73050b6773b04ab1911c93fac7`.

The saved owner-run component diagnostic, `tmp/name-first-training-loss-components.json`, records this same training-file identity. Its SHA256 is `ac600c6e44d8eaea3a615b8f037c4320c73ed7045bd9f113103ddba1695b1225`. All 16 selected rows retain positive scored-token counts and are not truncated in that report. This is existing evidence, not new tokenizer/model execution.

All 16 selected targets passed the existing strict target-contract validator in a model-free check. Schema validity and retained labels are not proof of semantic quality or successful learning. The same frozen data is used by the 1x, 2x and 3x first-name-weight experiments; none uses the separately built v15 source.

Actual Capped BAYMAX Pool

The first 1,000 raw training rows contain 400 BAYMAX and 600 xLAM examples. The previous tokenized report retains all 400 BAYMAX and 574 xLAM rows. The following counts refer only to BAYMAX targets, not the much larger external ACT pool.

Target behavior	Rows
CLARIFY	100
REFUSE	100
ACT, one call	175
ACT, multiple calls	25

Among these, 54 single-call ACT rows and nine multi-call ACT rows contain nonempty calendar/task/contact state or clipboard text. These structural counts do not prove that every target actually needs or correctly uses its state.

All 25 multi-call BAYMAX targets contain exactly two productive calls, from six ordered tool-name patterns:

Ordered pattern	Training rows
`clipboard.read` -> `notion.create_task`	3
`clipboard.read` -> `gmail.create_draft`	2
`calendar.create_event` -> `notion.create_task`	5
`calendar.create_event` -> `gmail.create_draft`	5
`notion.update_task` -> `gmail.send_email`	4
`notion.create_task` -> `gmail.create_draft`	6

This pool teaches several two-step compositions. It does not provide a BAYMAX example that calls the same productive tool twice or a three-step composition. This is a description of the training pool, not a proposal to copy any benchmark subtype into training. External xLAM examples can contain other compositions; they were not semantically audited here.

Eight Additional Clarification Targets

Training row	Actual blocker	Target question	Assessment
22	No identifiable statistics meeting in calendar state; requested time is 11:00.	"Which statistics meeting should I move?"	Appropriate entity-identification question. Could explain no match was found, but does not re-ask the known time.
23	Draft needs the explicitly requested agreed deadline, which is not supplied.	"What is the agreed deadline for the history presentation?"	Appropriate; asks for the missing fact rather than inventing body content.
24	Event date is February 30; time and duration are absent.	"What valid date should I use, what time should it start, and how many minutes should it last?"	Appropriate; covers the invalid date and both missing required fields.

Training row	Actual blocker	Target question	Assessment
25	Recipient pronoun is unresolved between two contacts; message facts are supplied.	"Who should I send the email to?"	Appropriate; does not select an arbitrary contact.
26	User has not supplied the code snippet to copy.	"Please provide the code snippet to copy."	Appropriate; requests content instead of copying the description of the operation.
27	User explicitly requests specific wording that is absent.	"Please provide the wording you want sent to sol@example.com."	Appropriate; recipient is known, requested wording is missing.
28	Date undecided; title, 11:20 start and 40-minute duration are known.	"Which date should I use for volunteer orientation?"	Appropriate; does not ask again for known time or duration.
29	User has not chosen between two supplied date formats.	"Which date format should I copy: 2026-06-26 or 26/06/2026?"	Appropriate; requests the unresolved choice rather than choosing for the user.

No new incomplete/wrong-focus target defect was found in these eight rows. This does not certify the remaining clarification targets. Earlier reviewed examples also teach asking for time and duration together when both are absent.

Eight State-Bearing ACT Targets

Single Call

Training row	State and requested operation	Target mapping	Assessment
206	One calculus planning review exists; move it to 13:00.	Uses the supplied event ID and `start_time: "13:00"`.	Correct state-bound entity selection; no invented duration or date.
211	One chemistry planning review exists; move it to 13:00.	Uses that event's supplied ID and requested time.	Correct, but closely repeats row 206's request/operation pattern.
214	One programming reading-notes task exists; mark it done.	Uses its supplied task ID and `status: "done"`.	Correct state-bound entity selection.
227	One programming problem-set task exists; mark it done.	Uses its supplied task ID and done status.	Correct, but closely repeats row 214's pattern.

Multiple Calls

Training row	State and requested operation	Target mapping	Assessment
224	Clipboard contains `statistics reading notes`; read it and create a task from it.	Reads clipboard, then sets task title to `statistics reading notes`.	Correct content transfer and complete two-call plan.

Training row	State and requested operation	Target mapping	Assessment
242	Clipboard contains `physics reading notes`; same request wording as row 224.	Reads clipboard, then uses that text as title.	Correct; varies content but not request wording.
246	Clipboard contains `chemistry notes for Nadia`; read and draft to the supplied recipient.	Reads clipboard, then uses that text as body and the supplied recipient.	Correct content transfer; subject is permissible composition.
248	One programming-outline task exists; mark it done and email the supplied recipient.	Updates the supplied task ID, then sends the supplied completion fact.	Correct entity mapping and complete sequence under the existing routine-send policy.

All eight reviewed ACT targets appear consistent with their own requests, state and tool schemas. This is target review, not a model generation or tool execution.

Coverage and Consistency Limits

The three `clipboard.read` -> `notion.create_task` examples are training rows 224, 242 and 285. All say exactly "Read my clipboard and create a task from it." Their clipboard texts differ, and all three targets correctly use the text as the task title. Thus the mapping is taught, but by only three content variations of one instruction template. The two clipboard-to-draft examples similarly vary content/recipient while using the same instruction pattern.

Some clarification target questions also repeat across different inputs/state. For example, six targets ask "Who should I send the email to?" and four ask "What time tonight should I schedule it, and how many minutes should it last?" Shared questions can be correct and useful; they are not automatically duplicate examples or evidence that the model memorized them. Counts alone do not measure semantic diversity.

The known bare-urgency policy error remains in the frozen v14 cap at rows 9, 11 and 81. These requests say to revise a subject "soon" without explicitly asking for a deadline, yet their targets ask for a due date. Under the owner's decision, the tool permits creating an undated task instead. Six other "due soon" requests explicitly request deadlines and are not the same defect.

Generator 15 corrects this generic policy rule, and its separately built source exists locally. The checkpoint's frozen v14 mix was not repointed or rebuilt. Updating generator code or building another source does not retroactively fix the data learned by an existing adapter. See the policy correction note.

The three known bad labels encourage unnecessary clarification, not failure to clarify. They cannot alone explain the checkpoint's ACT bias or all observed grounding/clarification errors. They were not silently removed from this audit's accounting or used to reinterpret historical results.

What This Means for More Training

The checked skills are not absent from training: reviewed examples teach asking for missing required fields and copying supplied state into arguments, and all 16 have nontruncated supervision in the saved diagnostic. The existing selected in-sample clarification probe nevertheless makes wrong decisions on correctly labeled seen inputs; review of coverage alone cannot explain that away.

The data also provides narrow coverage for some compositions and retains a known policy inconsistency. Repeating it for more epochs repeats both its limited patterns and its wrong labels. Therefore this audit does not justify prescribing more epochs as a fix, nor prove that adding data will solve the problem.

Keep the unweighted reference. A defensible future data experiment would use a new versioned artifact, resolve independently established policy contradictions, and test broader expression/state variation at approximately fixed class counts and training settings. Do not author validation/benchmark-specific solutions, combine that with a duration sweep, or claim improvement without a new run. No such experiment was implemented or trained in this batch.

Only local notes were added/linked. No data, source code, configuration, checkpoint, scorer or prior result was changed; nothing was staged or committed.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

Controlled Training-Data Revision: Policy and Grounding Variation

Status: Owner built, validated, trained and evaluated; candidate not promoted. Date: 2026-10-16.

Hypothesis and Scope

The training audit found correct but narrow state-grounding examples and a retained urgency-policy defect. Prepare a separate data-quality experiment before considering more epochs. This is a combined policy/wording/state intervention; an eventual result cannot identify which of those components caused any change.

Only eligible BAYMAX rows inside the existing 1,000-row training cap change. Preserve their positions and broad classes, all REFUSE targets, all xLAM rows, the rest of the training file, and validation/test bytes. No resplitting, new examples from held-out fixtures, oversampling, or new framework is introduced.

The reference is unweighted two-epoch QKVO name-first SFT, config hash `684a76cd39d845b6177bdaee495bb77bcb436406d384baf912f80b8164fb8b8b`. Its parent dataset revision is `685a6b67832a5c55f54f2fc529f5c5234a05baa93d3e0cf241c38c006b9bbc12`. The transform version is `training-grounding-v1`, not a claim that the whole dataset was regenerated by generator 15 or 16.

Previewed Changes

The model-free preview on the actual capped pool applies 49 changed rows:

Generic rule	Rows	Change
Explicit deadline clarification	3	Bare "soon" requests become explicit "due soon" requests, so their CLARIFY target is appropriate.
Missing-fields wording	4	Rephrase calendar requests and questions while retaining both missing time and duration.
Undated-task ACT	10	Replace eligible dated task requests with explicit undated urgency requests; omit due_date without inventing one.
Entity-state variation	27	Preserve the uniquely named target, vary its internal ID and position, and add an unrelated entity. Calendar distractors use a different date to avoid inventing a scheduling conflict.
Clipboard-content binding	5	Vary instruction wording and clipboard values while binding the task title or email body to that state. Task cases also include an unrelated existing task.

One additional undated-task candidate would duplicate an earlier revised input. Its original dated parent is retained unchanged. The change journal records this skip explicitly; provenance counts distinguish applied changes from skipped rows.

The urgency correction does not relabel the three CLARIFY rows into ACT, which would change quotas. It revises their requests to genuinely require a deadline; separate existing ACT slots teach the agreed undated behavior. This is a new training artifact, not a rewrite of evaluation targets or old results.

Task/clipboard variants use independent stock/equipment text and generic phrasing, not held-out scenario solutions. Unspecified draft subjects are omitted where the subject is not required; supplied recipients and content remain grounded.

No claim of sophisticated semantic diversity is made: the bank remains small, and structural row uniqueness is not a measure of conceptual independence.

Fixed Controls

Control	Retained setting
Raw cap	1,000 rows
BAYMAX contribution	400: 200 ACT, 100 CLARIFY, 100 REFUSE
External capped contribution	Same 600 xLAM rows in the same slots
Base model	Qwen/Qwen2.5-1.5B-Instruct
LoRA	r=8, alpha=16, Q/K/V/O, dropout=0.05
Loss weighting	Unweighted, first-name weight 1
Learning rate / epochs / seed	2e-4 / 2 / 1
Laptop settings for a later run	Context 1,024, batch 1, accumulation 16
Formatting / policy prompt	Same compact JSON, name-first targets and system prompt
Validation cap	Same first 200 rows; full validation/test files preserved

Raw counts are preserved, not promised post-tokenization counts. Changed wording and additional state change token exposure. Inspect the actual tokenized artifact before training; do not assume the old 974 usable rows still apply automatically.

Preparation and Integrity

`scripts/prepare_sft_grounding.py` uses a focused training-only derivation in `baymax.models.sft_grounding`. It rejects populated destinations, validates the reference identity, copies its training settings into a newly hashed config, and records source/output file hashes plus every changed row's before/after SFT messages. Parent source identities and split membership remain unchanged; the change report records transformed content rather than misrepresenting parent row hashes as new-content hashes.

Changed targets pass the existing strict contract validator and in-memory fake adapter/state validation. No live tool or external API executes. Full preparation rejects revised inputs that duplicate training inputs or overlap any protected validation/test input. It checks source counts, prompt hashes, source stability, and exact held-out byte preservation before publishing from a staging directory. Exact overlap checks do not prove semantic/template separation.

The preview validates the capped changes and counts only. It creates no dataset, config, model or tokenizer and does not replace the full preparation checks. Tiny fixture tests cover deterministic rules, grounded arguments, class/slot preservation, unchanged held-outs, duplicate/overlap rejection, and no-overwrite behavior. No full project build or training was run by the coding agent.

Owner-Run Next Step

From `project-baymax`:

```
.venv\Scripts\python scripts\prepare_sft_grounding.py

.venv\Scripts\python scripts\validate_sft_dataset.py `
  --dataset-dir data\sft\v1\name-first-grounding-v1
```

Preparation creates these new paths:

- `data/sft/v1/name-first-grounding-v1/`
- `configs/sft/qwen2_5_1_5b_lora_grounding_v1_epochs2.json`
- A future training output named `models/qwen2_5_1_5b_lora_grounding_v1_epochs2/`.

The dataset contains `grounding-changes.json` for before/after review and `grounding-provenance.json` for counts, rules and parent/output identity. Review the generated revision and post-tokenization report before starting training. Do not overwrite old artifacts or treat preview counts as benchmark improvements.

The existing unweighted checkpoint remains the reference. Any later comparison must retain compact inference, generation settings and scoring. The repeatedly inspected 150-case suite is development evidence, not an untouched final test. Local experiment Markdown and generated datasets are not staged or committed.

Owner-Reported Outcome

The owner validated revision `91ce08228633738365461a556668a6883b5faa806b84d969dddd8164114603a1`, with training config hash `f060ddbc1df6182840086ffc4b60f27785390ee2ee41938e2578bea3f3663986`. The 1,024-token artifact retained 974 training rows and 195 validation rows, including all 100 training clarification and 100 refusal rows, with zero exact duplicates. The owner then completed the two-epoch training and compact evaluation.

Run	Successes / 150	Macro-F1	Invalid / 150	Correct ACT / 81	Correct CLARIFY / 40	Correct REFUSE / 29
Unweighted reference	57	0.612	1	66	10	23
Grounding v1	53	0.591	0	66	10	21

Raw artifacts: unweighted and grounding v1. These figures come from the owner's comparison, not an invented estimate or a new comparison run by the coding agent. The revision did not improve the measured result; keep the unweighted model as the development reference. A single seed and four fewer successes do not establish a consistent regression or identify which revision component caused it. The corrections remain in a separate artifact.

The next 2k coverage experiment uses grounding v1 as its corrected 1k parent. Compare against that parent to measure expansion, while also retaining the stronger unweighted reference for context.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.

A Larger Unique Training Pool, Still Two Epochs

Status: Owner built, validated, trained and evaluated; four-epoch follow-up completed. Date: 2026-10-16.

Question

Does a larger pool with additional unique BAYMAX coverage improve on the corrected 1k model without increasing epochs or changing loss weighting?

The owner approved 2,000 raw training rows at two epochs, not 2,000 rows at four epochs. This is a size-and-coverage experiment, not an isolated test of wording diversity. At two epochs, twice as many usable rows means roughly twice the sample exposure and optimizer steps. The cosine schedule also spans the longer run.

Parent and Reference

The immediate parent is grounding v1:

- Config: `configs/sft/qwen2_5_1_5b_lora_grounding_v1_epochs2.json`.
- Dataset: `data/sft/v1/name-first-grounding-v1`.
- Dataset revision: `91ce08228633738365461a556668a6883b5faa806b84d969dddd8164114603a1`.
- Training config hash: `f060ddbc1df6182840086ffc4b60f27785390ee2ee41938e2578bea3f3663986`.

This retains the independently justified corrections, rather than restoring incorrect examples to recover a score. Grounding v1 scored 53/150 successes and 0.591 macro-F1. The unweighted reference remains the selected model at 57/150 and 0.612; do not silently replace it with this parent or the new experiment.

Exact Raw Quantities

Contribution	Parent 1k	Added	Expanded 2k
BAYMAX single productive calls	175	175	350
BAYMAX multi-tool sequences	25	25	50
BAYMAX clarification	100	100	200
BAYMAX refusal	100	100	200
xLAM productive examples	600	600	1,200
Total	1,000	1,000	2,000

The 2k raw mixture is 800 BAYMAX and 1,200 xLAM, with broad routing quantities 1,600 ACT / 200 CLARIFY / 200 REFUSE. These are not post-tokenization counts. The raw 1k parent was checked model-free: 1,000 unique inputs, zero exact duplicates, and the BAYMAX quantities above. The larger artifact still needs owner inspection.

Where the New Examples Come From

`sft_expansion_examples.py` independently authors 100 training-only groups. Each has two ACT examples, one CLARIFY and one REFUSE. Eight families cover:

- Clipboard-to-task content binding, including an empty-clipboard contrast.
- Calendar creation with explicit timing versus missing time and/or duration.
- Calendar updates using a unique state entity versus multiple matching entities.
- Task deadlines: known dates, explicitly undated urgency and unresolved deadlines.
- Task completion using state IDs versus ambiguous same-title tasks.
- Routine email with known/contact-resolved recipients versus missing addresses; indiscriminate bulk outreach is refused.
- Clipboard-to-draft body binding versus missing contents or unavailable tools.
- Draft recipients resolved from state; a send request is refused when only drafts are supported.

Instruction banks contain three to six phrasings per relevant family. Contents, dates, start times, durations, contact state, entity IDs and distractors vary. The pool is still templated: 400 unique inputs are not 400 independent reasoning patterns, and no claim of unrestricted linguistic generalization is made.

All additions are explicitly training-only. No evaluation fixture or held-out target supplies their content. Existing source validation executes only fake in-memory adapters, and strict schema checks validate all generated targets.

Sampling and Separation

`scripts/prepare_sft_expansion.py` derives a new dataset and a freshly hashed config. It does not change the default mixer, its fixed BAYMAX quotas, or any old artifact.

The original first 1,000 training lines are retained byte-for-byte. A new 1,000-row block is shuffled with the existing seed and placed after them. The 600 external additions are the next eligible unique xLAM rows from the parent's training tail; they move into the cap and are removed from that tail, not copied twice.

Retained parent rows keep their source identity and split. Full validation/test files remain byte-identical. New inputs must be distinct from every parent input and from each other. A frozen-prefix input overlapping any held-out input blocks publication; conflicting unused training-tail inputs are excluded and recorded. They cannot be selected as xLAM additions. Exact input checks are not proof of semantic/template independence. No new held-out set is created; inherited split-ratio metadata describes the parent split policy, not a new 70/20/10 split of the expanded data.

Before any tail-overlap exclusions, the full dataset grows by 400 BAYMAX rows, not 1,000, because external rows are relocated. Review `expansion-sources.jsonl` for all authored examples and `expansion-report.json` for source hashes, moved external identities, before/after cap counts, unique inputs and duplicate counts. Populated/overlapping destinations, quota mismatches, insufficient unique data and source drift fail explicitly.

Build Guard Correction

The first owner build stopped with an input overlap in the parent's training tail. The historical splitter/validator compares complete examples, including answers; identical inputs with different answers can therefore remain in different splits. This is not evidence that the separate 150 benchmark scenarios were used for training. A bounded read-only check found zero held-out input matches in the parent's first 2,000 training rows; the rest was not audited by the coding agent.

The corrected preparation preserves the frozen 1k prefix and excludes conflicts only outside it, with source references and held-out locations in `expansion-report.json:excluded_parent_tail` and manifest skipped rows. Source files and historical results are untouched. Full builds still check all inputs. Regression tests reproduce a same-input/different-answer overlap that passes the old validator and prove both safe tail exclusion and strict frozen-prefix failure. The new guard policy is included in the derivation hash.

Use new `coverage-2k-v2` output/config identities; the failed v1 folder contains only an invalid validation report and is not reused. Do not validate a failed build.

Fixed Training Settings

Qwen2.5-1.5B-Instruct; fresh LoRA r=8, alpha=16, Q/K/V/O; dropout 0.05; learning rate 2e-4; two epochs; seed 1; unweighted first-tool loss; same prompt, compact input serialization, name-first targets and generation/scoring settings. Laptop overrides remain context 1,024, batch one and accumulation 16. No adapter is resumed. Validation cap remains the same 200 rows.

Owner-Run Steps

```
.venv\Scripts\python scripts\prepare_sft_expansion.py
if ($LASTEXITCODE -ne 0) { throw "Preparation failed; do not validate or train." }

.venv\Scripts\python scripts\validate_sft_dataset.py `
  --dataset-dir data\sft\v1\name-first-coverage-2k-v2

$config = "configs\sft\qwen2_5_1_5b_lora_coverage_2k_v2_epochs2.json"

.venv\Scripts\python scripts\report_sft_artifact.py `
  --config $config --split train `
  --max-rows 2000 --max-sequence-length 1024

.venv\Scripts\python scripts\report_sft_artifact.py `
  --config $config --split validation `
  --max-rows 200 --max-sequence-length 1024
```

Review those counts and the source examples before training. In particular, both 200-row boundary classes should retain labels, and no duplicates should appear. After review, the intended fresh run is:

```
.venv\Scripts\python scripts\train_sft_lora.py `
  --config configs\sft\qwen2_5_1_5b_lora_coverage_2k_v2_epochs2.json `
```

```
  --local-low-vram `
  --max-train-rows 2000 --max-eval-rows 200 `
  --max-sequence-length 1024
```

The owner reviewed the tokenized artifact before training. The coding agent did not run the full build, tokenizer, training or evaluation. The repeatedly inspected 150-case suite is development evidence, not an untouched final test.

Completed Artifact and Results

Dataset revision: `43892841fad91d9c328f3577e831c5d192242f84ceabe82d18e7deb30d2c6ab4`. Config hash: `e48c75d01c9c14eb6ef1c24d4e8d1ae516fc3bf164a288918517c6f1ce2db524`. The owner build excluded four unused-tail overlaps. The final full dataset has 58,419 converted rows; train 40,821, validation 11,719 and test 5,879.

Tokenized training class	Rows	Trainable tokens
Direct tool call	885	27,313
Multi-tool sequence	667	34,244
Clarification	200	5,789
Refusal	200	5,515

The usable 1,952-row cap contains all 800 BAYMAX rows and 1,152 xLAM rows, with zero exact duplicates. The unchanged validation artifact retains 195 rows, including 30 clarification and 30 refusal examples.

Run	Successes / 150	Macro-F1	Invalid / 150	Argument accuracy	Clarification wording	Refusal wording
Grounding 1k	53	0.591	0	0.826	0.163	0.379
Coverage 2k / 2 epochs	64	0.615	2	0.806	0.163	0.603

The saved two-epoch run is timestamped 2026-10-16T19:43:41Z. Its matrix has expected rows and predicted columns:

Expected	ACT	CLARIFY	REFUSE	Invalid/unknown
ACT	59	15	6	1
CLARIFY	24	12	4	0
REFUSE	3	0	25	1

The observed size/coverage change improves task completion and refusal compared with the corrected 1k parent, but still acts prematurely in 24/40 clarification cases. It does not isolate data diversity from additional training exposure. The four-epoch follow-up changes duration separately.

The Same 2k Pool at Four Epochs

Status: Owner training/evaluation completed; final reported local candidate at project closeout. Date: 2026-10-17.

Question and Controls

The owner requested a fresh four-epoch run on the same expanded 2k dataset. Compare against that dataset's two-epoch run, not the earlier 1k duration trial. The unweighted compact/name-first checkpoint remains the historical development reference. The four-epoch candidate's measured result is recorded below; no statistical significance or untouched-test generalization is established.

Only `num_train_epochs` changes from 2 to 4. Run/config identity, output paths and config hash also change so the two-epoch artifact is not overwritten. Data, split membership, prompt, LoRA, learning rate, seed, loss weighting and laptop overrides stay fixed. This starts from the base model with a fresh LoRA adapter, not from the saved two-epoch adapter. The cosine schedule spans the longer training horizon, so this is not an identical first two epochs followed by extra updates.

Identity

Item	Value
Config	`configs/sft/qwen2_5_1_5b_lora_coverage_2k_v2_epochs4.json`
Config SHA256	`da38eb64730f8db30b1bf1a5d6f5296a805351598ca2057a1f70e6874141f2e5`
Dataset	`data/sft/v1/name-first-coverage-2k-v2`
Dataset revision	`43892841fad91d9c328f3577e831c5d192242f84ceabe82d18e7deb30d2c6ab4`
Base model	`Qwen/Qwen2.5-1.5B-Instruct`
LoRA	r=8, alpha=16, Q/K/V/O, dropout=0.05
Learning rate / epochs / seed	2e-4 / 4 / 1
First-tool loss weighting	1x, unweighted
Laptop settings	Context 1,024, batch one, accumulation 16
Output	`models/qwen2_5_1_5b_lora_coverage_2k_v2_epochs4/adapter`

The owner reported 1,952 usable training rows from a 2,000-row raw cap: 800 BAYMAX, 1,152 xLAM, 200 clarification and 200 refusal rows, with zero exact duplicates. The same validation cap yields 195 usable rows. These are existing artifact counts, not a new four-epoch measurement. At the same usable counts, four epochs

give approximately twice the sample exposure of two epochs. Model quality is measured separately by the saved evaluation below.

Owner Command

```
.venv\Scripts\python scripts\train_sft_lora.py `
  --config configs\sft\qwen2_5_1_5b_lora_coverage_2k_v2_epochs4.json `
  --local-low-vram `
  --max-train-rows 2000 `
  --max-eval-rows 200 `
  --max-sequence-length 1024
```

No dataset rebuild or additional examples are needed. The coding agent did not run training or evaluation. Preserve raw results and compare both durations under the same compact JSON input, greedy decoding, 150-scenario suite and scoring. Report task success, routing macro-F1, invalid outputs and the decision matrix; lower training loss alone is not a model-quality win. This repeatedly inspected suite is development evidence, not an untouched final test.

Saved Evaluation Results

Sources: two epochs and four epochs. The four-epoch result is timestamped 2026-10-17T16:52:19Z. Both use the same 150 scenarios, compact JSON, temperature zero, at most 256 generated tokens and fake adapters. Values below are transcribed from saved artifacts, not re-evaluated.

Measurement	Two epochs	Four epochs
Task successes / 150	64	73
Task success rate	0.427	0.487
Routing macro-F1	0.615	0.681
Invalid outputs / 150	2	0
Tool-call accuracy	0.667	0.753
Argument accuracy	0.806	0.836
Clarification wording accuracy	0.163	0.350
Refusal wording accuracy	0.603	0.690
Conditional decision accuracy	0.649	0.707
Decision coverage	0.987	1.000
Hallucination rate	0.000	0.007
Mean inference latency, ms	2,758	5,652
Reported API cost, USD	0.0000	0.0000

Four epochs, expected rows and predicted columns:

Expected	ACT	CLARIFY	REFUSE	Invalid/unknown
ACT	62	13	6	0
CLARIFY	17	17	6	0
REFUSE	1	1	27	0

The recorded four-epoch run completes nine additional tasks and has better macro-F1 than the two-epoch counterpart. Compared with the historical unweighted 1k reference, it completes 73 versus 57 tasks and records macro-F1 0.681 versus 0.612. This is a useful development result for closing the project, not proof that four epochs always help: the earlier 1k four-epoch trial regressed, and no repeated training seeds were measured here.

Clarification remains incomplete: 17/40 requests that should clarify still cause ACT, and 6/40 cause REFUSE. ACT recall is 62/81, and refusal recall is 27/29. The recorded latency is higher; hardware/background conditions were not controlled well enough to attribute that solely to the training duration. A zero local API cost field does not mean zero hardware, energy or training cost.

The project owner accepted the work as complete on 2026-10-17. This does not imply production readiness, real integrations, a new MLX checkpoint release or an untouched final evaluation. Model weights and generated training data remain local and are not part of the Git closeout.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.